MINISTRY OF EDUCATION AND TRAINING

TRƯỜNG ĐẠI HỌC FPT

FPT UNIVERSITY

Software Requirement Specification

Sport Center Management System

Group Members	Nguyễn Văn Trọng Hiếu Đặng Trang Thanh Sử Nguyễn Trọng Toàn Phạm Duy Tân Nguyễn Hoàng Phúc
Supervisor	Tôn Thất Hoàng Minh

- HCM, 10/2026 -

Record of Changes

Date	Version	A/M/D	In charge	Description
2026-10-12	2.3	A, M	Arena3 team	Restructured to the FPT SRS/SDD template: front matter, heading tree, table formats, the five-block function specification, Message IDs and the software design part. Added Definition and Acronyms, screen layouts, the logical ERD, the system architecture, the package diagram and two sequence diagrams; split the class diagrams and the conceptual ERD in two. Restated requirements with "must" and gave each actor one name.
2026-10-12	2.2.1	M	Arena3 team	Redrew the context, use case, activity, screen flow and sequence diagrams to the FPT templates and Learning UML 2.0. Aligned the secondary actors of 19 use cases with their flows, and removed Gemini AI from UC-32 to match the code.
2026-10-12	2.2	M	Arena3 team	Brought the figures in line with Learning UML 2.0: merge nodes in the activity diagrams, goal-phrase ovals without UC codes in the use case diagrams, and unnumbered messages in the sequence diagrams. Redrew the context diagram as a level-0 data flow diagram.
2026-10-12	2.1	M	Arena3 team	Consolidated 86 use cases into 44, each with a full specification. Restated the business rules as BR-01 to BR-53, grouped by category and typed by rule type. Redrew the use case, activity and sequence figures they affect.
2026-10-11	2.0	M	Arena3 team	Redrew all 28 UML figures to the FPT template and Learning UML 2.0 rules. Aligned every section with the deployed app.
2026-10-04	1.4.1	A, D	Arena3 team	Added QR gate check-in, promo codes, at-risk list, deposits, fixed weekly bookings, day passes and loyalty points. Removed member debt and court conversion.
2026-09-27	1.4	M	Arena3 team	Traced each requirement to its endpoint or business rule. Added online payment by payOS.
2026-09-20	1.1	A	Arena3 team	Added receipts, customer care requests and the training module (feature flag F4).

**A – Added, M – Modified, D – Deleted*

Table of Contents

I. Software Requirement Specification 8
1. Product Overview 8
2. User Requirements 10
2.1 Actors 10
2.2 Use Cases 10
2.2.1 Diagram(s) 10
2.2.1.1 Account and Access 10
2.2.1.2 Member Court Booking and Payment 11
2.2.1.3 Member Classes and Training 12
2.2.1.4 Front Desk Sales 13
2.2.1.5 Front Desk Operations 14
2.2.1.6 Coach 15
2.2.1.7 Manager Setup 16
2.2.1.8 Manager Oversight 17
2.2.2 Descriptions 18
3. Functional Requirements 21
3.1 System Functional Overview 21
3.1.1 Screens Flow 21
3.1.1.1 Public Site 21
3.1.1.2 Member Portal 21
3.1.1.3 Front Desk Portal 22
3.1.1.4 Coach Portal 23
3.1.1.5 Manager Portal 23
3.1.2 Screen Descriptions 24
3.1.2.1 Public Site 24
3.1.2.2 Member Portal 24
3.1.2.3 Front Desk Portal 25
3.1.2.4 Coach Portal 25
3.1.2.5 Manager Portal 26
3.1.3 Screen Authorization 26
3.1.4 Non-Screen Functions 29
3.1.5 Entity Relationship Diagram 30
3.1.5.1 Conceptual Entity Relationship Diagram 30
3.1.5.2 Logical Entity Relationship Diagram 31
Entities Description 34
3.2 Account and Access 35
3.2.1 Browse Courts, Classes and Prices 36
3.2.2 Register a Member Account 38
3.2.3 Sign In and Sign Out 42
3.2.4 Reset a Forgotten Password 44
3.2.5 Manage the Personal Profile 46
3.2.6 View Notifications 48

3.3 Member Court Booking and Payment 50
3.3.1 View My Schedule 50
3.3.2 Book a Court and Pay 52
3.3.3 Reschedule or Cancel a Booking 56
3.3.4 Order a Membership Plan 58
3.3.5 View Receipts and Redeem Points 61
3.3.6 Check In by Self-Service 63
3.4 Member Classes and Training 65
3.4.1 Enrol in a Class 65
3.4.2 Accept a Waitlist Offer 68
3.4.3 Leave a Class 70
3.4.4 Follow My Training Progress 71
3.4.5 Get Help from the AI Assistant or Staff 73
3.5 Front Desk Sales 75
3.5.1 Open and Close a Cashier Shift 75
3.5.2 Manage a Member Record at the Desk 77
3.5.3 Manage a Member Plan at the Desk 79
3.5.4 Book a Court at the Desk 82
3.5.5 Settle Pending Payments 84
3.5.6 Sell and Redeem a Day Pass 87
3.5.7 Raise a Refund 89
3.6 Front Desk Operations 92
3.6.1 Check In a Court Booking 92
3.6.2 Check a Member In at the Gate 94
3.6.3 Rent Out Gear and Record Its Return 96
3.6.4 Handle Maintenance Work Orders 99
3.6.5 Reply to a Support Request 101
3.7 Coach Teaching and Earnings 103
3.7.1 View the Teaching Schedule and Rosters 103
3.7.2 Take Session Attendance 104
3.7.3 Plan a Training Session 106
3.7.4 Record Session Results and Homework 108
3.7.5 Assess a Student's Progress 110
3.7.6 View Monthly Earnings 112
3.8 Manager Setup 114
3.8.1 Manage User Accounts 114
3.8.2 Manage Membership Plans and Prices 116
3.8.3 Manage Promo Codes 118
3.8.4 Schedule Classes and Sessions 120
3.8.5 Configure the Centre 123
3.9 Manager Oversight 125
3.9.1 View Reports and Export Data 125
3.9.2 Approve or Reject a Refund 127
3.9.3 Close and Pay Coach Commission 129
3.9.4 Review the Audit Log 131

4. Non-Functional Requirements 133
4.1 External Interfaces 133
4.1.1 User Interface 133
4.1.2 Communications Interfaces 133
4.2 Quality Attributes 134
4.2.1 Usability 134
4.2.2 Reliability 134
4.2.3 Performance 134
4.2.4 Security 134
4.2.5 Auditability 135
4.2.6 Maintainability 135
5. Requirement Appendix 136
5.1 Business Rules 136
5.2 Common Requirements 138
Error Responses 139
Configurable Settings 139
5.3 Application Messages List 140
II. Software Design Description 142
1. System Design 142
1.1 System Architecture 142
1.2 Package Diagram 142
Package Descriptions 143
2. Database Design 144
Table Descriptions 144
3. Detailed Design 149
3.1 Booking and Payment 149
3.1.1 Class Diagram 149
3.1.2 Sequence Diagram 152
3.2 Membership and Classes 154
3.2.1 Class Diagram 154
3.2.2 Sequence Diagram 156
3.3 Coaching and Training 157
3.3.1 Class Diagram 157
3.3.2 Sequence Diagram 159
3.4 Centre Operations 160
3.4.1 Class Diagram 160
3.4.2 Sequence Diagram 162

List of Figures

Figure 1. Context diagram of the Arena3 Sports Centre Management System 8
Figure 2. Use case diagram: Account and Access 11
Figure 3. Use case diagram: Member Court Booking and Payment 12
Figure 4. Use case diagram: Member Classes and Training 13
Figure 5. Use case diagram: Front Desk Sales 14
Figure 6. Use case diagram: Front Desk Operations 15
Figure 7. Use case diagram: Coach 16
Figure 8. Use case diagram: Manager Setup 17
Figure 9. Use case diagram: Manager Oversight 18
Figure 10. Screens flow: Public Site 21
Figure 11. Screens flow: Member Portal 22
Figure 12. Screens flow: Front Desk Portal 22
Figure 13. Screens flow: Coach Portal 23
Figure 14. Screens flow: Manager Portal 24
Figure 15. Conceptual ERD: courts, bookings and payments 30
Figure 16. Conceptual ERD: members, plans and classes 31
Figure 17. Logical ERD, part 1: accounts and platform 32
Figure 18. Logical ERD, part 2: courts and bookings 32
Figure 19. Logical ERD, part 3: payments, plans and promotions 33
Figure 20. Logical ERD, part 4: classes, attendance and training 33
Figure 21. Logical ERD, part 5: coaching records and pay 34
Figure 22. Activity diagram: Register a Member Account 41
Figure 23. Activity diagram: Book a Court and Pay 55
Figure 24. Activity diagram: Enrol in a Class or Join the Waitlist 67
Figure 25. Activity diagram: Raise and Approve a Refund 91
Figure 26. System architecture 142
Figure 27. Package diagram 143
Figure 28. Class diagram: courts and bookings 150
Figure 29. Class diagram: payments 151
Figure 30. Sequence diagram: book a court and pay by payOS QR 153
Figure 31. Sequence diagram: raise and approve a refund 154
Figure 32. Class diagram: membership plans 155
Figure 33. Class diagram: classes and attendance 156
Figure 34. Sequence diagram: enrol in a class 157
Figure 35. Class diagram: coach and member roles 158
Figure 36. Class diagram: coach pay, plans and homework 159
Figure 37. Sequence diagram: take session attendance 160
Figure 38. Class diagram: accounts and member care 161
Figure 39. Class diagram: centre and equipment 162
Figure 40. Sequence diagram: register a member account 163
Figure 41. Sequence diagram: check a member in at the gate 164

Definition and Acronyms

Acronym	Definition
AI	Artificial Intelligence
API	Application Programming Interface
BR	Business Rule
ERD	Entity Relationship Diagram
FK	Foreign Key
FPT	FPT University (the issuing university)
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ID	Identifier
JSON	JavaScript Object Notation
NFR	Non-Functional Requirement
OTP	One-Time Password
PDF	Portable Document Format
PK	Primary Key
QR	Quick Response (code)
REST	Representational State Transfer
SDD	Software Design Description
SHA	Secure Hash Algorithm
SMS	Short Message Service
SMTP	Simple Mail Transfer Protocol
SRS	Software Requirement Specification
UC	Use Case
UI	User Interface
UML	Unified Modeling Language
VAT	Value-Added Tax
VND	Vietnamese đồng, the currency (also written ₫)
XML	Extensible Markup Language

I. Software Requirement Specification

1. Product Overview

Arena3 is one web application that runs a single sports centre in Ho Chi Minh City, from court booking to the end-of-day till. It serves badminton, basketball and volleyball courts, membership plans, coached classes and the front desk.

The context diagram in Figure 1 shows Arena3 as one process exchanging data with five human actors and three external systems.

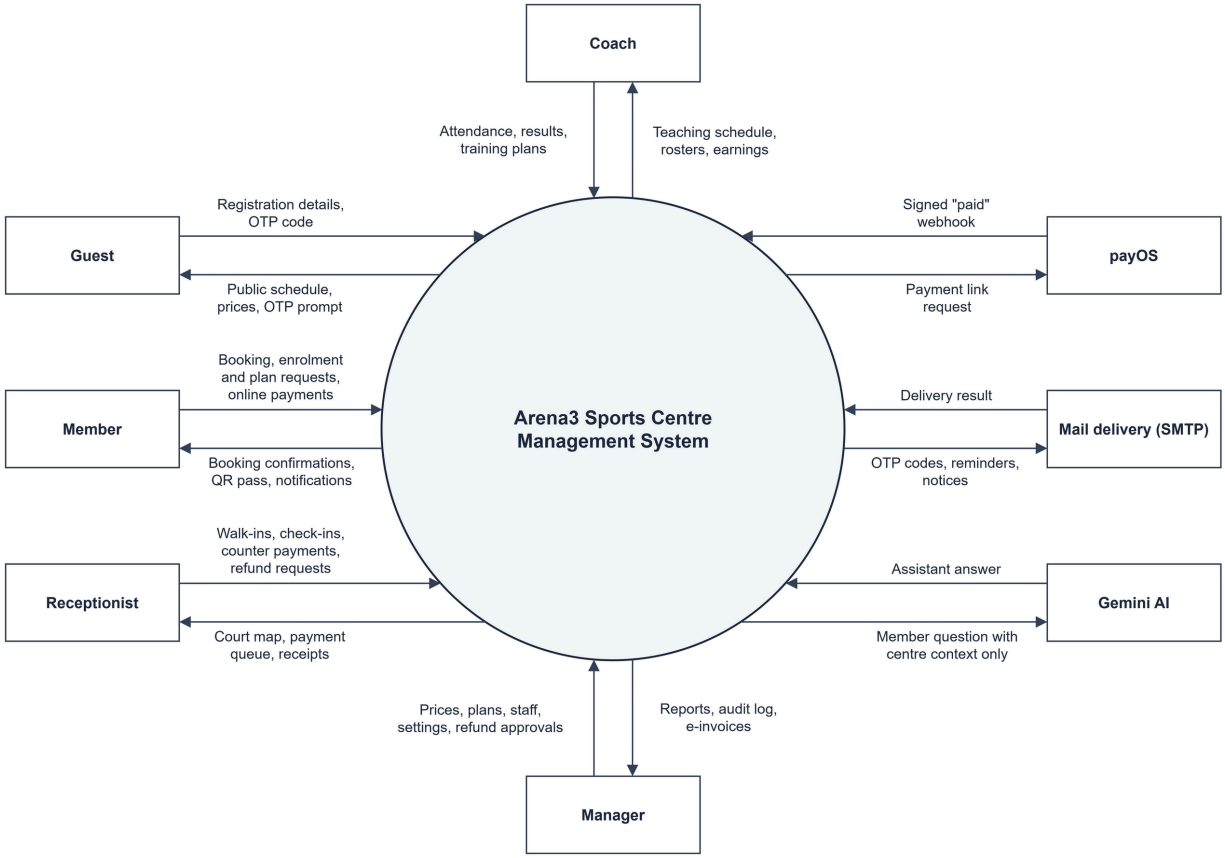


Figure 1. *Context diagram of the Arena3 Sports Centre Management System*

External entity	Data into Arena3	Data out of Arena3
Guest	Registration details, OTP code	Public schedule, prices, OTP prompt
Member	Booking, enrolment and plan requests, online payments	Booking confirmations, QR pass, notifications
Receptionist	Walk-ins, check-ins, counter payments, refund requests	Court map, payment queue, receipts
Coach	Attendance, results, training plans	Teaching schedule, rosters, earnings
Manager	Prices, plans, staff, settings, refund approvals	Reports, audit log, e-invoices

External entity	Data into Arena3	Data out of Arena3
payOS	Signed "paid" webhook	Payment link request
Mail delivery (SMTP)	Delivery result	OTP codes, reminders, notices
Gemini AI	Assistant answer	Member question with centre context only

Each portal of the application serves one group of users.

Portal	Route	Who uses it	Main purpose
Public site	`/`, `/login`, `/register`	Guest	Browse prices and classes, sign up, sign in
Member portal	`/app`	Member	Book courts, pay, enrol in classes, show the gate pass
Front desk	`/desk`	Receptionist	Serve walk-ins, take payments, run the gate and the till
Coach portal	`/coach`	Coach	Teach classes, take attendance, track student progress
Manager portal	`/manager`	Manager	Set prices and rules, staff the centre, read reports

The app is built with TanStack Start and React, with a JSON API under `/v1` in the same deployment. Production runs on Vercel with Neon PostgreSQL. Development runs offline on an embedded PGLite database that migrates and seeds itself.

Out of scope. Multi-branch operation, payroll, native mobile apps and SMS delivery are not part of this release. Class fees are covered by membership plans and are not charged per session.

2. User Requirements

2.1 Actors

Arena3 has five human actors and three external system actors. The four staff and member roles specialise one abstract Registered User, who can sign in.

#	Actor	Description
1	Guest	Human actor. Anyone not signed in. Browses prices and classes and registers a member account.
2	Registered User	Abstract actor. Any signed-in person. Owns the sign-in, profile, password and notification use cases.
3	Member	Human actor. A customer with an account. Books courts, buys plans, joins classes and enters with a QR pass.
4	Receptionist	Staff actor. Runs the front desk and the till: walk-ins, payments, gate, gear, refunds up to the limit.
5	Coach	Staff actor. Teaches classes. Takes attendance and records results, plans, homework and reviews.
6	Manager	Staff actor. Sets prices, plans, classes and rules. Issues staff accounts, approves large refunds and reads reports.
7	payOS	System actor. Vietnamese payment gateway. Issues VietQR payment links and sends signed "paid" webhooks.
8	Mail delivery (SMTP)	System actor. Sends one-time codes, reminders and notices by email.
9	Gemini AI	System actor. Google's model. Answers member questions from centre data only.

Staff accounts are issued by a manager; there is no public staff sign-up and no admin role.

2.2 Use Cases

2.2.1 Diagram(s)

Each diagram covers one feature area inside one system boundary. Every oval is a complete goal of its actor, named by a short verb phrase; table 2.2.2 gives each goal its UC number, which the specifications (3.2 to 3.9) and the activity and sequence diagrams use. Steps such as verifying a one-time code or paying by payOS QR are flows inside their use case, so the diagrams need no «include» or «extend» relationships. Associations carry no arrowheads. Primary actors stand left and secondary actors right. An actor is linked only to the use cases in which it interacts with Arena3; people who are only served at the counter or notified later are not actors of those use cases.

2.2.1.1 Account and Access

Figure 2 shows the guest and registered-user goals, with the four roles that specialise Registered User.

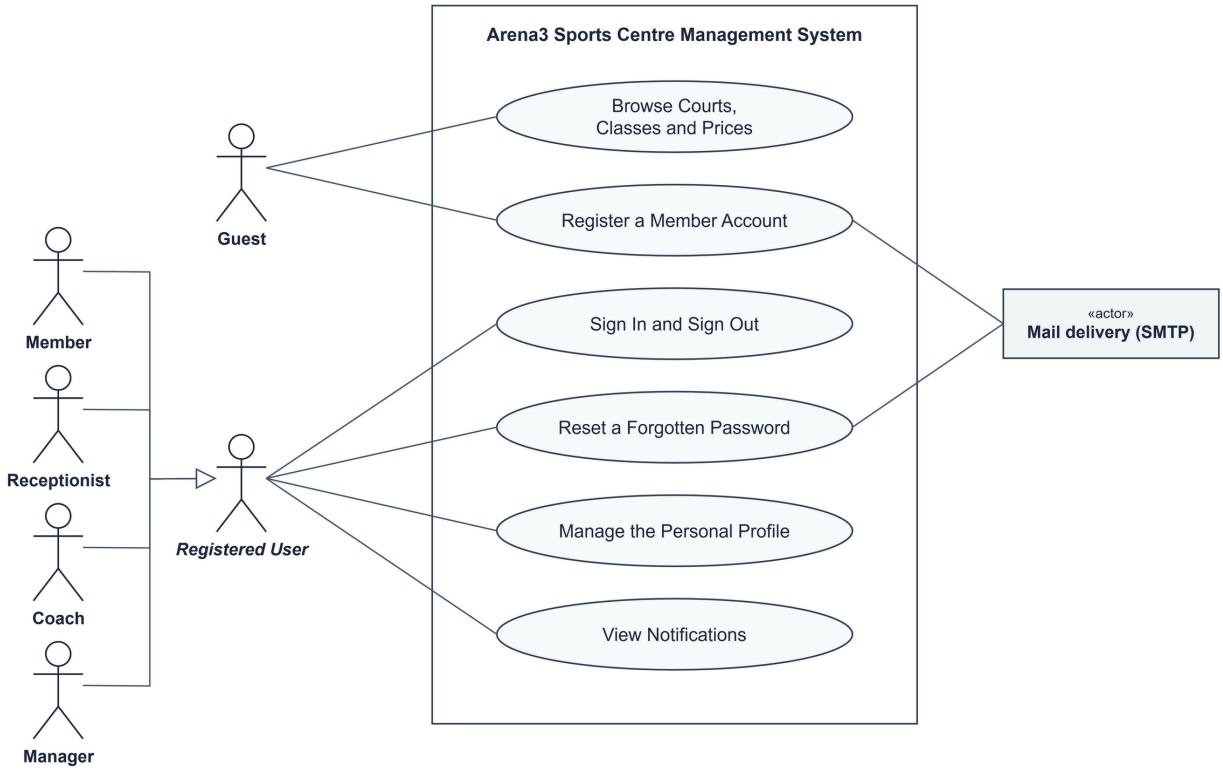


Figure 2. *Use case diagram: Account and Access*

2.2.1.2 Member Court Booking and Payment

Figure 3 shows the member's court booking, plan and payment goals.

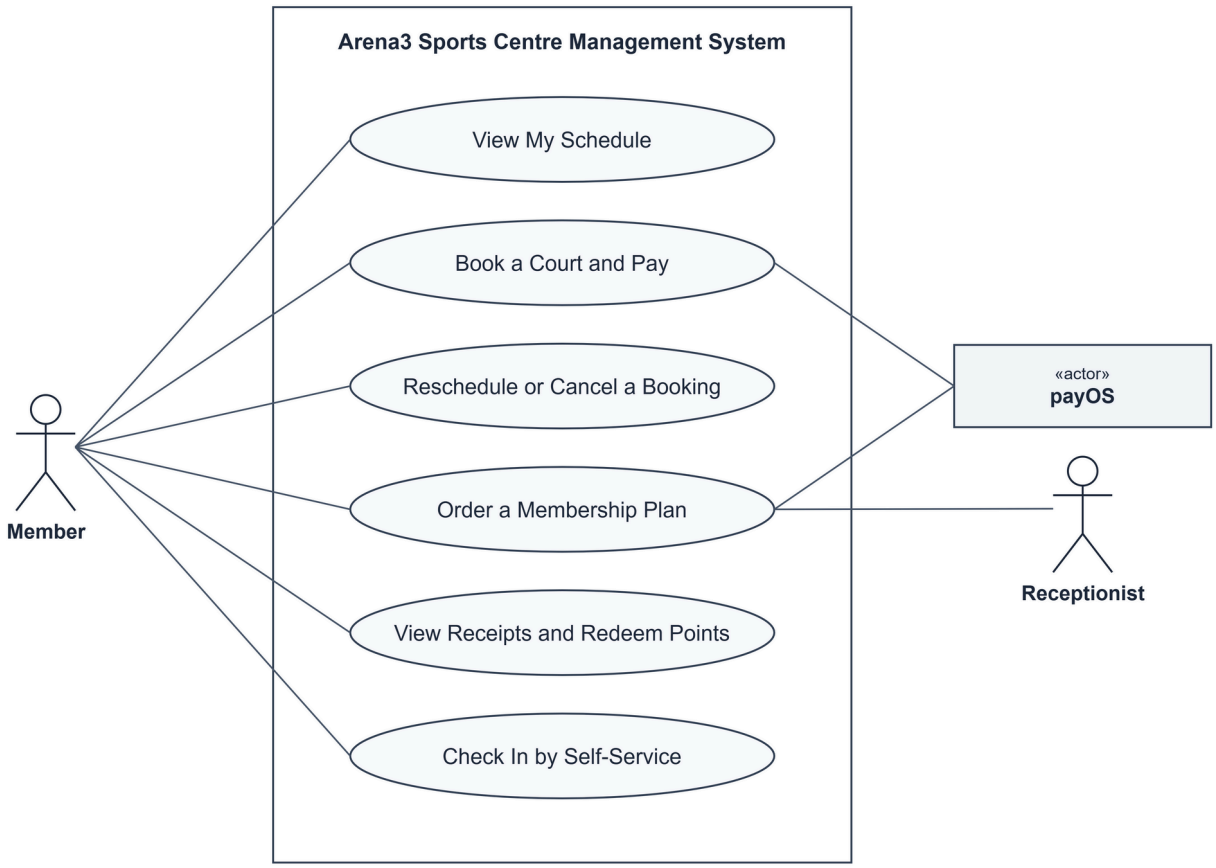


Figure 3. *Use case diagram: Member Court Booking and Payment*

2.2.1.3 Member Classes and Training

Figure 4 shows the member's class, training and help goals.

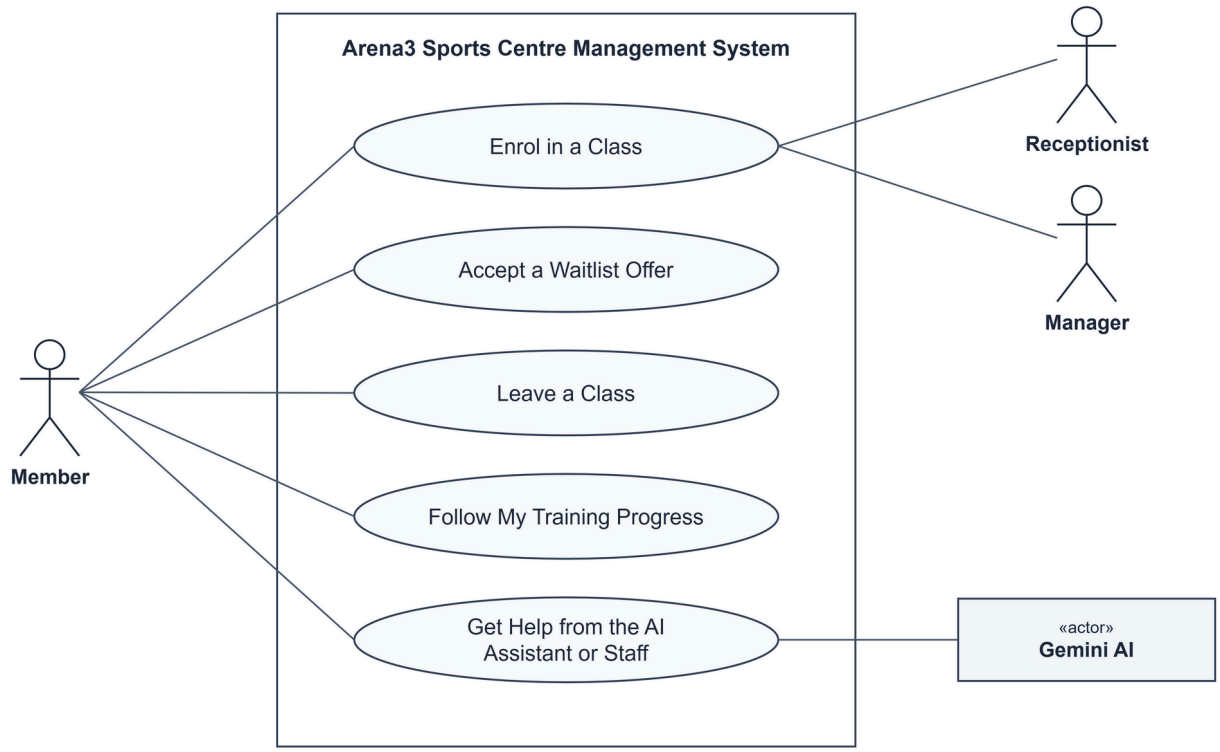


Figure 4. *Use case diagram: Member Classes and Training*

2.2.1.4 Front Desk Sales

Figure 5 shows the receptionist's selling and till goals.

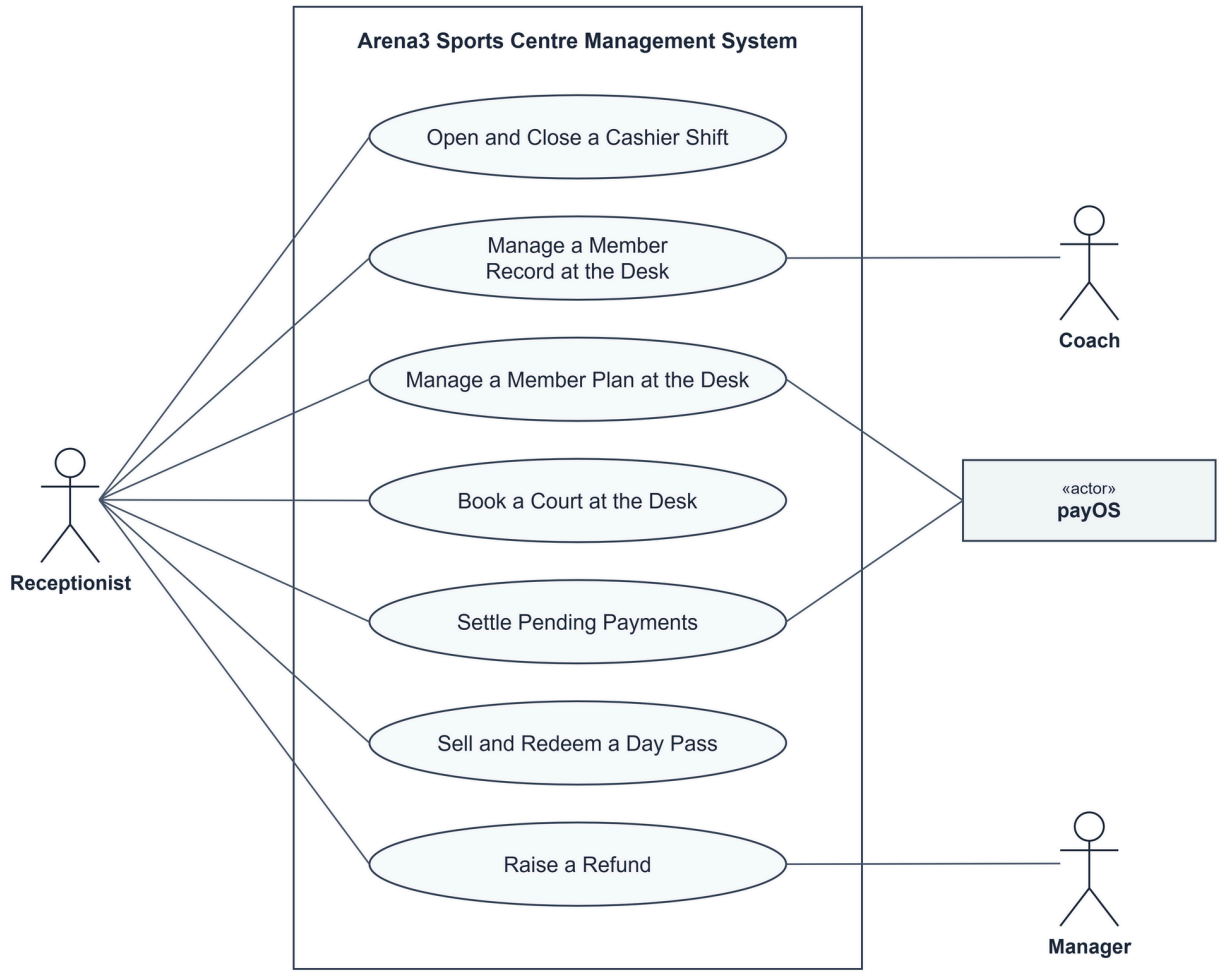


Figure 5. *Use case diagram: Front Desk Sales*

2.2.1.5 Front Desk Operations

Figure 6 shows the receptionist's daily operations, with the manager on maintenance work orders.

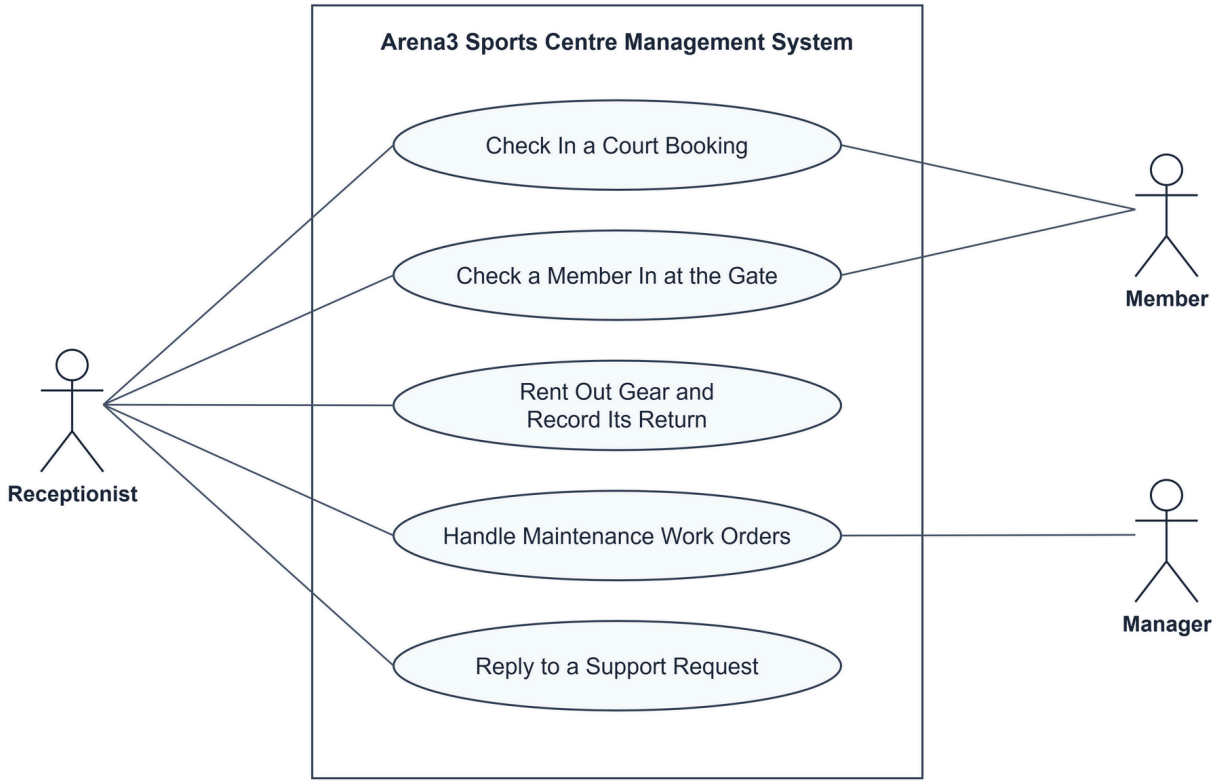


Figure 6. *Use case diagram: Front Desk Operations*

2.2.1.6 Coach

Figure 7 shows the coach's teaching and earnings goals.

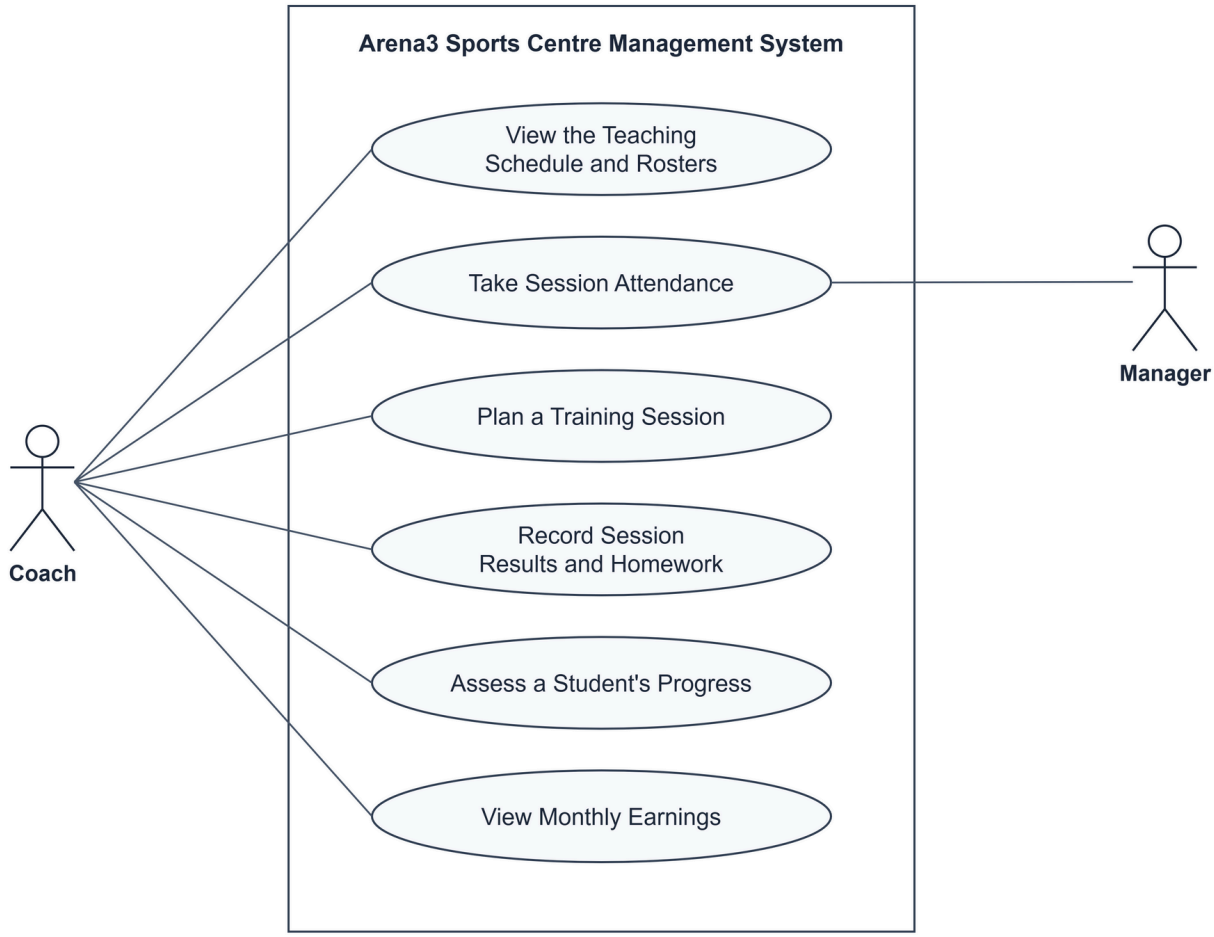


Figure 7. *Use case diagram: Coach*

2.2.1.7 Manager Setup

Figure 8 shows the manager's set-up goals.

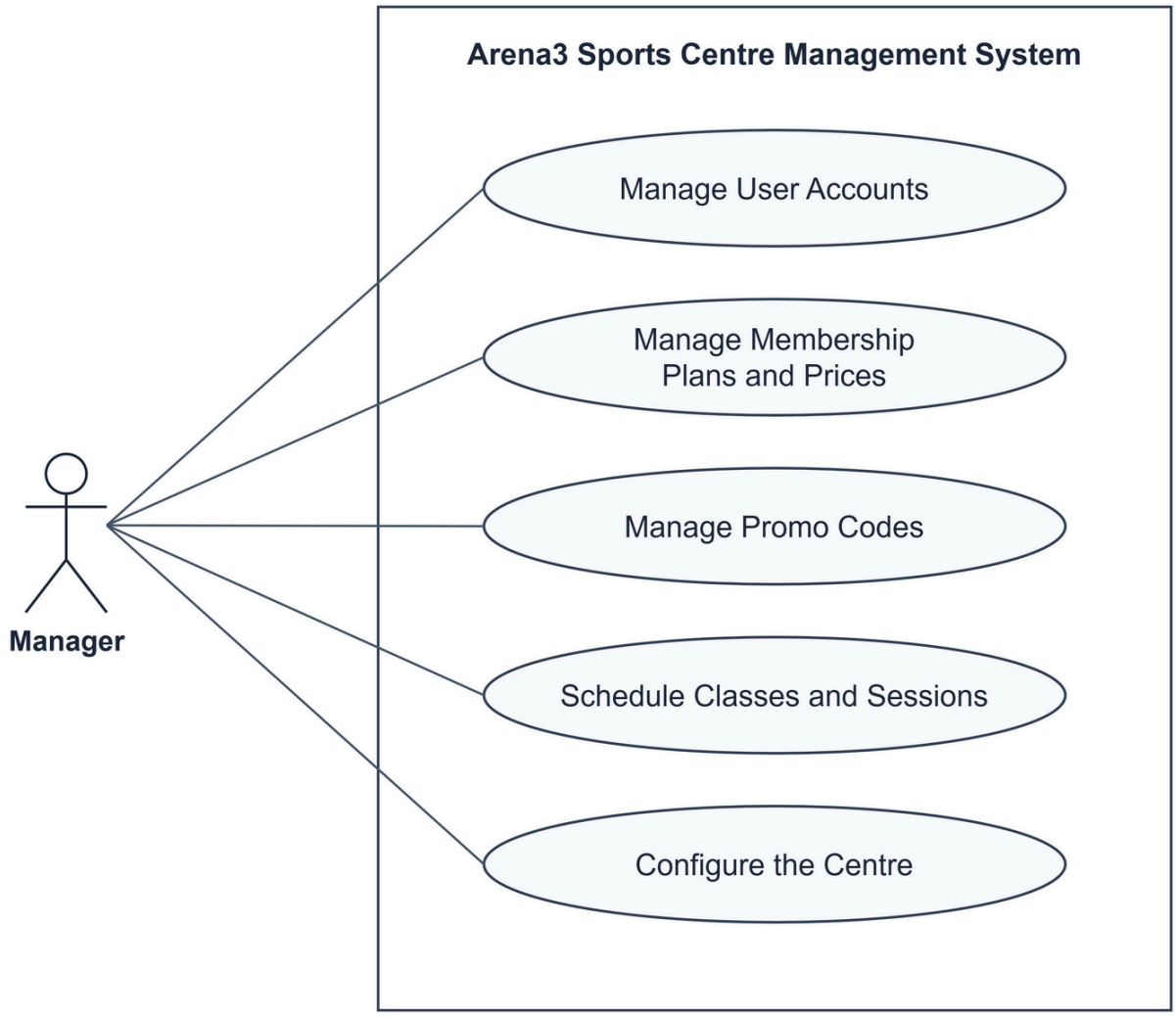


Figure 8. *Use case diagram: Manager Setup*

2.2.1.8 Manager Oversight

Figure 9 shows the manager's oversight goals.

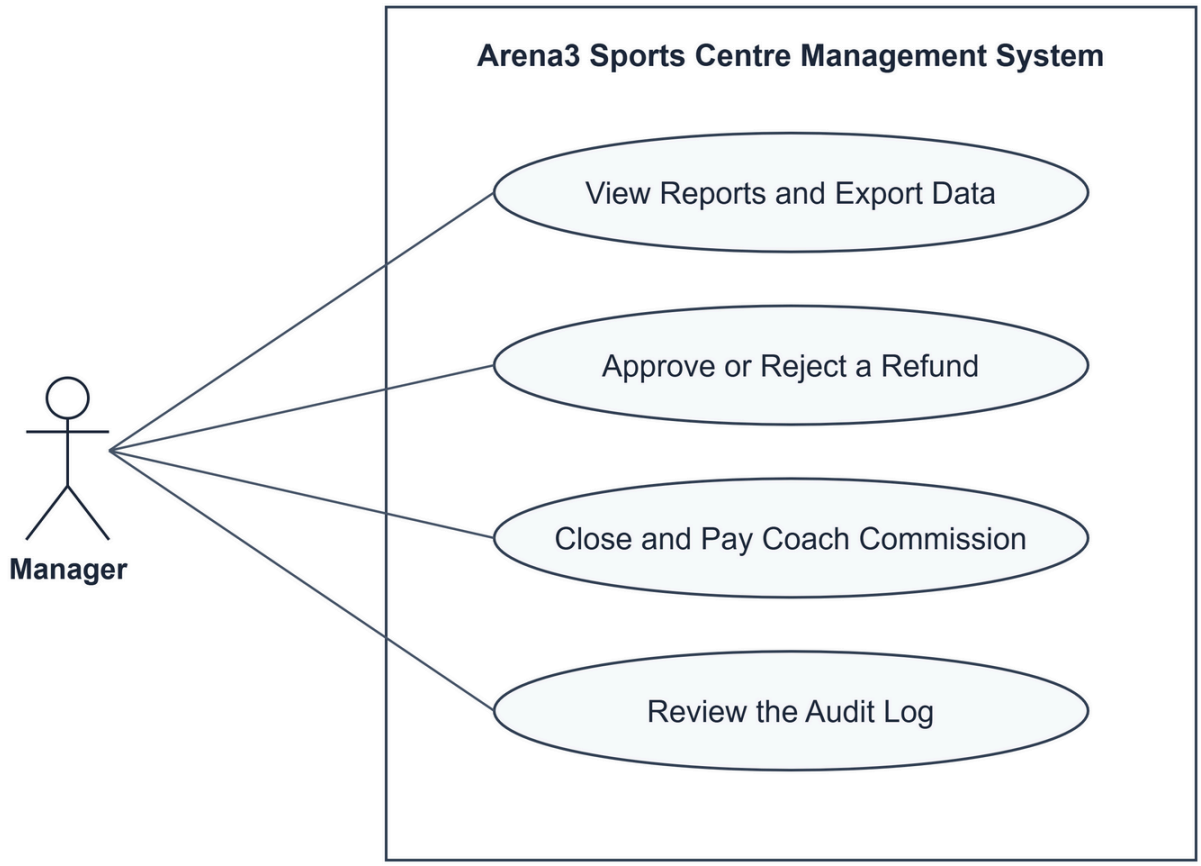


Figure 9. *Use case diagram: Manager Oversight*

2.2.2 Descriptions

The list numbers all 44 use cases in the order of the diagrams. Each is one complete goal of its primary actor and has a full specification in 3.2 to 3.9. Version 2.1 folded the 86 cases of v2.0 into these 44: sub-steps and variants of the same goal became flows of that goal.

ID	Use Case	Actors	Use Case Description
UC-01	Browse Courts, Classes and Prices	Guest	A guest reads the sports, courts, opening hours, plan prices and open classes before deciding to join.
UC-02	Register a Member Account	Guest, Mail delivery (SMTP)	A guest creates a member account and proves the email address with a one-time code.
UC-03	Sign In and Sign Out	Registered User	A registered user opens a session with phone or email and password, and closes it when done.
UC-04	Reset a Forgotten Password	Registered User, Mail delivery (SMTP)	A registered user who has forgotten the password proves control of the account email with a one-time code and sets a new password.
UC-05	Manage the Personal Profile	Registered User	A registered user updates the profile details and changes the password.
UC-06	View Notifications	Registered User	The registered user reads booking, payment, class, plan and support notices, and marks them as read.

ID	Use Case	Actors	Use Case Description
UC-07	View My Schedule	Member	The member sees upcoming court bookings, class sessions, plan balance and recent attendance in one place.
UC-08	Book a Court and Pay	Member, payOS	The member holds a court slot for five minutes and pays the amount due to confirm it.
UC-09	Reschedule or Cancel a Booking	Member	The member moves a confirmed booking to another slot, or cancels it and receives a refund by the notice given.
UC-10	Order a Membership Plan	Member, Receptionist, payOS	The member orders a membership plan online and pays for it at the front desk.
UC-11	View Receipts and Redeem Points	Member	The member views paid receipts and e-invoices, and turns loyalty points into a single-use discount code.
UC-12	Check In by Self-Service	Member	The member shows a short-lived QR pass at the desk, or scans the desk code to check in without staff.
UC-13	Enrol in a Class	Member, Receptionist, Manager	The member takes a seat in an open class, or joins its waitlist when the class is full.
UC-14	Accept a Waitlist Offer	Member	A waitlisted member accepts a freed seat before the offer expires.
UC-15	Leave a Class	Member	The member leaves a class in time, which frees the seat for the waitlist.
UC-16	Follow My Training Progress	Member	The member follows training plans, coach feedback, reviews and homework, and sets a personal goal.
UC-17	Get Help from the AI Assistant or Staff	Member, Gemini AI	The member asks the AI assistant about the centre, or sends a support request to the front desk.
UC-18	Open and Close a Cashier Shift	Receptionist	The receptionist opens a cashier shift with a float and closes it by declaring the cash counted.
UC-19	Manage a Member Record at the Desk	Receptionist, Coach	The receptionist registers a walk-in customer, views and edits a member record, and follows up members at risk of leaving.
UC-20	Manage a Member Plan at the Desk	Receptionist, payOS	The receptionist sells, renews, freezes and resumes a member's plan.
UC-21	Book a Court at the Desk	Receptionist	The receptionist books a court for a walk-in customer, or a fixed weekly series of 4 to 12 weeks at 10% off.
UC-22	Settle Pending Payments	Receptionist, payOS	The receptionist settles money that is still open: a booking balance, a bank transfer to confirm, or a payment for an order.
UC-23	Sell and Redeem a Day Pass	Receptionist	The receptionist sells a day pass for entry on a chosen day and marks it used when the customer arrives.
UC-24	Raise a Refund	Receptionist, Manager	The receptionist gives back all or part of a posted payment, and a large refund waits for a manager's approval.

ID	Use Case	Actors	Use Case Description
UC-25	Check In a Court Booking	Receptionist, Member	The receptionist checks a court booking in on the live court map, inside the check-in window.
UC-26	Check a Member In at the Gate	Receptionist, Member	The receptionist admits a member and the system records the visit, flagging any override.
UC-27	Rent Out Gear and Record Its Return	Receptionist	The receptionist rents out gear, charges the hire to the till with an e-invoice, and records the return.
UC-28	Handle Maintenance Work Orders	Receptionist, Manager	A receptionist reports a fault as a work order, and a manager tracks it to done with its cost.
UC-29	Reply to a Support Request	Receptionist	The receptionist answers a member's support request, which closes it and notifies the member.
UC-30	View the Teaching Schedule and Rosters	Coach	The coach sees the week's sessions and the roster of each class taught.
UC-31	Take Session Attendance	Coach, Manager	The coach marks each student present, late, absent or excused for a session.
UC-32	Plan a Training Session	Coach	The coach builds the plan of a session from blocks, a saved template or an AI suggestion.
UC-33	Record Session Results and Homework	Coach	The coach records how each student did in the session and sets homework.
UC-34	Assess a Student's Progress	Coach	The coach sets a student's level, writes a 2- or 4-week review and keeps private notes.
UC-35	View Monthly Earnings	Coach	The coach sees the sessions taught and the commission earned for a month, and whether it is paid.
UC-36	Manage User Accounts	Manager	The manager creates and maintains staff accounts and looks after member accounts.
UC-37	Manage Membership Plans and Prices	Manager	The manager sets up the plans on sale and the court price rules.
UC-38	Manage Promo Codes	Manager	The manager creates promo codes with their scope, limits and dates, and pauses them.
UC-39	Schedule Classes and Sessions	Manager	The manager creates a class with its coach, court, schedule and capacity, opens it, and changes or cancels sessions.
UC-40	Configure the Centre	Manager	The manager sets the centre's hours, booking limits, refund tiers, check-in and other settings, and switches modules on or off.
UC-41	View Reports and Export Data	Manager	The manager reviews revenue, occupancy, capacity, members and attendance, and exports reports and e-invoices.
UC-42	Approve or Reject a Refund	Manager	A manager other than the person who raised a large refund approves or rejects it.
UC-43	Close and Pay Coach Commission	Manager	The manager sets coach rates, closes a finished month so its figures are frozen, and marks each coach paid.
UC-44	Review the Audit Log	Manager	The manager reviews who changed what and when, to investigate a dispute or a mistake.

3. Functional Requirements

3.1 System Functional Overview

3.1.1 Screens Flow

Each portal has one screen flow that starts at its entry screen. A box is a routed screen with its route; the dialogs and in-page steps that open over a screen sit inside its box, and arrows lead from one screen to the next.

3.1.1.1 Public Site

Figure 10 shows the screens of the public site.

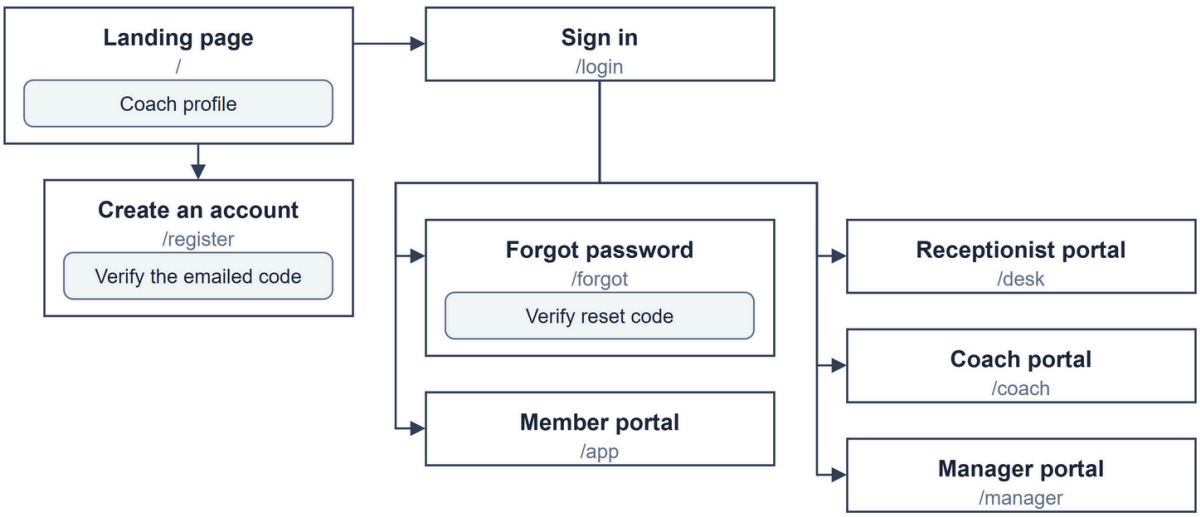


Figure 10. *Screens flow: Public Site*

3.1.1.2 Member Portal

Figure 11 shows the screens of the member portal.

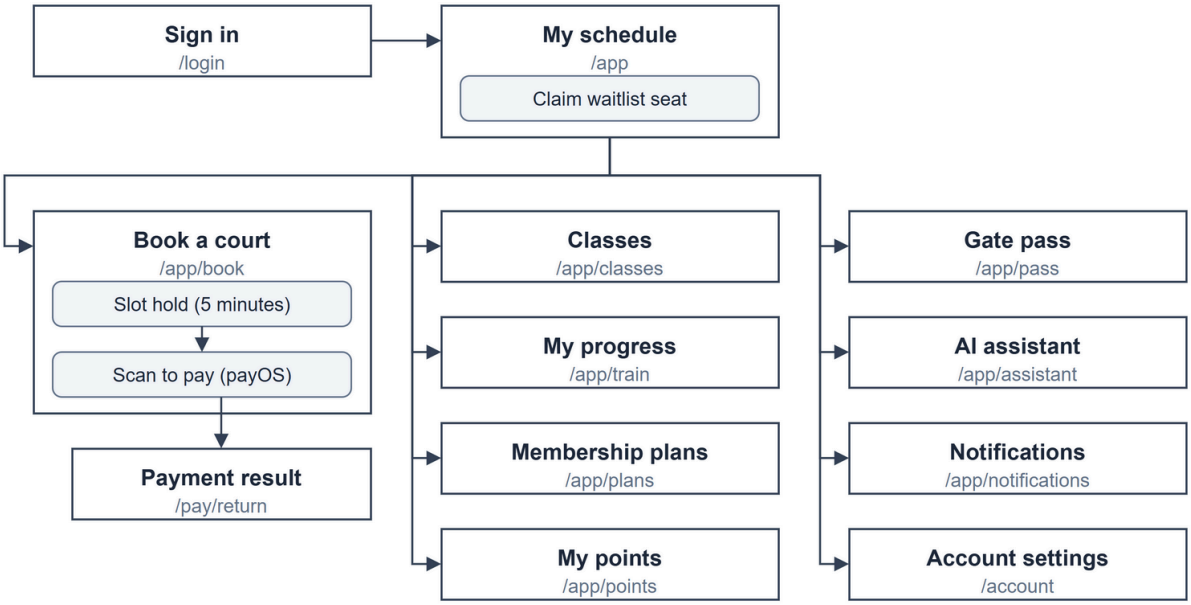


Figure 11. *Screens flow: Member Portal*

3.1.1.3 Front Desk Portal

Figure 12 shows the screens of the front desk.

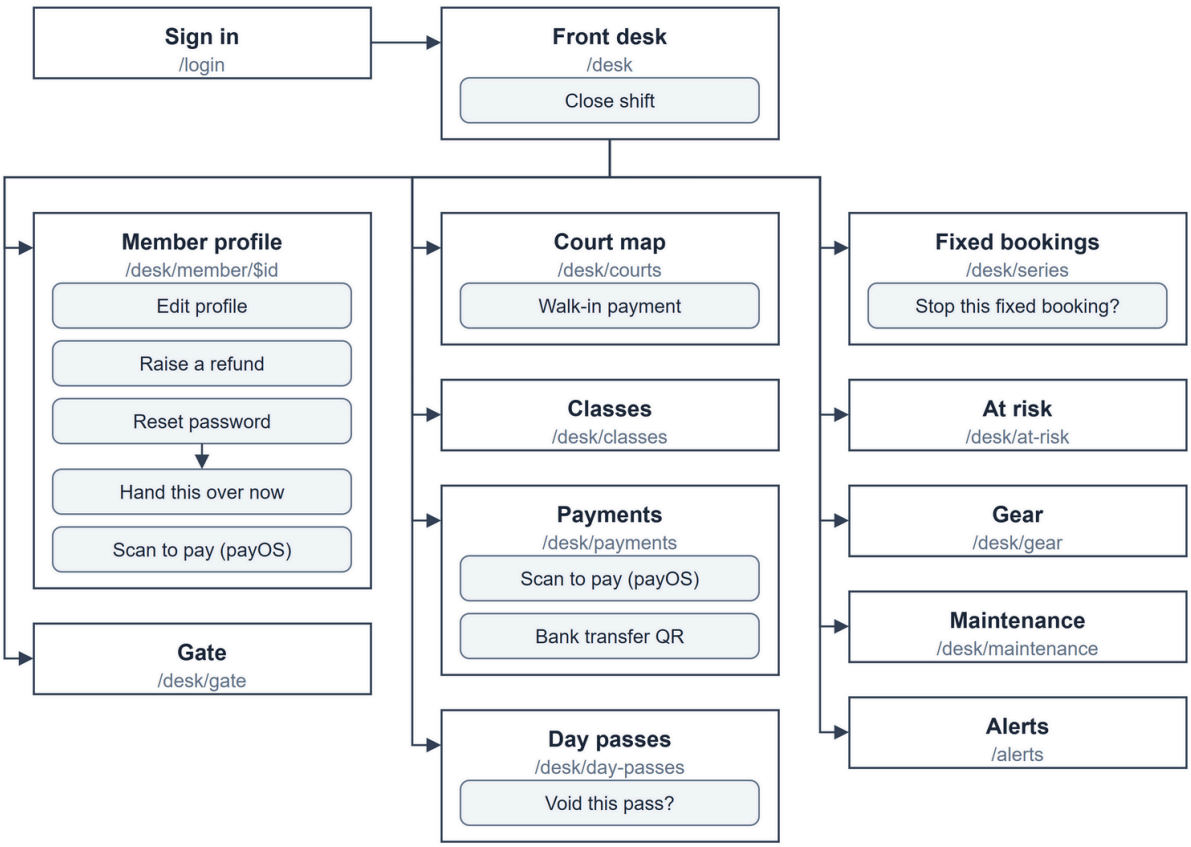


Figure 12. *Screens flow: Front Desk Portal*

3.1.1.4 Coach Portal

Figure 13 shows the screens of the coach portal.

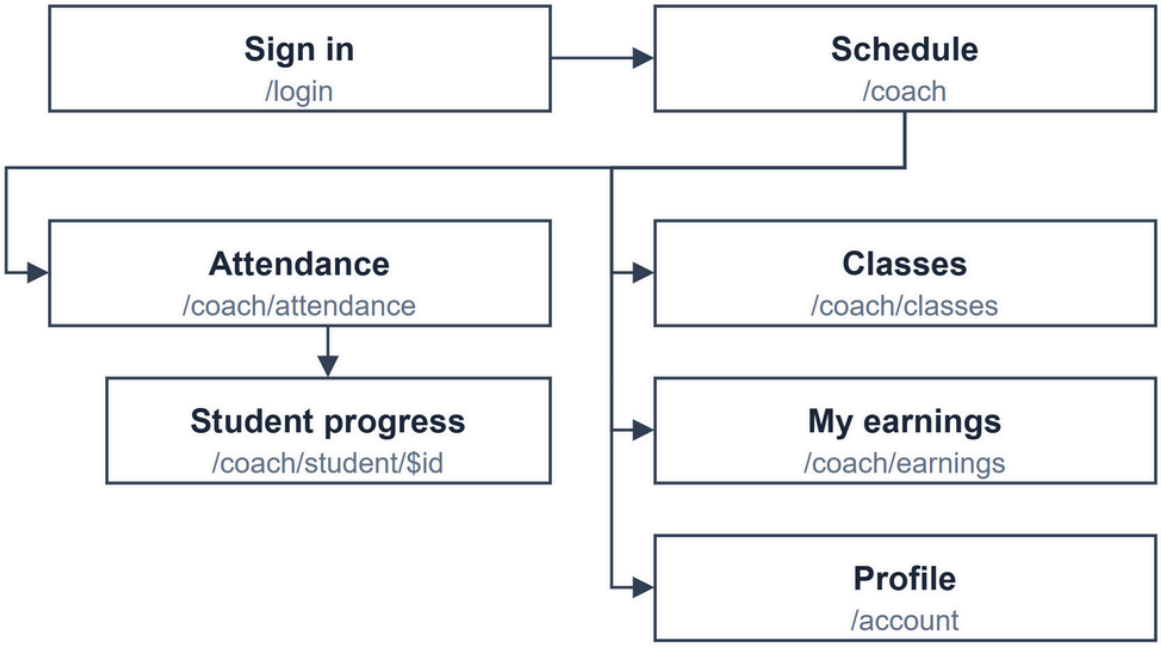


Figure 13. *Screens flow: Coach Portal*

3.1.1.5 Manager Portal

Figure 14 shows the screens of the manager portal.

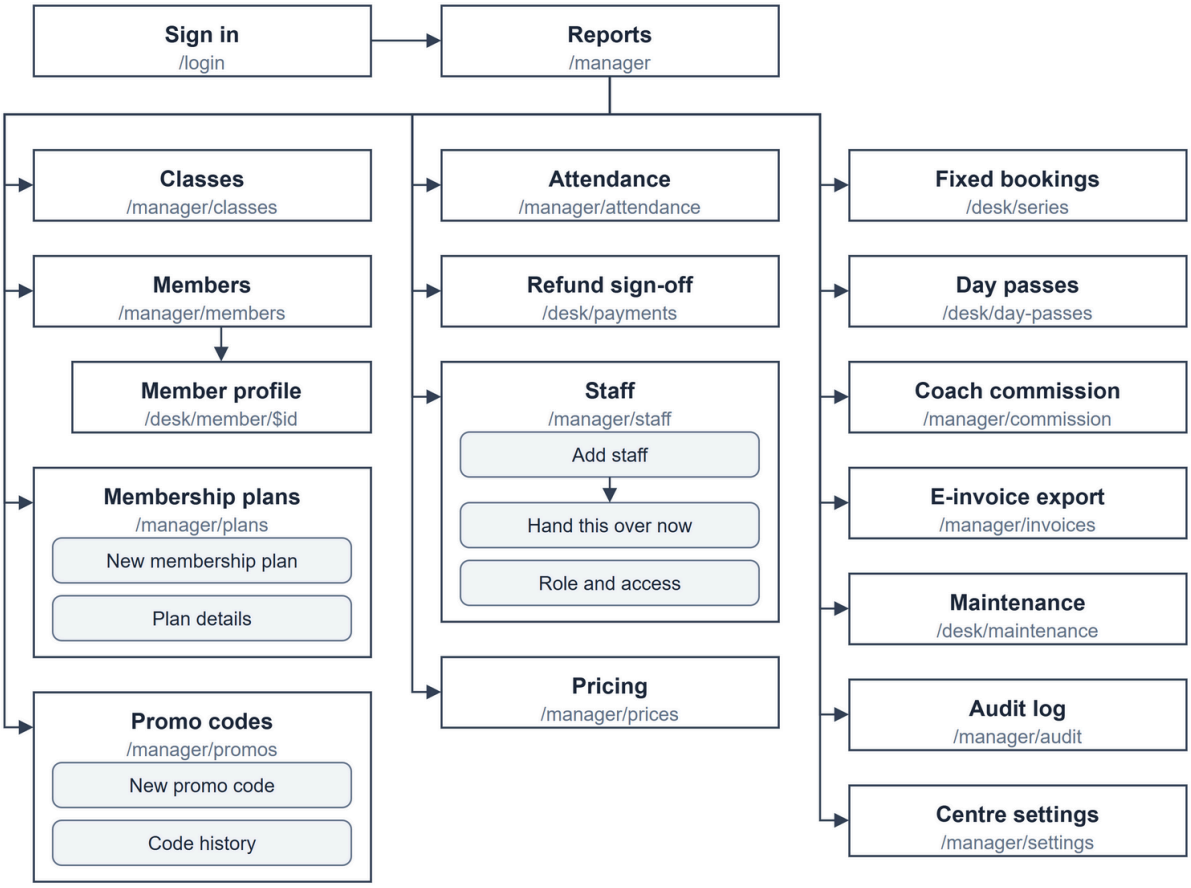


Figure 14. *Screens flow: Manager Portal*

3.1.2 Screen Descriptions

The application has 44 routed screens. Each screen name carries the route exactly as the application serves it.

3.1.2.1 Public Site

#	Feature	Screen	Description
1	Public	Landing page (/)	Centre introduction, sports, prices, open classes and coach profiles.
2	Account	Sign in (/login)	Sign in by phone or email and password. Each role lands on its own portal.
3	Account	Create an account (/register)	Member sign-up with consent, confirmed by a one-time code.
4	Account	Forgot password (/forgot)	Sends a reset code and sets a new password.

3.1.2.2 Member Portal

#	Feature	Screen	Description
1	Account	Account settings (/account)	Personal details, password, receipts and support requests.

#	Feature	Screen	Description
2	Payment	Payment result (/pay/return)	Return page after payOS. Asks Arena3 to verify the payment, then shows the outcome.
3	Member	My schedule (/app)	Upcoming bookings and classes, plan balance and waitlist offers.
4	Booking	Book a court (/app/book)	Pick a date and sport, tap a free slot, hold it for 5 minutes and pay.
5	Classes	Classes (/app/classes)	Enrol by sport. A full class offers a first-come waitlist.
6	Training	My progress (/app/train)	Attendance, results, reviews, training plans and homework.
7	Plans	Membership plans (/app/plans)	Order a plan here and pay at the desk.
8	Loyalty	My points (/app/points)	Points balance and history.
9	Gate	Gate pass (/app/pass)	QR pass whose code changes every minute.
10	Assistant	AI assistant (/app/assistant)	Gemini answers grounded in the timetable, plans and coaches.
11	Notifications	Notifications (/app/notifications)	In-app notices for the member.

3.1.2.3 Front Desk Portal

#	Feature	Screen	Description
1	Account	Alerts (/alerts)	Staff alerts, such as an absence streak.
2	Desk	Front desk (/desk)	Find a member, sell a plan, take a payment, open and close the shift.
3	Desk	Member profile (/desk/member/$id)	Plans, bookings and payments. Edit, refund, reset password, take a payOS payment.
4	Gate	Gate (/desk/gate)	Scan a member's QR pass, or look the member up by phone.
5	Courts	Court map (/desk/courts)	Live court grid. Sells a walk-in slot.
6	Classes	Classes (desk) (/desk/classes)	Every class with its sessions, court, coach and roster.
7	Payments	Payments (/desk/payments)	Plans awaiting payment, transfers to reconcile, refunds awaiting a manager, receipts.
8	Day passes	Day passes (/desk/day-passes)	Sell, mark as used or void a day pass.
9	Bookings	Fixed bookings (/desk/series)	Fixed weekly bookings over 4–12 weeks at 10% off.
10	Care	At risk (/desk/at-risk)	Members whose attendance has dropped, with a contact log.
11	Gear	Gear (/desk/gear)	Lend and take back equipment against stock.
12	Maintenance	Maintenance (/desk/maintenance)	Work orders. An open order can block a court.

3.1.2.4 Coach Portal

#	Feature	Screen	Description
1	Coach	Schedule (/coach)	The coach's own sessions by day, with court, class code and enrolment.

#	Feature	Screen	Description
2	Coach	Attendance (/coach/attendance)	Session register: Present, Late, Absent or Excused, plus results.
3	Coach	Student progress (/coach/student/$id)	Goal, level, attendance totals, reviews, notes and homework.
4	Coach	Classes (coach) (/coach/classes)	Classes the coach leads or assists, with plans and homework.
5	Coach	My earnings (/coach/earnings)	Fee per class plus a fee per student present, by month.

3.1.2.5 Manager Portal

#	Feature	Screen	Description
1	Reports	Reports (/manager)	Revenue by day, method, source and shift; capacity and members reports; export.
2	Classes	Classes (manager) (/manager/classes)	Create, publish, edit and cancel classes. Clashes are blocked.
3	Members	Members (/manager/members)	Every member with filters, plan and class count.
4	Plans	Membership plans (/manager/plans)	Create, edit and withdraw plans. A plan is never deleted.
5	Promotions	Promo codes (/manager/promos)	Create codes and read their redemption history.
6	Attendance	Attendance report (/manager/attendance)	Attendance across classes, sessions and members.
7	Staff	Staff (/manager/staff)	Issue, lock and re-role staff accounts; reset passwords.
8	Pricing	Pricing (/manager/prices)	Price rules by sport, court, day type and hour band. A new price applies only to new sales.
9	Commission	Coach commission (/manager/commission)	Close a month's coach commission and mark it paid.
10	Invoices	E-invoice export (/manager/invoices)	Export invoices as a spreadsheet or XML.
11	Audit	Audit log (/manager/audit)	Audit entries filtered by actor, action, entity and date.
12	Settings	Centre settings (/manager/settings)	Opening hours, slot length, windows, limits, VAT, rounding and feature flags.

3.1.3 Screen Authorization

Each portal layout checks the signed-in role before it renders. A wrong role is sent to its own home; no session is sent to Sign in. A row in angle brackets is a screen; the rows under it are the functions on that screen.

Screen	Guest	Member	Receptionist	Coach	Manager
<<Landing page>>	x	x	x	x	x
Browse Courts, Classes and Prices	x	x	x	x	x
<<Sign in>>	x	x	x	x	x
Sign In and Sign Out	x	x	x	x	x

Screen	Guest	Member	Receptionist	Coach	Manager
<<Create an account>>	x	x	x	x	x
Register a Member Account	x				
<<Forgot password>>	x	x	x	x	x
Reset a Forgotten Password	x				
<<Account settings>>		x	x	x	x
Manage the Personal Profile		x	x	x	x
<<Payment result>>	x	x	x	x	x
Book a Court and Pay		x			
<<My schedule>>		x			
View My Schedule		x			
Reschedule or Cancel a Booking		x			
Leave a Class		x			
<<Book a court>>		x			
Book a Court and Pay		x			
<<Classes>>		x			
Enrol in a Class		x			
Accept a Waitlist Offer		x			
<<My progress>>		x			
Follow My Training Progress		x			
<<Membership plans>>		x			
Order a Membership Plan		x			
<<My points>>		x			
View Receipts and Redeem Points		x			
<<Gate pass>>		x			
Check In by Self-Service		x			
<<AI assistant>>		x			
Get Help from the AI Assistant or Staff		x			
<<Notifications>>		x			
View Notifications		x			
Accept a Waitlist Offer		x			
<<Alerts>>			x	x	x
View Notifications			x	x	x
<<Front desk>>			x		x
Manage a Member Record at the Desk			x		x
Reply to a Support Request			x		x
<<Member profile>>			x		x
Manage a Member Record at the Desk			x		x

Screen	Guest	Member	Receptionist	Coach	Manager
Manage a Member Plan at the Desk			x		x
Raise a Refund			x		x
<<Gate>>			x		x
Check a Member In at the Gate			x		x
<<Court map>>			x		x
Book a Court at the Desk			x		x
Check In a Court Booking			x		x
<<Classes (desk)>>			x		x
Enrol in a Class			x		x
<<Payments>>			x		x
Open and Close a Cashier Shift			x		
Settle Pending Payments			x		x
Approve or Reject a Refund					x
<<Day passes>>			x		x
Sell and Redeem a Day Pass			x		x
<<Fixed bookings>>			x		x
Book a Court at the Desk			x		x
<<At risk>>			x		x
Manage a Member Record at the Desk			x		x
<<Gear>>			x		x
Rent Out Gear and Record Its Return			x		x
<<Maintenance>>			x		x
Handle Maintenance Work Orders			x		x
<<Schedule>>				x	x
View the Teaching Schedule and Rosters				x	x
<<Attendance>>				x	x
Take Session Attendance				x	x
<<Student progress>>				x	x
Assess a Student's Progress				x	x
<<Classes (coach)>>				x	x
View the Teaching Schedule and Rosters				x	x
Plan a Training Session				x	x
Record Session Results and Homework				x	x
<<My earnings>>				x	x
View Monthly Earnings				x	
<<Reports>>					x
View Reports and Export Data					x

Screen	Guest	Member	Receptionist	Coach	Manager
<<Classes (manager)>>					x
Schedule Classes and Sessions					x
<<Members>>					x
Manage User Accounts					x
<<Membership plans>>					x
Manage Membership Plans and Prices					x
<<Promo codes>>					x
Manage Promo Codes					x
<<Attendance report>>					x
View Reports and Export Data					x
<<Staff>>					x
Manage User Accounts					x
<<Pricing>>					x
Manage Membership Plans and Prices					x
<<Coach commission>>					x
Close and Pay Coach Commission					x
<<E-invoice export>>					x
View Reports and Export Data					x
<<Audit log>>					x
Review the Audit Log					x
<<Centre settings>>					x
Configure the Centre					x

3.1.4 Non-Screen Functions

Ten background jobs and the payOS webhook keep holds, bookings, plans and messages correct without anyone pressing a button. A single probe query checks every 15 seconds whether any job has work.

#	Feature	System Function	Description
1	Booking	Expire holds	Runs every 15 seconds when due. Cancels court holds past their hold time and frees the court.
2	Booking	Mark no-shows	Runs every 15 seconds when due. Marks a confirmed booking as a no-show 10 minutes after its start without check-in.
3	Booking	Complete bookings	Runs every 15 seconds when due. Closes an in-use booking 10 minutes after it ends.
4	Notifications	Send notifications	Runs every 15 seconds when due. Sends up to 50 queued messages. A message is marked sent only on delivery and stops after 5 tries.

#	Feature	System Function	Description
5	Payment	Withdraw payment links	Runs every 15 seconds when due. Cancels a payOS link whose court hold has lapsed. Runs only when payOS is configured.
6	Classes	Expire waitlist offers	Runs every 15 seconds when due. Expires an unanswered offer and offers the seat to the next member.
7	Attendance	Lock attendance	Runs every 15 seconds when due. Closes a session's register 2 hours after it ends.
8	Plans	Expire plans	Runs every 15 seconds when due. Marks a plan expired the day after its end date.
9	Plans	Plan expiry reminders	Runs every 15 seconds when due. Sends one in-app reminder 7 days, 3 days and 0 days before a plan ends.
10	Classes	Generate sessions	Runs every 10 minutes. Creates upcoming sessions for every open class.
11	Payment	payOS webhook	Runs when payOS calls the webhook. Verifies the signature and amount, then posts the payment through the normal settle path.

The job loop starts with the first API request on each server instance. Each job with work runs in its own transaction, so one failure cannot roll back another.

3.1.5 Entity Relationship Diagram

3.1.5.1 Conceptual Entity Relationship Diagram

The conceptual ERD shows 17 core entities in Chen notation, with a verb on each relationship and 1, N or M cardinality. Figure 15 covers courts, bookings and payments, and Figure 16 covers members, plans and classes; User, Payment and Court appear in both.

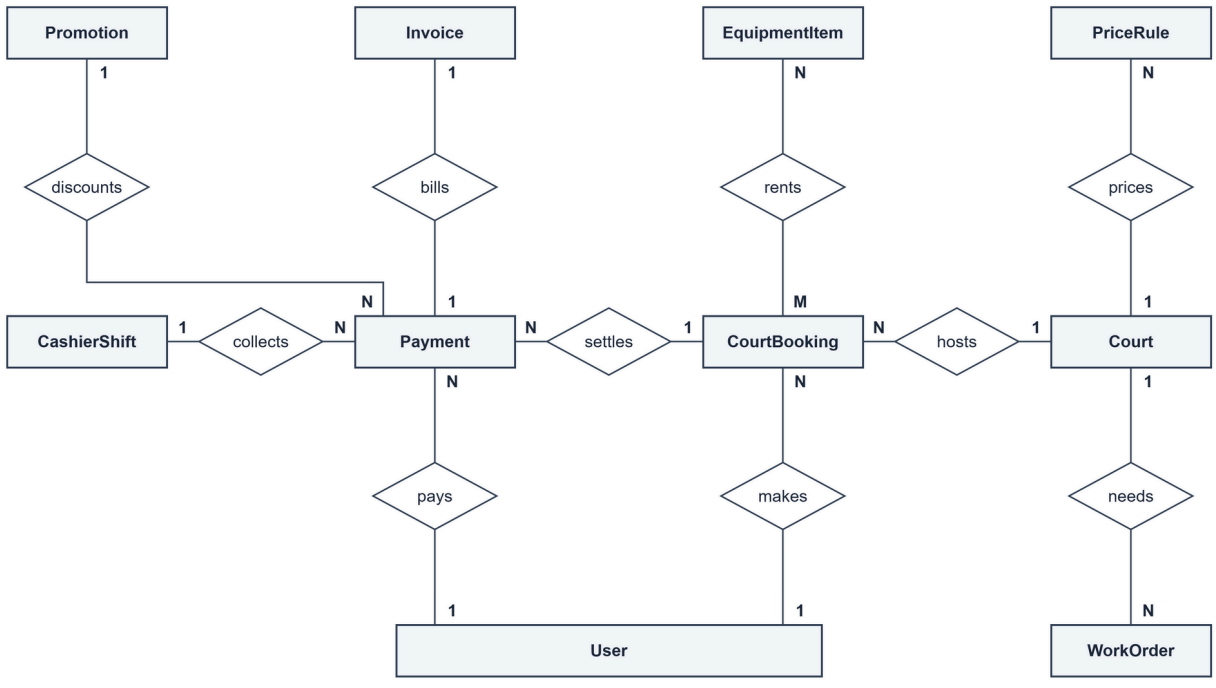


Figure 15. *Conceptual ERD: courts, bookings and payments*

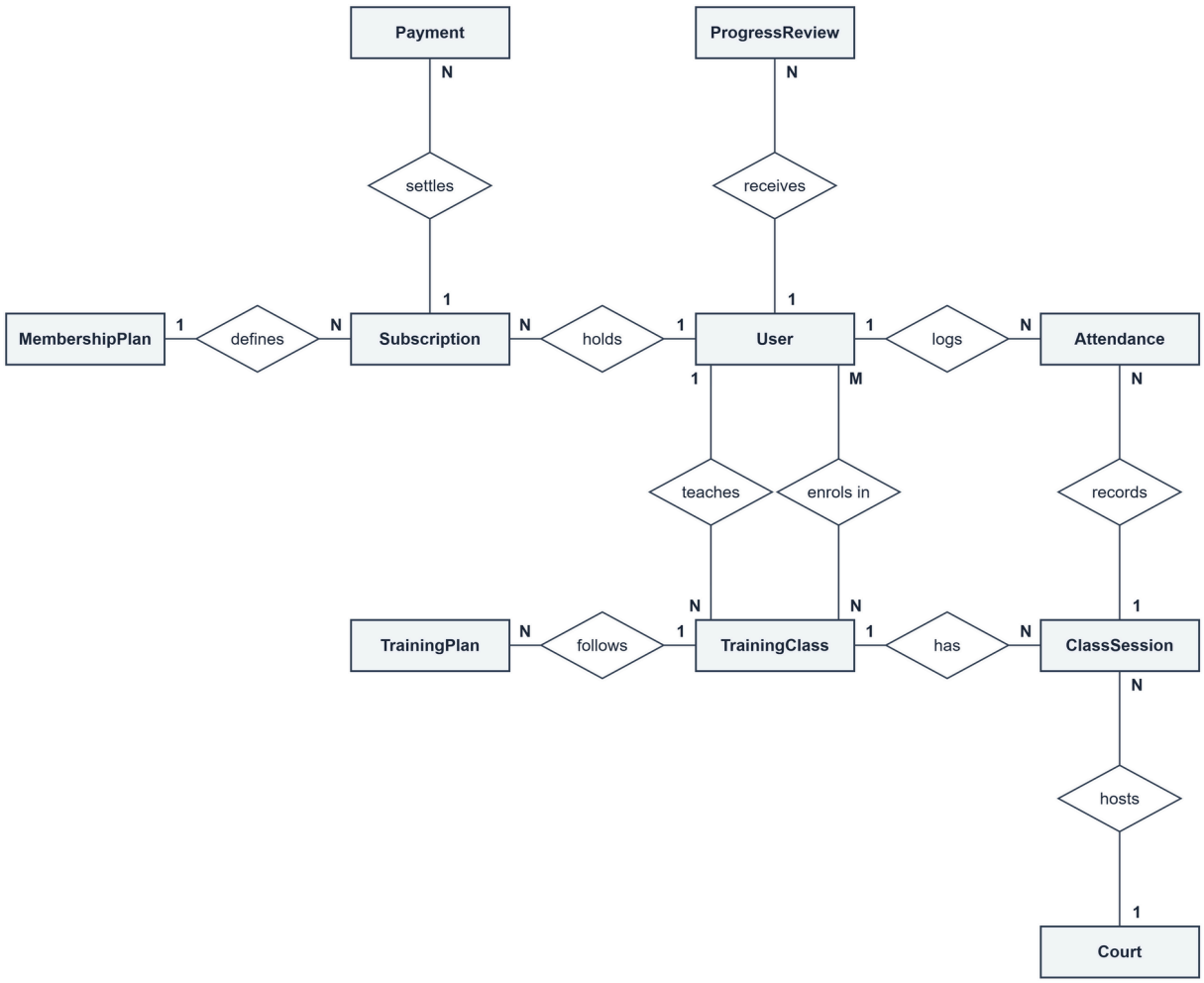


Figure 16. *Conceptual ERD: members, plans and classes*

3.1.5.2 Logical Entity Relationship Diagram

The logical ERD shows every one of the 49 tables in migrations/*.sql with its primary key and foreign keys, in crow's foot notation. It is drawn in five parts so each stays readable on the page: Figure 17 shows accounts and the platform, Figure 18 courts and bookings, Figure 19 payments, plans and promotions, Figure 20 classes, attendance and training, and Figure 21 coaching records and pay. Users, courts, occupancies and promotions repeat where a part needs them, and a foreign key to a table in another part stays in the box.

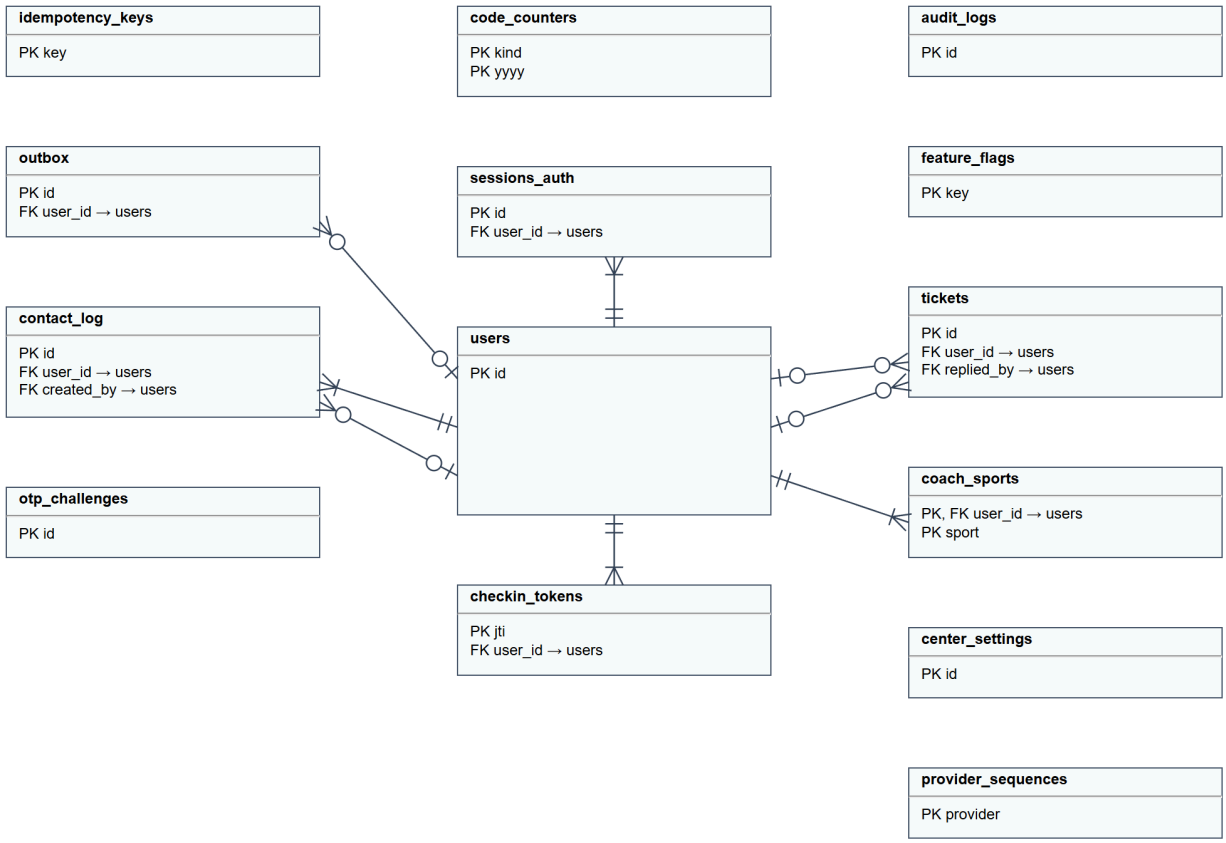


Figure 17. *Logical ERD, part 1: accounts and platform*

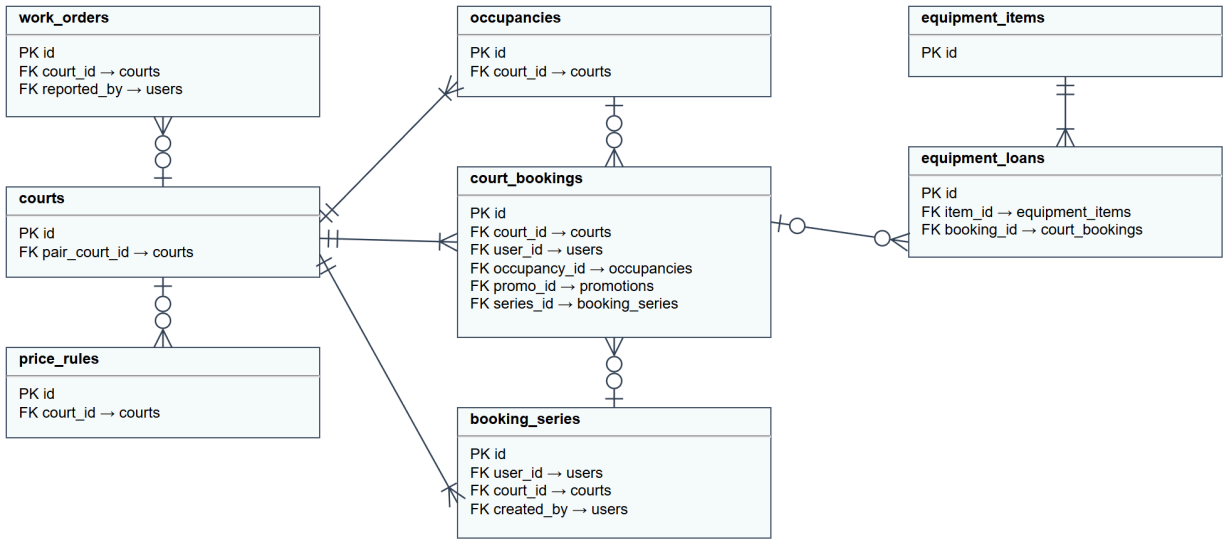


Figure 18. *Logical ERD, part 2: courts and bookings*

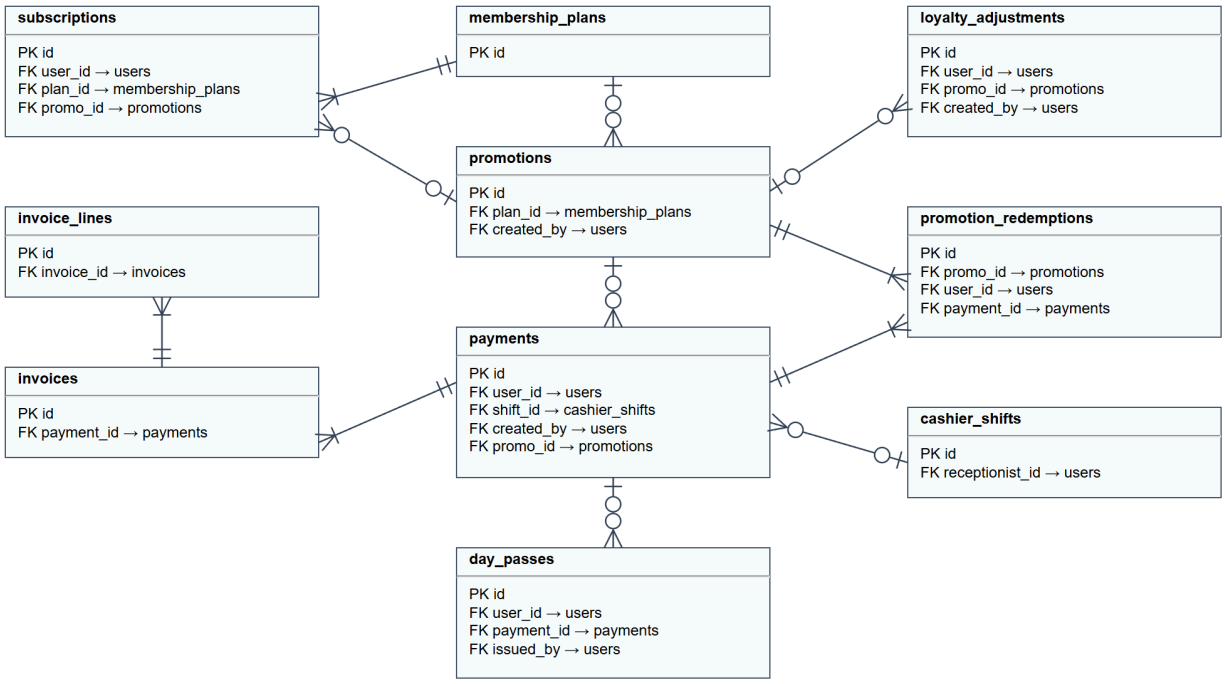


Figure 19. *Logical ERD, part 3: payments, plans and promotions*

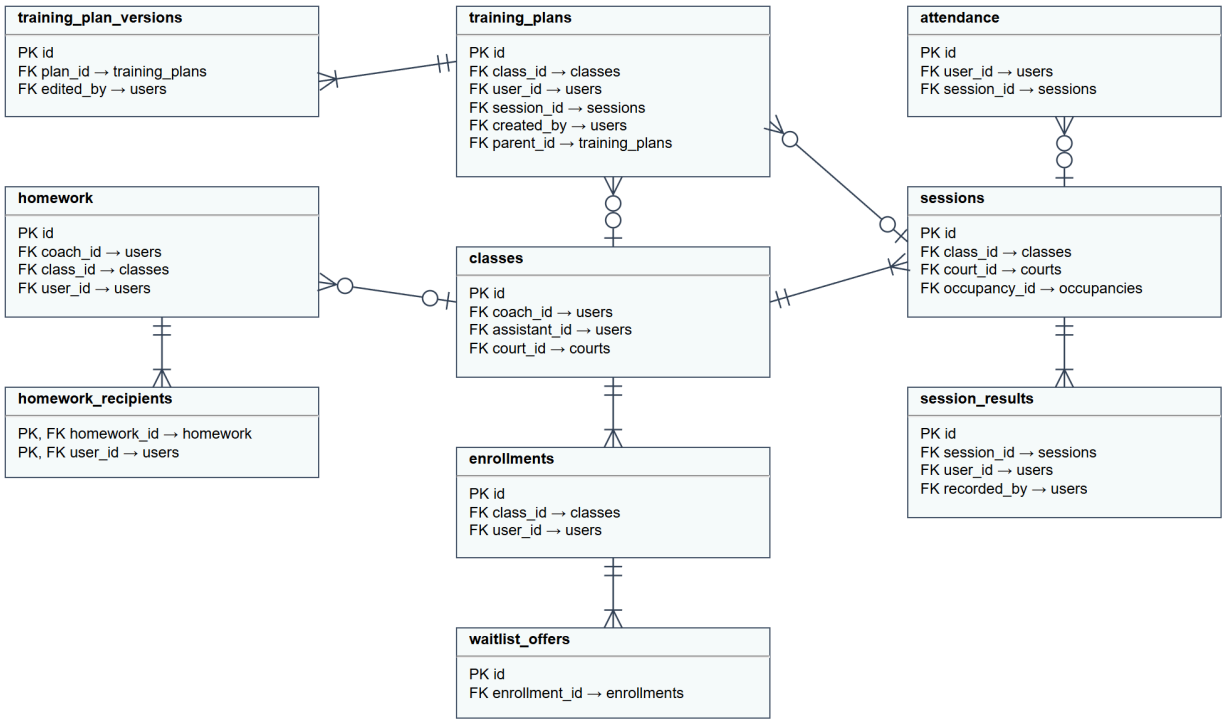


Figure 20. *Logical ERD, part 4: classes, attendance and training*

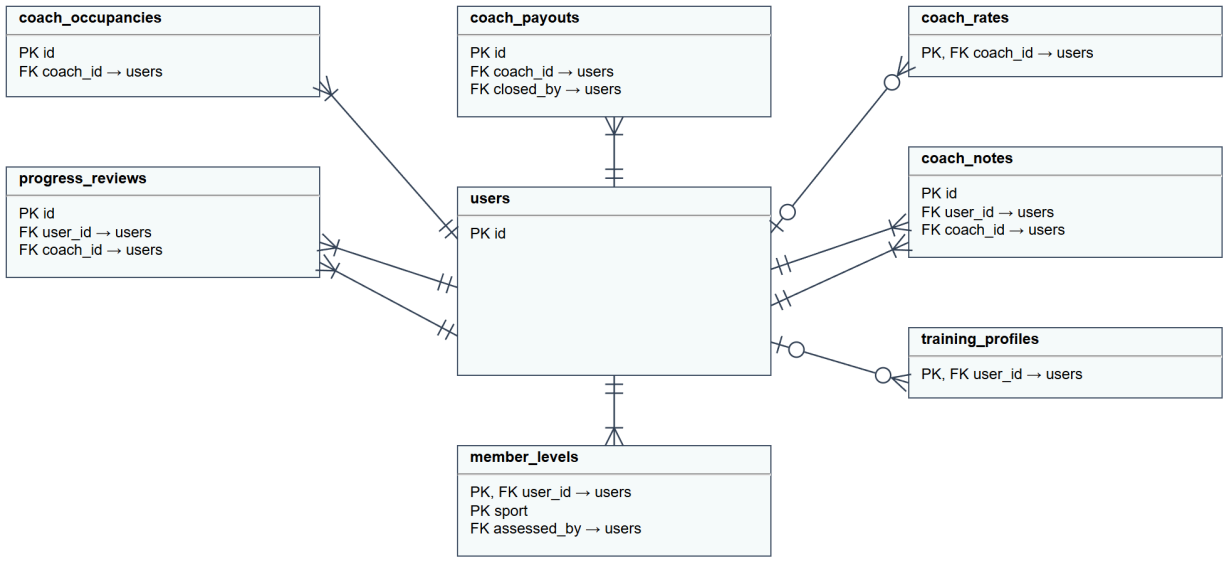


Figure 21. *Logical ERD, part 5: coaching records and pay*

Entities Description

#	Entity	Description
1	User	Store every person with an account, with one role: member, receptionist, coach or manager (table `users`). - Primary Key: id - Foreign Key: None
2	Court	Store a playable court for one sport, with a status of ready, maintenance or closed (table `courts`). - Primary Key: id - Foreign Key: pair_court_id → courts
3	PriceRule	Store price per hour by sport, court class, day type and hour band (table `price_rules`). - Primary Key: id - Foreign Key: court_id → courts
4	CourtBooking	Store a court-time owned by a member or a walk-in, from hold to completed (table `court_bookings`). - Primary Key: id - Foreign Key: court_id → courts, user_id → users, occupancy_id → occupancies, promo_id → promotions, series_id → booking_series
5	Payment	Store money in or out, with method, amount, VAT, shift and capture mode (table `payments`). - Primary Key: id - Foreign Key: user_id → users, shift_id → cashier_shifts, created_by → users, promo_id → promotions
6	Invoice	Store the receipt issued for a payment, with a gapless code (table `invoices`). - Primary Key: id - Foreign Key: payment_id → payments
7	Promotion	Store a promo code with its discount, validity and usage limits (table `promotions`). - Primary Key: id - Foreign Key: plan_id → membership_plans, created_by → users

#	Entity	Description
8	CashierShift	Store one receptionist's till session, from opening float to counted close (table `cashier_shifts`). - Primary Key: id - Foreign Key: receptionist_id → users
9	MembershipPlan	Store a plan on sale: price, duration, court hours or sessions, sport scope (table `membership_plans`). - Primary Key: id - Foreign Key: None
10	Subscription	Store a member's copy of a plan, with start and end dates and balances (table `subscriptions`). - Primary Key: id - Foreign Key: user_id → users, plan_id → membership_plans, promo_id → promotions
11	TrainingClass	Store a coached class with a coach, a court, a weekly timetable and a capacity (table `classes`). - Primary Key: id - Foreign Key: coach_id → users, assistant_id → users, court_id → courts
12	ClassSession	Store one dated meeting of a class (table `sessions`). - Primary Key: id - Foreign Key: class_id → classes, court_id → courts, occupancy_id → occupancies
13	Attendance	Store one member's mark for one session (table `attendance`). - Primary Key: id - Foreign Key: user_id → users, session_id → sessions
14	TrainingPlan	Store up to 12 timed blocks for a class, a session or one student (table `training_plans`). - Primary Key: id - Foreign Key: class_id → classes, user_id → users, session_id → sessions, created_by → users, parent_id → training_plans
15	ProgressReview	Store an append-only review scoring technique, fitness and attitude (table `progress_reviews`). - Primary Key: id - Foreign Key: user_id → users, coach_id → users
16	EquipmentItem	Store rentable gear with stock on hand (table `equipment_items`). - Primary Key: id - Foreign Key: None
17	WorkOrder	Store a maintenance job that can block a court while it is open (table `work_orders`). - Primary Key: id - Foreign Key: court_id → courts, reported_by → users

3.2 Account and Access

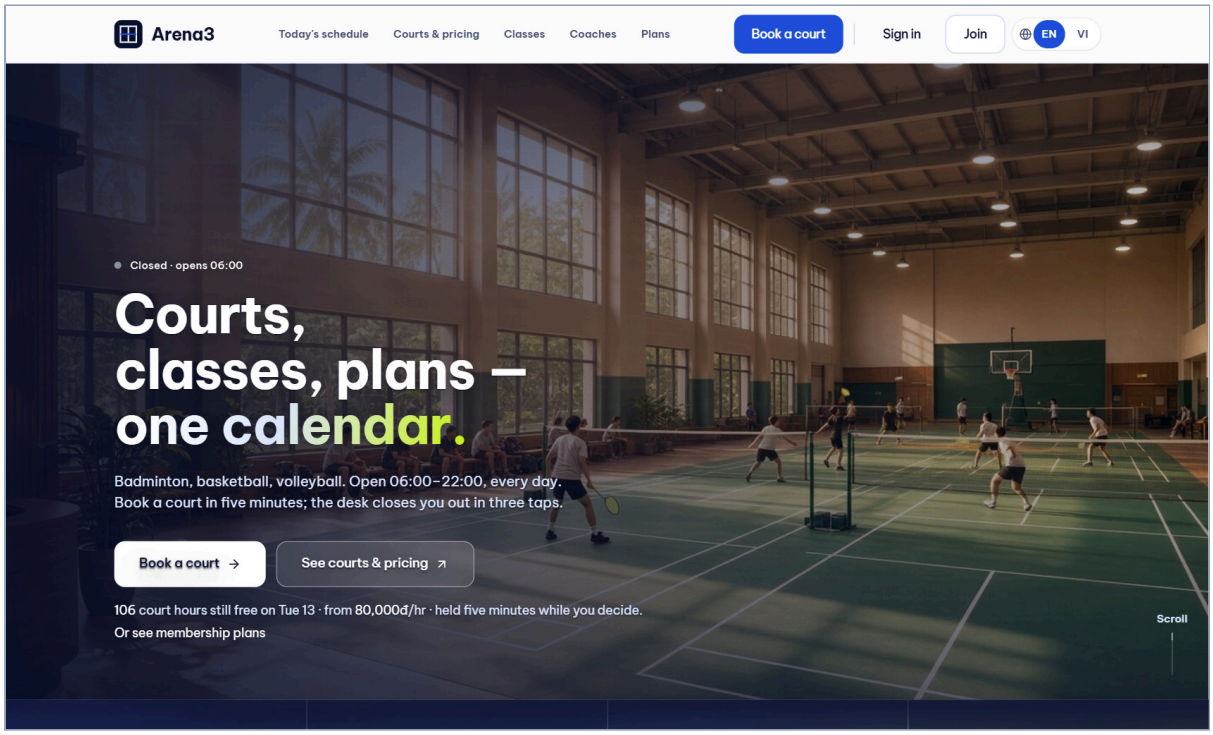
Each use case below follows the template: Function Trigger, Function Description, Data Processing, Screen Layout and Function Details. The Function Details table is followed by the

normal flow, the alternative flows and the exceptions; each refusal names its Message ID (5.3), its business rule (5.1) or its error response (5.2).

Every refusal named below reaches the screen as a sentence, with its error code and, where a business rule applies, the rule code from 5.1. The common requirements in 5.2 (idempotency keys, audit rows, sessions and rate limits) apply to every use case and are not repeated.

3.2.1 Browse Courts, Classes and Prices

- **Function Trigger:** The guest opens the public site (/).
- **Function Description:**
  - **Use Case ID:** UC-01
  - **Actors:** Guest; Secondary: None
  - **Purpose:** A guest reads the sports, courts, opening hours, plan prices and open classes before deciding to join.
  - **Interface:** Landing page (/)
- **Data Processing:**
  - **Data Collect:** The language chosen in the header.
  - **Data Computation:** Selects the sports, courts, plans on sale and open classes; every price shown includes VAT (BR-38).
  - **Data Binding:** Reads courts, membership_plans and classes with their coaches and seats left, and shows them on the landing page.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-01 Browse Courts, Classes and Prices
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Guest
Secondary Actors	None
Trigger	The guest opens the public site (/).
Description	A guest reads the sports, courts, opening hours, plan prices and open classes before deciding to join.
Preconditions	None.
Postconditions	POST-1. No data changes.
Priority	Must have
Frequency of Use	Many times a day
Business Rules	BR-38

Normal Flow

1.0 Browse Courts, Classes and Prices

1. The guest opens the landing page.
2. The system shows the sports, the opening hours, the court list and the plan prices.
3. The guest opens the class list.
4. The system shows each open class with its sport, coach, schedule and seats left. (see 1.0.E1)
5. The guest selects Create an account to continue (UC-02).

Alternative Flows

1.1 The guest switches the language

1. The page is shown again in English or Vietnamese.
   Return to step 3 of the Normal Flow.

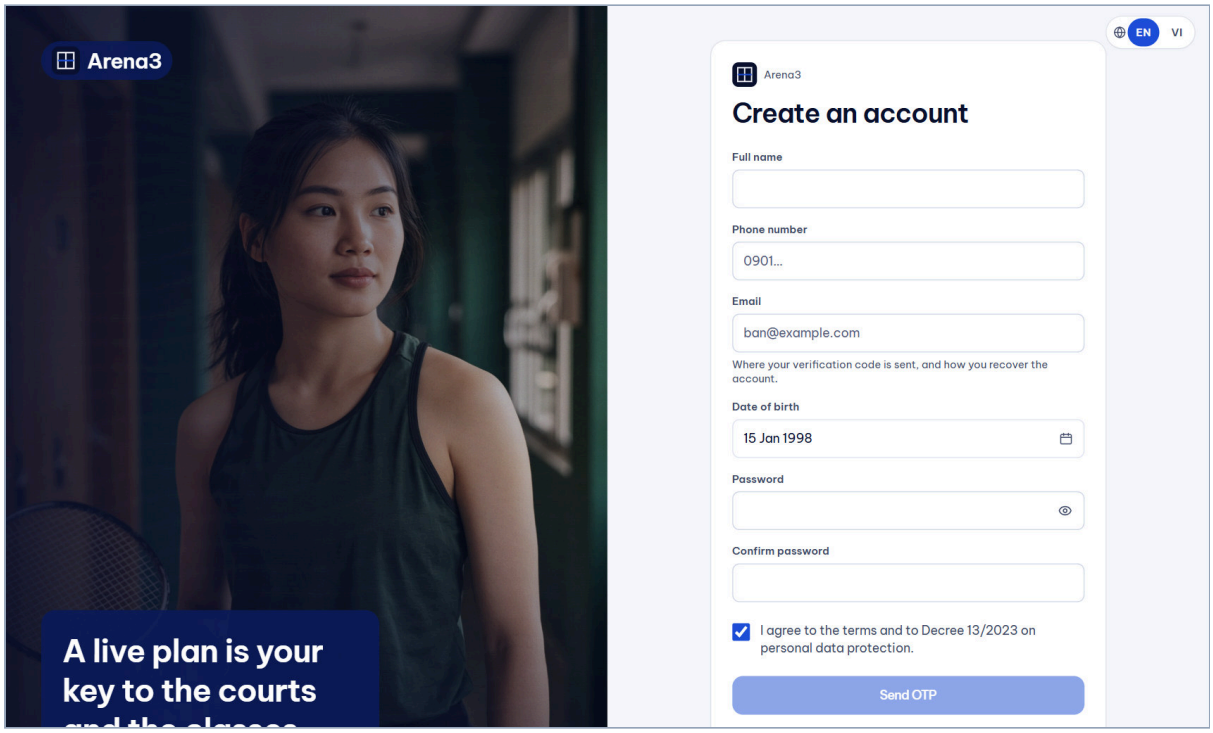
Exceptions

1.0.E1 No class is open

1. The system says so and shows the plan prices only.
   The use case ends.

3.2.2 Register a Member Account

- **Function Trigger:** The guest selects Create an account on /register.
- **Function Description:**
  - **Use Case ID:** UC-02
  - **Actors:** Guest; Secondary: Mail delivery (SMTP)
  - **Purpose:** A guest creates a member account and proves the email address with a one-time code.
  - **Interface:** Create an account (/register)
- **Data Processing:**
  - **Data Collect:** Full name, phone, email, password, date of birth, terms consent and, for a minor, the guardian's name and phone; then the one-time code.
  - **Data Computation:** Checks every field, that the phone and email are unused (BR-01), the password rule (BR-02), the guardian rule (BR-03) and consent (BR-04); hashes the code, keeps it five minutes and counts wrong attempts.
  - **Data Storage:** Writes the sign-up challenge to otp_challenges, then the member to users, a session to sessions_auth and an audit row to audit_logs. The code is sent through Mail delivery (SMTP).
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-02 Register a Member Account
Created By	Arena3 team

Field	Detail
Date Created	2026-10-12
Primary Actor	Guest
Secondary Actors	Mail delivery (SMTP)
Trigger	The guest selects Create an account on /register.
Description	A guest creates a member account and proves the email address with a one-time code.
Preconditions	PRE-1. The guest is signed out. PRE-2. The phone and email have no account yet.
Postconditions	POST-1. A member account exists, the guest is signed in for seven days, and an audit row is written.
Priority	Must have
Frequency of Use	Once per new member
Business Rules	BR-01, BR-02, BR-03, BR-04

Normal Flow

2.0 Register a Member Account

1. The guest enters full name, phone, email, password and date of birth.
2. The guest accepts the terms and the data-privacy notice, then submits.
3. The system checks every field and confirms the phone and email are unused. (see 2.0.E1, 2.0.E2, 2.0.E3, 2.0.E4)
4. The system stores a sign-up challenge for five minutes and emails a one-time code.
5. The guest enters the code on the same page.
6. The system checks the code, creates the member account and returns a session token. (see 2.0.E5, 2.0.E6, 2.0.E7)
7. The system opens My schedule.

Alternative Flows

2.1 The guest is under 16

1. The guest also enters a guardian's name and phone (BR-03).
   Return to step 2 of the Normal Flow.

Exceptions

2.0.E1 A field is malformed

1. The system refuses the request with VALIDATION (5.2 Error Responses), and the field to correct is marked.
   Return to step 1 of the Normal Flow.

2.0.E2 The phone or email already has an account

1. The system refuses the request with BR-01 (ERR_AUTH_01).
   Return to step 1 of the Normal Flow.

2.0.E3 The password is shorter than 8 characters or lacks letters and digits

1. The system refuses the request with BR-02 (ERR_AUTH_02).
   Return to step 1 of the Normal Flow.

2.0.E4 The terms are not accepted

1. The system refuses the request with BR-04.
   Return to step 2 of the Normal Flow.

2.0.E5 The code is wrong

1. The attempt is counted and the guest may try again.
   Return to step 5 of the Normal Flow.

2.0.E6 The code is older than five minutes

1. The guest submits the form again to receive a new code.
   Return to step 2 of the Normal Flow.

2.0.E7 Five wrong attempts

1. The challenge is locked for 15 minutes (RATE_LIMITED, ERR_AUTH_03).
   The use case ends.

A guest becomes a member only after proving the email address with a one-time code. Figure 22 shows this use case as an activity diagram.

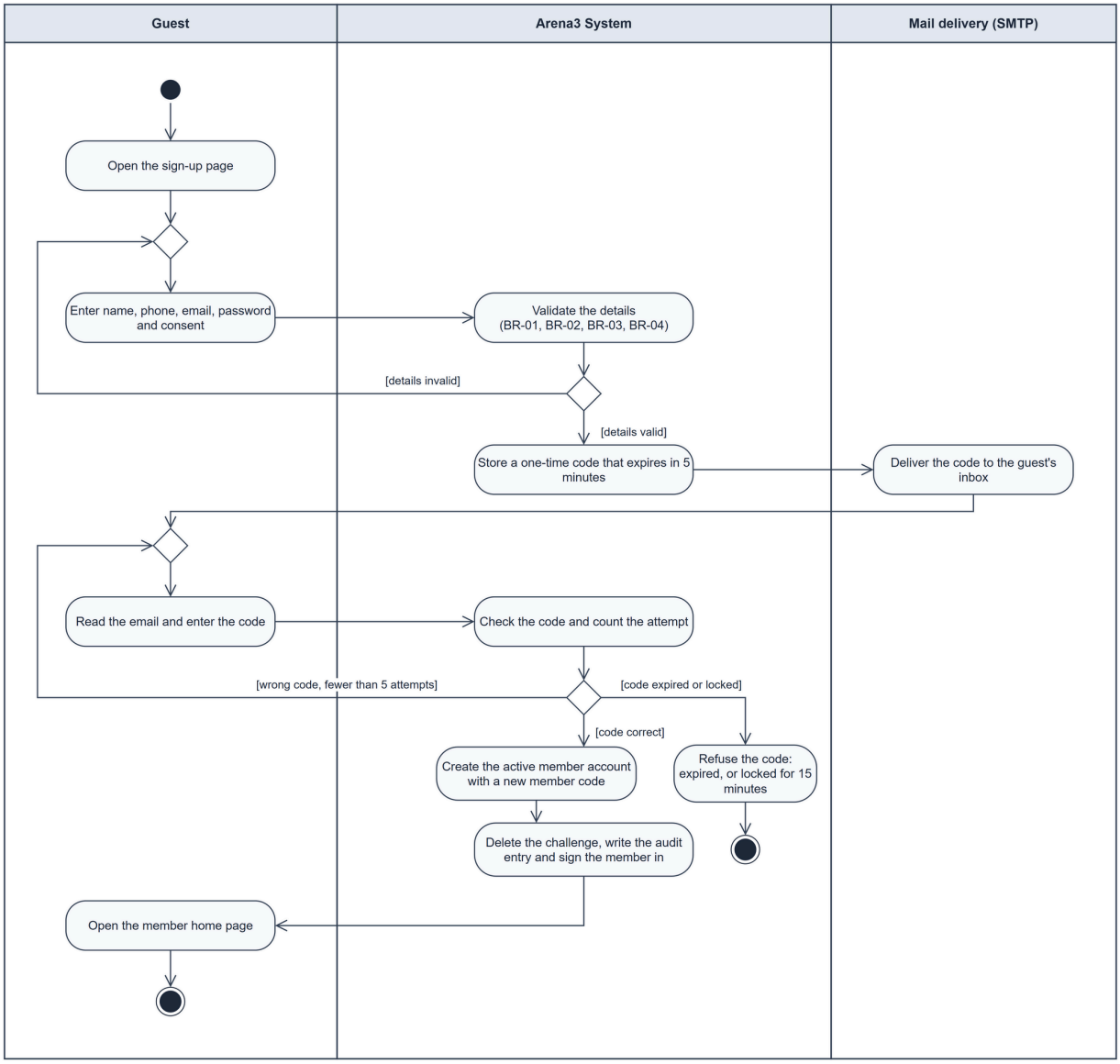
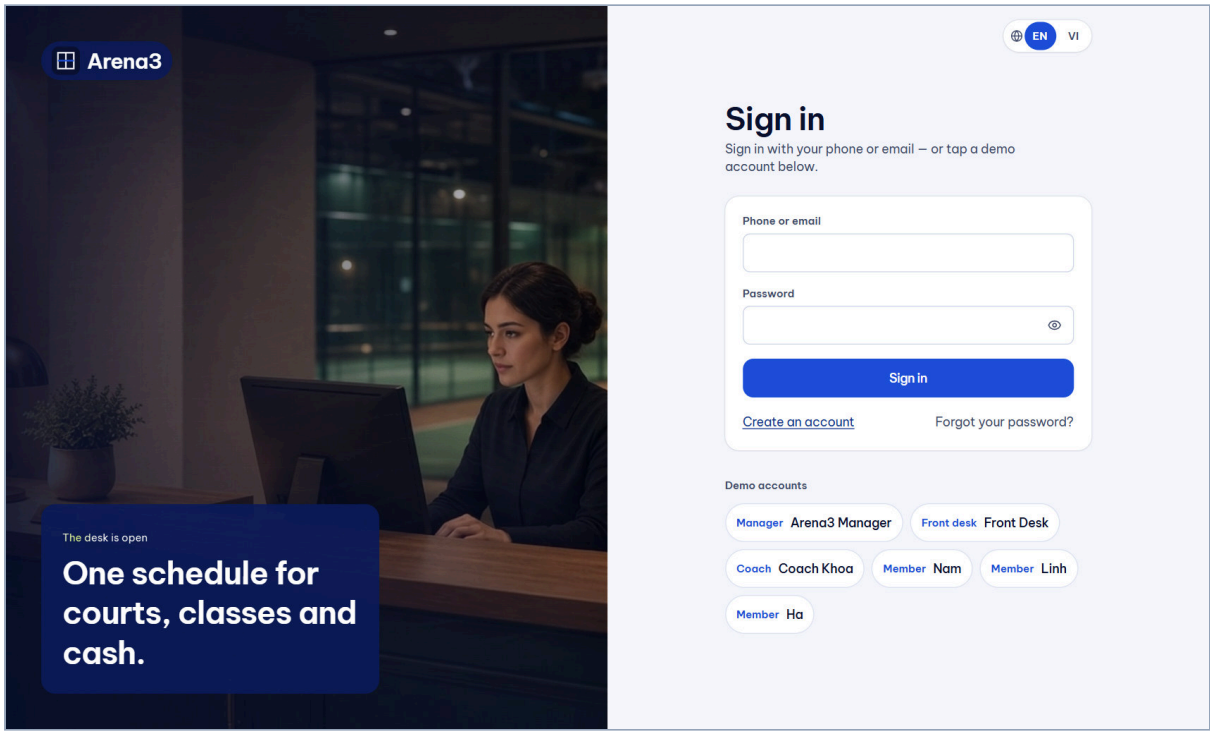


Figure 22. *Activity diagram: Register a Member Account*

The code expires after 5 minutes. Too many wrong attempts lock the challenge for 15 minutes. A successful check creates the account, writes an audit entry and signs the member in.

3.2.3 Sign In and Sign Out

- **Function Trigger:** The registered user selects Sign in on /login, or Sign out in any portal.
- **Function Description:**
  - **Use Case ID:** UC-03
  - **Actors:** Registered User; Secondary: None
  - **Purpose:** A registered user opens a session with phone or email and password, and closes it when done.
  - **Interface:** Sign in (/login)
- **Data Processing:**
  - **Data Collect:** Phone or email and password; the Sign out action.
  - **Data Computation:** Compares the password with its salted hash, counts failed sign-ins and locks sign-in for 15 minutes after five; sets the session length by role.
  - **Data Storage:** Writes a hashed session token to sessions_auth at sign-in and revokes it at sign-out; updates the failed sign-in count and lock time in users.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-03 Sign In and Sign Out
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Registered User
Secondary Actors	None

Field	Detail
Trigger	The registered user selects Sign in on /login, or Sign out in any portal.
Description	A registered user opens a session with phone or email and password, and closes it when done.
Preconditions	PRE-1. The registered user has an active account.
Postconditions	POST-1. Sign in: a session token is stored as a hash and the registered user lands on the home screen of their role. POST-2. Sign out: the session is revoked.
Priority	Must have
Frequency of Use	Many times a day
Business Rules	None

Normal Flow

3.0 Sign In and Sign Out

1. The registered user enters a phone or email and the password.
2. The system checks the password against the stored salted hash. (see 3.0.E1, 3.0.E2, 3.0.E3)
3. The system creates a session: seven days for a member, twelve hours for staff.
4. The system opens the portal for the registered user's role: /app, /desk, /coach or /manager.
5. Later, the registered user selects Sign out.
6. The system revokes the session and opens the landing page.

Alternative Flows

3.1 The registered user opened a protected page before signing in

1. The system returns the registered user to that page.

   Return to step 5 of the Normal Flow.

Exceptions

3.0.E1 The password is wrong

1. The system refuses the request with UNAUTHENTICATED (5.2 Error Responses), and the attempt is counted.

   The use case ends.

3.0.E2 Five wrong passwords

1. Sign-in is locked for 15 minutes (RATE_LIMITED, ERR_AUTH_04).

   The use case ends.

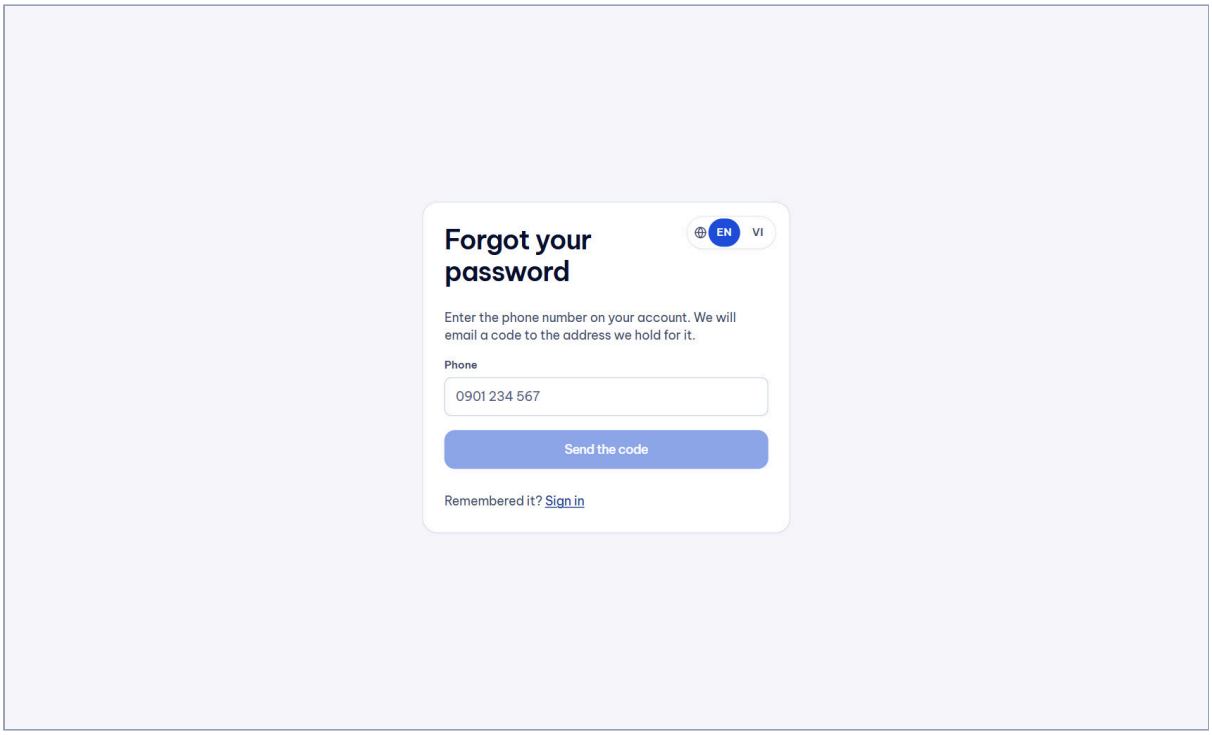
3.0.E3 The account is locked or disabled

1. The system refuses the request with FORBIDDEN (ERR_AUTH_05), and the registered user is told to contact the centre.

   The use case ends.

3.2.4 Reset a Forgotten Password

- **Function Trigger:** The registered user selects Forgot password on /login.
- **Function Description:**
  - **Use Case ID:** UC-04
  - **Actors:** Registered User; Secondary: Mail delivery (SMTP)
  - **Purpose:** A registered user who has forgotten the password proves control of the account email with a one-time code and sets a new password.
  - **Interface:** Forgot password (/forgot)
- **Data Processing:**
  - **Data Collect:** The account's phone number, the one-time code and the new password entered twice.
  - **Data Computation:** Answers the same way whether or not the phone has an account; checks the code, its five-minute life and the attempts, then the password rule (BR-02).
  - **Data Storage:** Writes the challenge to otp_challenges, the new password hash to users, revokes the other rows in sessions_auth and writes audit_logs. The code is sent through Mail delivery (SMTP).
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-04 Reset a Forgotten Password
Created By	Arena3 team
Date Created	2026-10-12

Field	Detail
Primary Actor	Registered User
Secondary Actors	Mail delivery (SMTP)
Trigger	The registered user selects Forgot password on /login.
Description	A registered user who has forgotten the password proves control of the account email with a one-time code and sets a new password.
Preconditions	PRE-1. The registered user is signed out.
Postconditions	POST-1. The password is replaced, every other session of the registered user is revoked, and an audit row is written.
Priority	Must have
Frequency of Use	A few times a week
Business Rules	BR-02

Normal Flow

4.0 Reset a Forgotten Password

1. The registered user enters the phone number of the account on /forgot.
2. The system emails a one-time code valid for five minutes to the address on the account. (see 4.0.E1)
3. The registered user enters the code and a new password twice.
4. The system checks the code and the password rules. (see 4.0.E2, 4.0.E3, 4.0.E4)
5. The system stores the new password and signs the registered user in.

Exceptions

4.0.E1 The phone has no account

1. The system shows the same message as for a real account, so it never reveals who is a member.

   The use case ends.

4.0.E2 The code is wrong or expired

1. The registered user may try again or request a new code.

   Return to step 3 of the Normal Flow.

4.0.E3 Five wrong codes

1. The reset is locked for 15 minutes (RATE_LIMITED, ERR_AUTH_03).

   The use case ends.

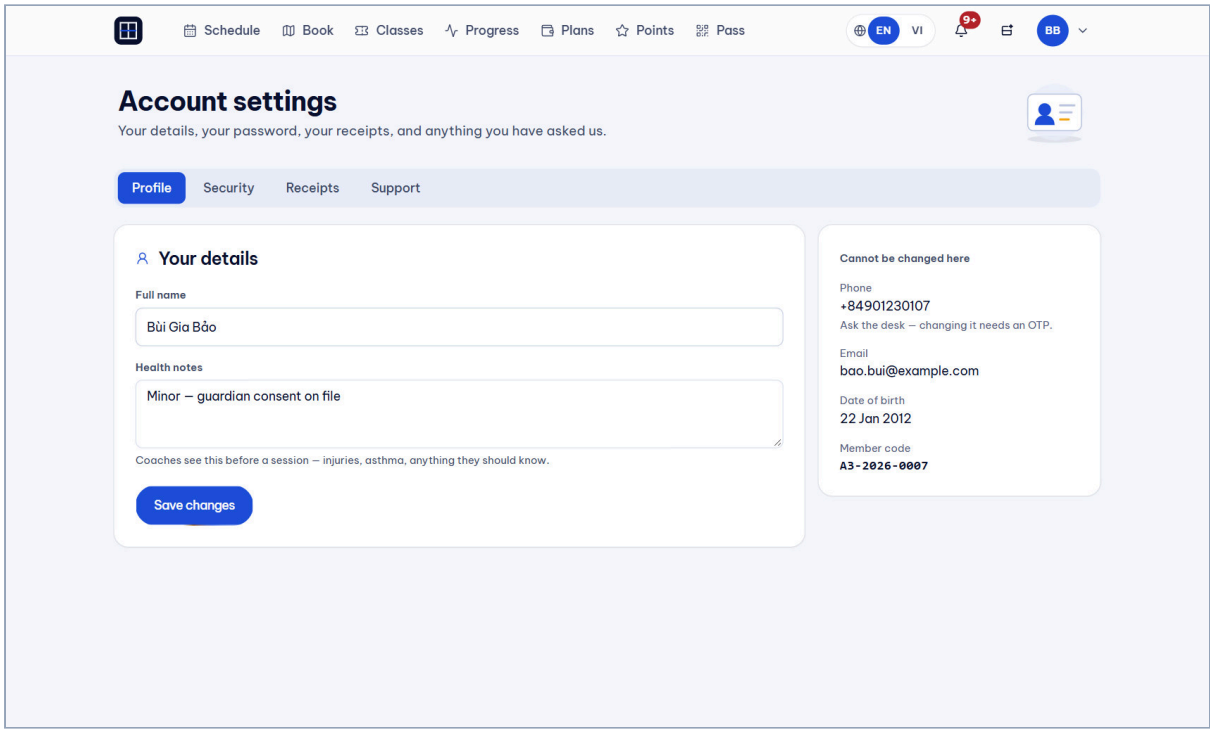
4.0.E4 The new password breaks BR-02, or the two entries differ

1. The system refuses the request with VALIDATION (5.2 Error Responses).

   Return to step 3 of the Normal Flow.

3.2.5 Manage the Personal Profile

- **Function Trigger:** The registered user opens Account (/account).
- **Function Description:**
  - **Use Case ID:** UC-05
  - **Actors:** Registered User; Secondary: None
  - **Purpose:** A registered user updates the profile details and changes the password.
  - **Interface:** Account settings (/account)
- **Data Processing:**
  - **Data Collect:** Name, phone, email, date of birth and language; for a password change, the current and the new password.
  - **Data Computation:** Checks the fields, that the phone and email are unused (BR-01) and the password rule (BR-02); refuses reuse of the current password.
  - **Data Storage:** Updates the registered user's row in users and writes audit_logs.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-05 Manage the Personal Profile
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Registered User
Secondary Actors	None
Trigger	The registered user opens Account (/account).

Field	Detail
Description	A registered user updates the profile details and changes the password.
Preconditions	PRE-1. The registered user is signed in.
Postconditions	POST-1. The profile or password is updated and an audit row is written.
Priority	Should have
Frequency of Use	Occasionally
Business Rules	BR-01, BR-02

Normal Flow

5.0 Manage the Personal Profile

1. The registered user opens Account.
2. The system shows the name, phone, email, date of birth and language.
3. The registered user edits the details and saves. (see 5.0.E1, 5.0.E2)
4. The system checks the fields and stores them. (see 5.0.E3)

Alternative Flows

5.1 The registered user changes the password

1. The registered user enters the current password and the new one twice. The system checks the current password, rejects reuse of the same password and stores the new hash.
   The use case ends.

Exceptions

5.0.E1 The current password is wrong

1. The system refuses the request with VALIDATION (5.2 Error Responses).
   Return to step 3 of the Normal Flow.

5.0.E2 The new password breaks BR-02 or the two entries differ

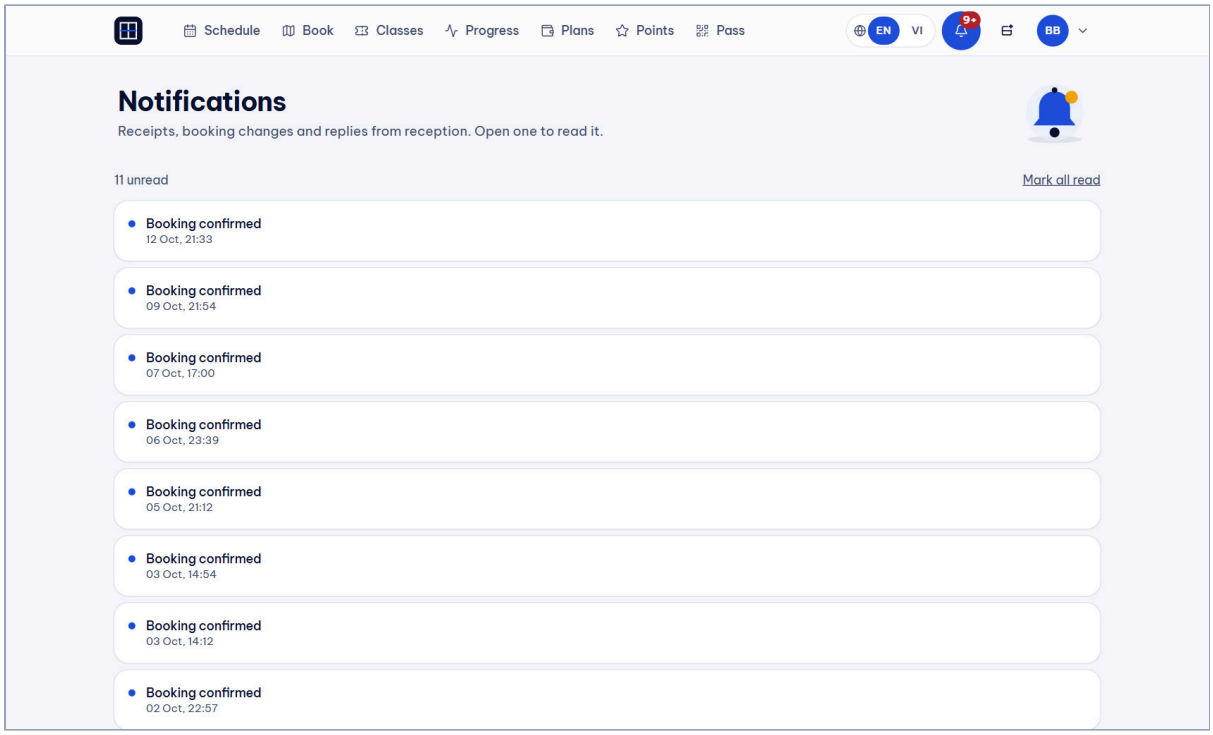
1. The system refuses the request with VALIDATION (5.2 Error Responses).
   Return to step 3 of the Normal Flow.

5.0.E3 The new phone or email belongs to another account

1. The system refuses the request with BR-01 (ERR_AUTH_01).
   Return to step 3 of the Normal Flow.

3.2.6 View Notifications

- **Function Trigger:** The registered user opens Notifications (/app/notifications or /alerts), or a new badge appears.
- **Function Description:**
  - **Use Case ID:** UC-06
  - **Actors:** Registered User; Secondary: None
  - **Purpose:** The registered user reads booking, payment, class, plan and support notices, and marks them as read.
  - **Interface:** Notifications (/app/notifications); Alerts (/alerts)
- **Data Processing:**
  - **Data Collect:** The notice opened, or Mark all as read.
  - **Data Computation:** Orders the notices newest first and counts the unread ones for the badge.
  - **Data Storage:** Sets the read time on the registered user's rows in outbox.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-06 View Notifications
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Registered User
Secondary Actors	None

Field	Detail
Trigger	The registered user opens Notifications (/app/notifications or /alerts), or a new badge appears.
Description	The registered user reads booking, payment, class, plan and support notices, and marks them as read.
Preconditions	PRE-1. The registered user is signed in.
Postconditions	POST-1. The notices shown are marked as read.
Priority	Should have
Frequency of Use	Daily
Business Rules	None

Normal Flow

6.0 View Notifications

1. The registered user opens Notifications.
2. The system lists the registered user's notices, newest first, with unread ones highlighted. (see 6.0.E1)
3. The registered user opens a notice; the system marks it as read and links to the related screen.

Alternative Flows

6.1 The registered user selects Mark all as read

1. Every notice is marked as read.

   The use case ends.

Exceptions

6.0.E1 The registered user has no notices

1. The system shows an empty list.

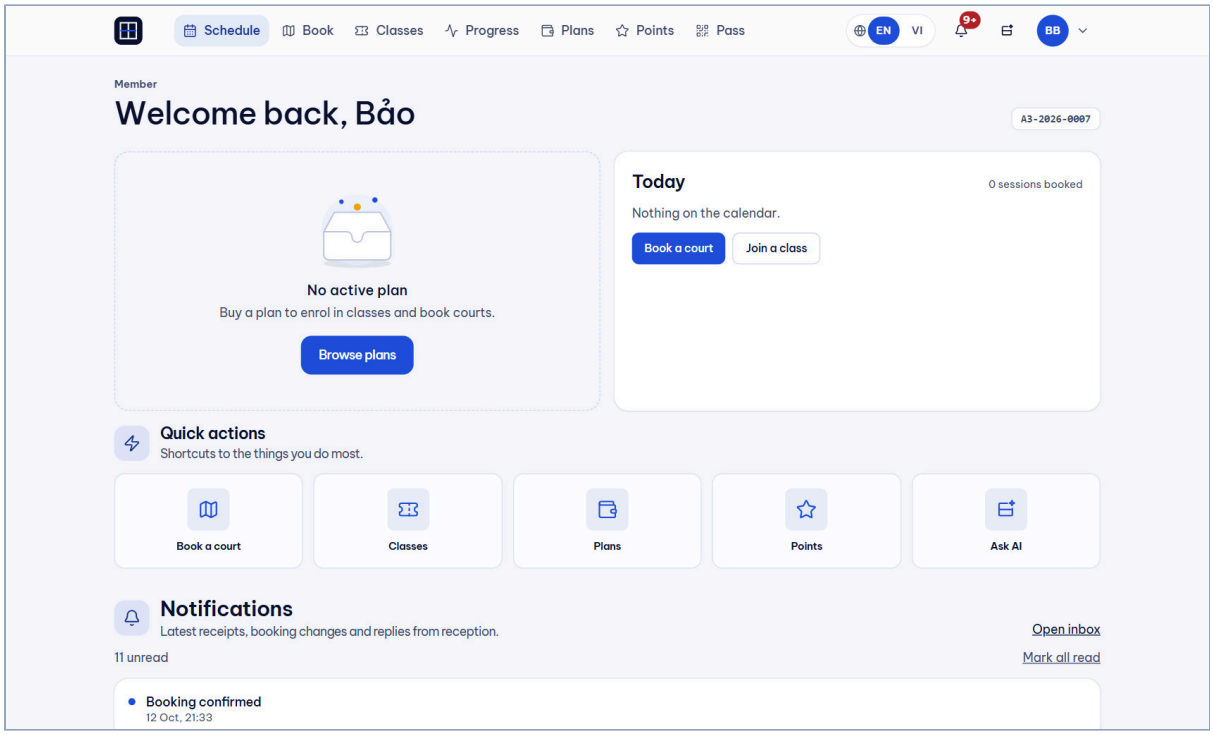
   The use case ends.

Other information: Notices are queued and sent in batches of 50, with up to five attempts each. Plan reminders go 7, 3 and 0 days before the end date.

3.3 Member Court Booking and Payment

3.3.1 View My Schedule

- **Function Trigger:** The member opens My schedule (/app).
- **Function Description:**
  - **Use Case ID:** UC-07
  - **Actors:** Member; Secondary: None
  - **Purpose:** The member sees upcoming court bookings, class sessions, plan balance and recent attendance in one place.
  - **Interface:** My schedule (/app)
- **Data Processing:**
  - **Data Collect:** None; the screen opens for the signed-in member.
  - **Data Computation:** Keeps upcoming items only and flags a plan that ends within seven days (BR-12).
  - **Data Binding:** Reads the member's court_bookings, enrollments with their sessions, subscriptions and attendance, and shows them in time order on My schedule.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-07 View My Schedule
Created By	Arena3 team
Date Created	2026-10-12

Field	Detail
Primary Actor	Member
Secondary Actors	None
Trigger	The member opens My schedule (/app).
Description	The member sees upcoming court bookings, class sessions, plan balance and recent attendance in one place.
Preconditions	PRE-1. The member is signed in.
Postconditions	POST-1. No data changes.
Priority	Must have
Frequency of Use	Daily
Business Rules	BR-12

Normal Flow

7.0 View My Schedule

1. The member opens My schedule.
2. The system lists upcoming bookings and class sessions in time order. (see 7.0.E1)
3. The system shows the active plan with its end date, sessions left and court hours left.
4. The system shows the attendance recorded for recent sessions.
5. The member opens a booking to reschedule or cancel it (UC-09).

Alternative Flows

7.1 The plan ends within seven days

1. The system shows a renewal notice.

   Return to step 4 of the Normal Flow.

Exceptions

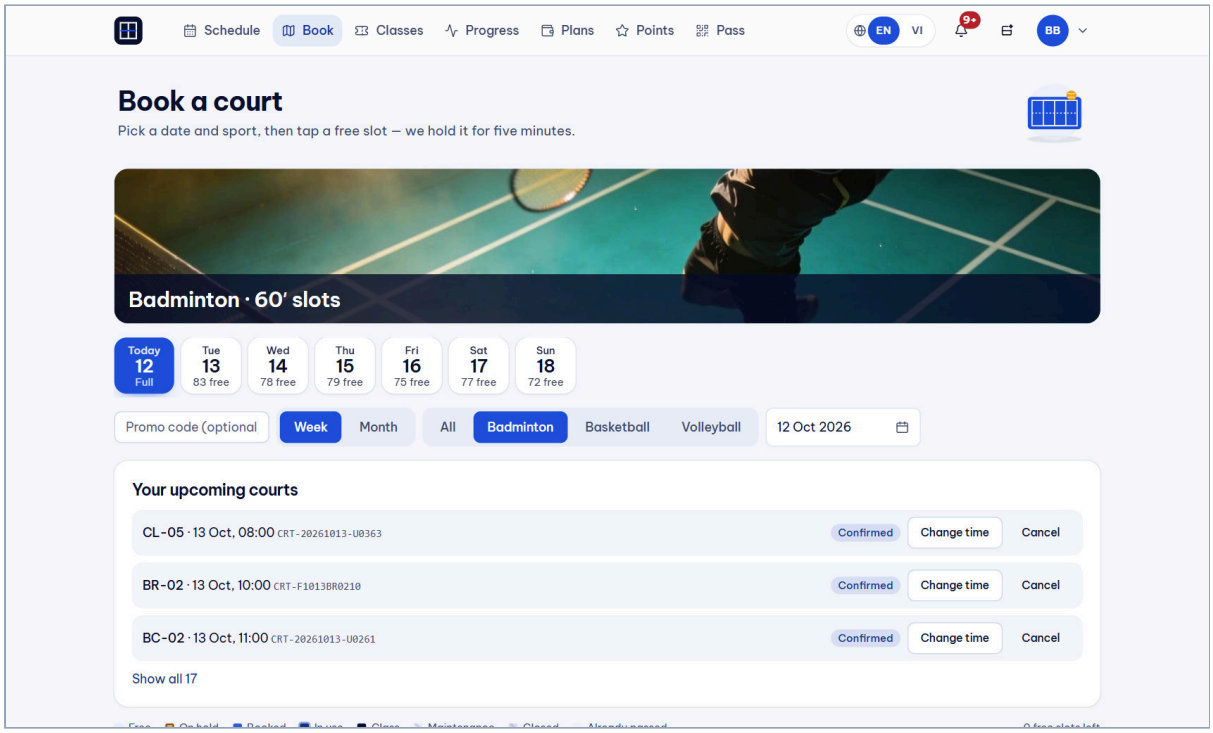
7.0.E1 The member has nothing booked

1. The system suggests Book a court and Classes.

   The use case ends.

3.3.2 Book a Court and Pay

- **Function Trigger:** The member selects a free slot on Book a court (/app/book).
- **Function Description:**
  - **Use Case ID:** UC-08
  - **Actors:** Member; Secondary: payOS
  - **Purpose:** The member holds a court slot for five minutes and pays the amount due to confirm it.
  - **Interface:** Book a court (/app/book); Payment result (/pay/return)
- **Data Processing:**
  - **Data Collect:** Sport, date and slot; an optional promo code; the payment method: payOS QR, bank transfer or plan hours.
  - **Data Computation:** Checks the booking window, opening hours, court status, daily cap and class clash (BR-05 to BR-10); prices the slot from price_rules (BR-37); applies the promo code (BR-40, BR-41, BR-42); rounds once (BR-39); verifies the payOS signature and the amount paid.
  - **Data Storage:** Writes the hold to court_bookings and occupancies, the payment to payments, the receipt to invoices and the code use to promotion_redemptions; deducts plan hours from subscriptions when paid by quota.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-08 Book a Court and Pay

Field	Detail
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Member
Secondary Actors	payOS
Trigger	The member selects a free slot on Book a court (/app/book).
Description	The member holds a court slot for five minutes and pays the amount due to confirm it.
Preconditions	PRE-1. The member is signed in. PRE-2. The slot is in the future, within seven days and inside opening hours. PRE-3. The court is ready.
Postconditions	POST-1. Success: the booking is confirmed and the payment is posted with a receipt code. POST-2. Failure: the hold lapses and the slot is free again.
Priority	Must have
Frequency of Use	Many times a day
Business Rules	BR-05, BR-06, BR-07, BR-08, BR-09, BR-10, BR-13, BR-37, BR-39, BR-40, BR-41, BR-42

Normal Flow

8.0 Book a Court and Pay

1. The member chooses a sport and a date, then a free slot on the court grid.
2. The system checks the booking window, the court status and the daily cap of two slots. (see 8.0.E1, 8.0.E2)
3. The system prices the slot from the price rules and holds it for five minutes. (see 8.0.E3, 8.0.E4)
4. The member may enter a promo code; the system applies it to the amount due. (see 8.0.E5)
5. The member chooses Pay online. The system creates a payOS link and shows the VietQR code. (see 8.0.E6, 8.0.E7)
6. The member scans the QR with a banking app and pays.
7. payOS confirms by a signed webhook, or the screen's check every three seconds reads the link as paid.
8. The system compares the amount with the amount due, posts the payment and confirms the booking. (see 8.0.E8)
9. The system shows the receipt, and the booking appears in My schedule.

Alternative Flows

8.1 The slot clashes with one of the member's classes

1. The system asks for confirmation (BR-09, WARN_BOOK_01), and the member may hold it anyway.

   Return to step 3 of the Normal Flow.

8.2 The slot is at peak time

1. The member pays a 30% deposit now, and the desk collects the balance on arrival (UC-22).

   Return to step 4 of the Normal Flow.

8.3 The member already holds a slot

1. The new hold is checked in full before it replaces the old one (BR-10).

   Return to step 4 of the Normal Flow.

8.4 The member pays from plan court hours

1. The system deducts the hours and confirms at once (BR-13).

   Return to step 9 of the Normal Flow.

Exceptions

8.0.E1 The slot is in the past, too far ahead or outside opening hours

1. The system refuses the request: BR-07, BR-05 (ERR_BOOK_01), BR-06.

   The use case ends.

8.0.E2 The court is not ready, or the member already holds two slots that day

1. The system refuses the request: BR-06, BR-05 (ERR_BOOK_01).

   The use case ends.

8.0.E3 No price rule matches the slot

1. The system refuses the request with BR-37 (ERR_BOOK_04); the system never invents a price.

   The use case ends.

8.0.E4 Another member took the slot first

1. The system refuses the request with CONFLICT_SLOT (ERR_BOOK_02).

   The use case ends.

8.0.E5 The promo code cannot be used

1. The system refuses the request: BR-40 (ERR_PAY_01), BR-41, BR-42.

   The use case ends.

8.0.E6 The plan has too few court hours left

1. The system refuses the request with BR-13.

   The use case ends.

8.0.E7 Online payment is switched off or cannot start

1. The system refuses the request with C-08 or C-09 (5.2 Error Responses), and the member pays at the desk.

   The use case ends.

8.0.E8 The hold lapsed before payment was confirmed

1. The system refuses the request with HOLD_EXPIRED (ERR_BOOK_03), and the member books again.

   Return to step 1 of the Normal Flow.

A booking is confirmed only when money is known to have arrived or plan hours are spent. Figure 23 shows this use case as an activity diagram.

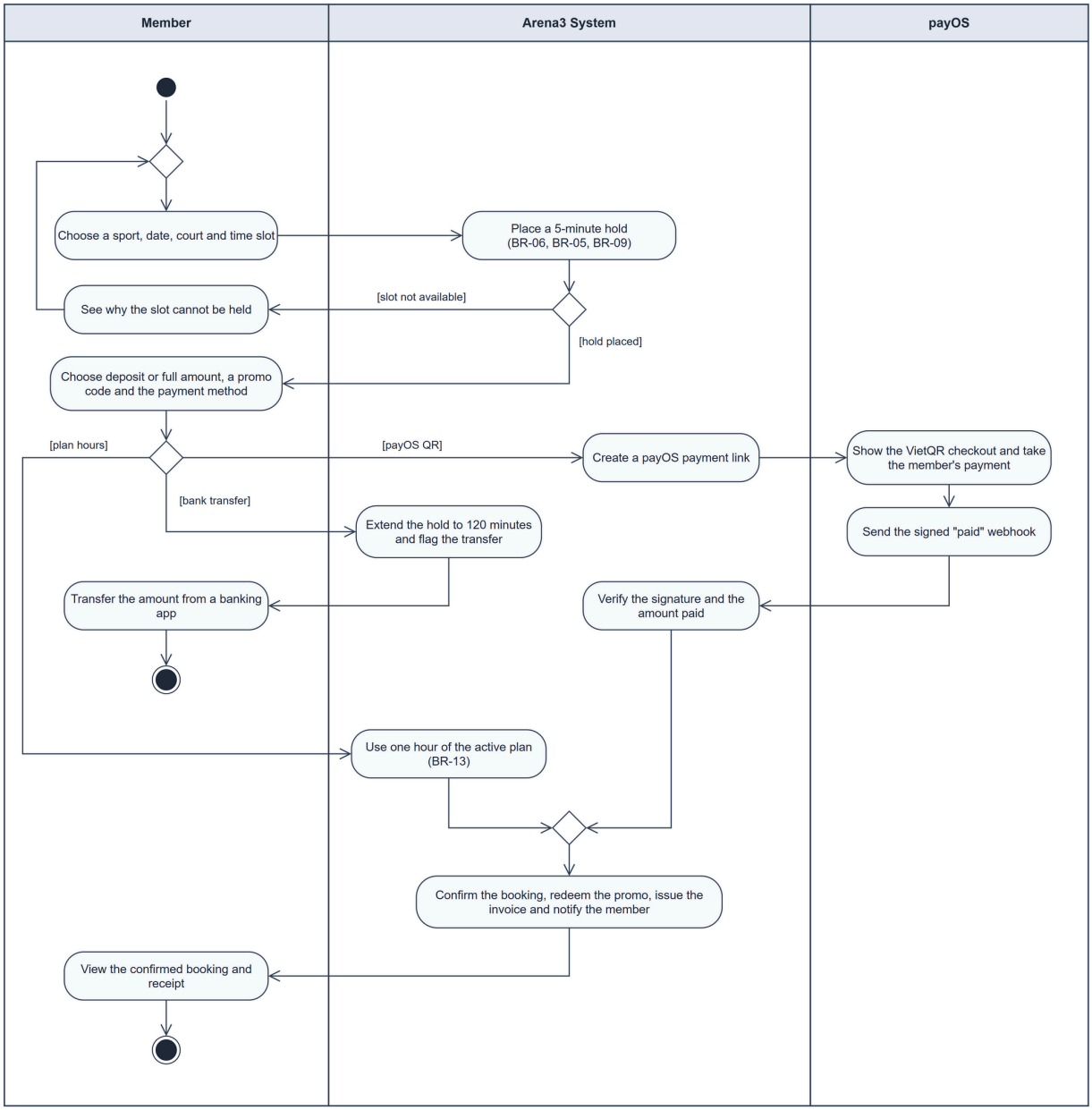


Figure 23. *Activity diagram: Book a Court and Pay*

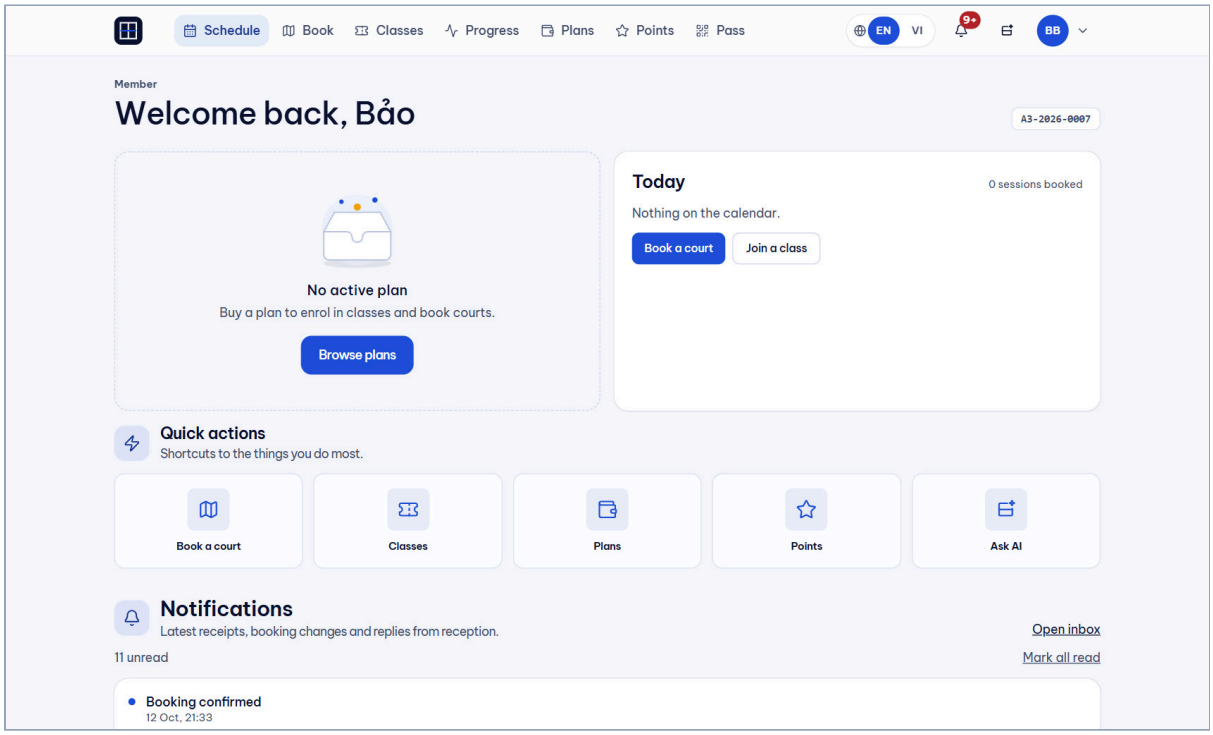
A slot is held for 5 minutes while the member pays. Three paths follow:

- **payOS QR:** the booking is confirmed only after Arena3 verifies payOS's signed result and the amount.

- **Bank transfer:** the hold stretches to 120 minutes and the transfer waits for the front desk to reconcile it.
- **Plan quota:** one court hour is taken from the member's active plan.

3.3.3 Reschedule or Cancel a Booking

- **Function Trigger:** The member opens a booking in My schedule and selects Reschedule or Cancel.
- **Function Description:**
  - **Use Case ID:** UC-09
  - **Actors:** Member; Secondary: None
  - **Purpose:** The member moves a confirmed booking to another slot, or cancels it and receives a refund by the notice given.
  - **Interface:** My schedule (/app)
- **Data Processing:**
  - **Data Collect:** The booking, then the new slot or the Cancel choice.
  - **Data Computation:** Checks the new slot against the booking rules (BR-05, BR-06, BR-07, BR-09) and the price paid; quotes the refund by notice: 100%, 50% or 0%.
  - **Data Storage:** Moves the booking in court_bookings and occupancies, or cancels it, frees the slot, writes the refund as a negative row in payments and returns plan hours to subscriptions.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-09 Reschedule or Cancel a Booking
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Member
Secondary Actors	None
Trigger	The member opens a booking in My schedule and selects Reschedule or Cancel.
Description	The member moves a confirmed booking to another slot, or cancels it and receives a refund by the notice given.
Preconditions	PRE-1. The member owns the booking. PRE-2. The booking is confirmed and has not started.
Postconditions	POST-1. Reschedule: the booking owns the new slot and the old slot is free. POST-2. Cancel: the booking is cancelled, the slot is free, and any refund or plan hour is returned.
Priority	Must have
Frequency of Use	Several times a day
Business Rules	BR-05, BR-06, BR-07, BR-09

Normal Flow

9.0 Reschedule or Cancel a Booking

1. The member opens the booking and selects Reschedule. (see 9.0.E1, 9.0.E2)
2. The member picks a new slot for the same sport.
3. The system checks the new slot with the booking rules and that its price equals the price paid. (see 9.0.E3, 9.0.E4, 9.0.E5)
4. The system moves the booking and frees the old slot.

Alternative Flows

9.1 Cancel the booking

1. The system quotes the refund by notice (100% at 24 hours or more, 50% at 4 to 24 hours, 0% under 4 hours).
2. The member confirms: the system cancels the booking, frees the slot and raises the refund. Plan hours come back when the refund is 50% or more.

   The use case ends.

9.2 The booking is still an unpaid hold

1. It is released at no cost.

   The use case ends.

Exceptions

9.0.E1 The booking is in a state that cannot be cancelled or moved

1. The system refuses the request with CONFLICT_STATE (5.2 Error Responses).

The use case ends.

9.0.E2 The booking starts in less than 2 hours

1. It can no longer be moved.

   The use case ends.

9.0.E3 The new slot breaks a booking rule or clashes with a class

1. The system refuses the request: BR-05 (ERR_BOOK_01), BR-06, BR-07, BR-09 (WARN_BOOK_01).

   The use case ends.

9.0.E4 The new slot has a different price

1. The member cancels and books the new slot instead.

   The use case ends.

9.0.E5 Another member took the new slot first

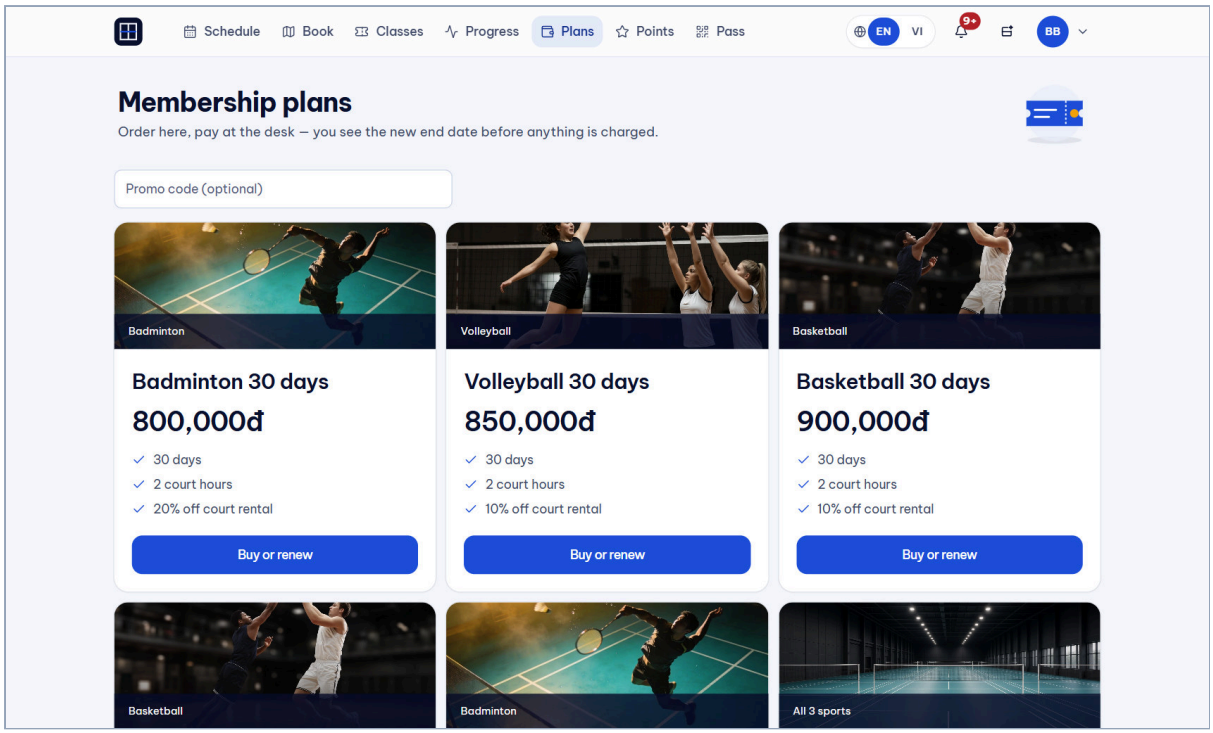
1. The system refuses the request with CONFLICT_SLOT (ERR_BOOK_02).

   The use case ends.

3.3.4 Order a Membership Plan

- **Function Trigger:** The member selects Order on a plan in Plans (/app/plans).
- **Function Description:**
  - **Use Case ID:** UC-10
  - **Actors:** Member; Secondary: Receptionist, who takes payment; payOS
  - **Purpose:** The member orders a membership plan online and pays for it at the front desk.
  - **Interface:** Membership plans (/app/plans)
- **Data Processing:**
  - **Data Collect:** The plan chosen and an optional promo code.
  - **Data Computation:** Checks the plan is on sale (BR-16), the current plan is not frozen (BR-14), the promo code (BR-40) and the order rate limit.
  - **Data Storage:** Writes a pending row to subscriptions; once the desk takes payment, the payment goes to payments and the receipt to invoices.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-10 Order a Membership Plan
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Member
Secondary Actors	Receptionist, who takes payment; payOS
Trigger	The member selects Order on a plan in Plans (/app/plans).
Description	The member orders a membership plan online and pays for it at the front desk.
Preconditions	PRE-1. The member is signed in. PRE-2. The plan is on sale.
Postconditions	POST-1. A plan order is pending payment. POST-2. Once paid, the plan becomes active from its start date.
Priority	Must have
Frequency of Use	Daily
Business Rules	BR-14, BR-16, BR-40

Normal Flow

10.0 Order a Membership Plan

1. The member opens Plans and compares the plans on sale.
2. The member selects a plan and may enter a promo code.
3. The system checks the plan and the code and creates a pending order. (see 10.0.E1, 10.0.E2, 10.0.E3, 10.0.E4, 10.0.E5)

4. The system tells the member to pay at the front desk.
5. The receptionist takes payment for the order (UC-20), and the plan becomes active.

Alternative Flows

10.1 The receptionist raises a payOS link for the order, and the member pays by VietQR

1. The receptionist raises a payOS link for the order, and the member pays by VietQR.
   The use case ends.

Exceptions

10.0.E1 The plan has been withdrawn

1. The system refuses the request with BR-16.
   The use case ends.

10.0.E2 The member's current plan is frozen

1. The system refuses the request with BR-14.
   The use case ends.

10.0.E3 The promo code does not apply

1. The system refuses the request with BR-40 (ERR_PAY_01).
   The use case ends.

10.0.E4 The member tries to pay for a plan online

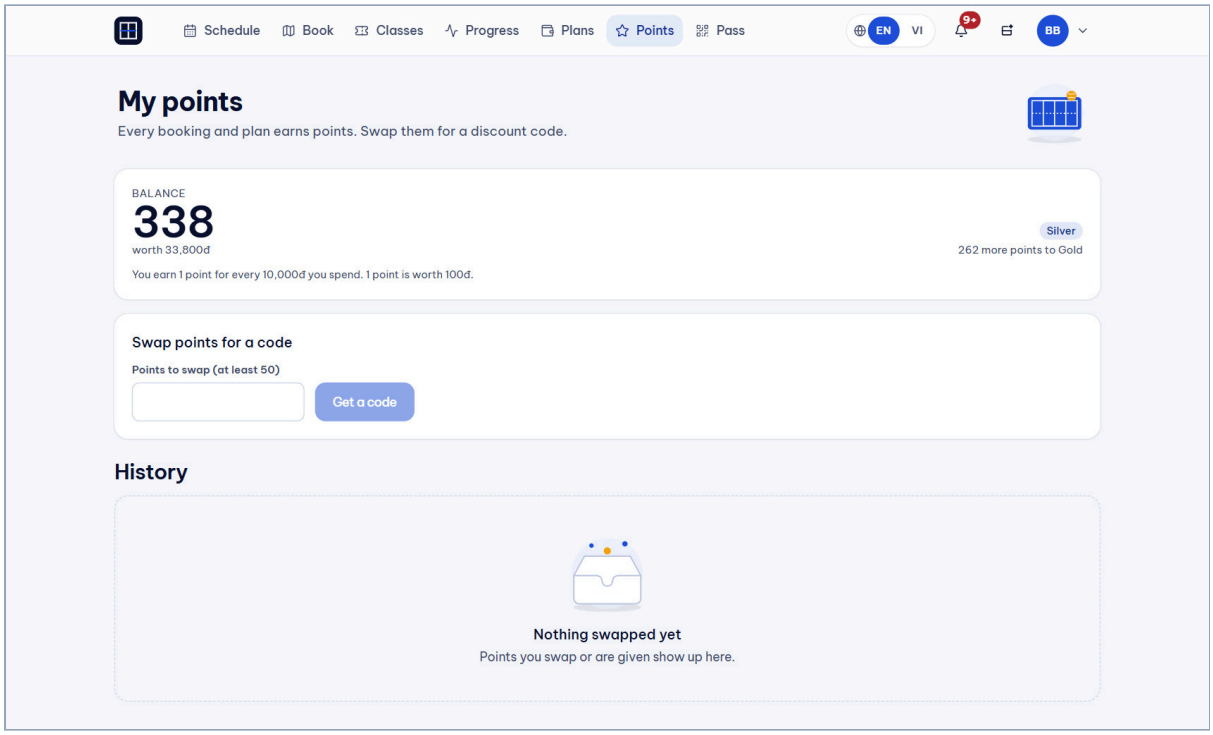
1. The system answers that the front desk takes payment for plans.
   The use case ends.

10.0.E5 More than 3 orders a minute or 10 an hour

1. The system refuses the request with RATE_LIMITED (ERR_AUTH_06).
   The use case ends.

3.3.5 View Receipts and Redeem Points

- **Function Trigger:** The member opens Points (/app/points) or a receipt from My schedule.
- **Function Description:**
  - **Use Case ID:** UC-11
  - **Actors:** Member; Secondary: None
  - **Purpose:** The member views paid receipts and e-invoices, and turns loyalty points into a single-use discount code.
  - **Interface:** My points (/app/points)
- **Data Processing:**
  - **Data Collect:** The number of points to redeem, or the receipt opened.
  - **Data Computation:** Totals the points, checks the minimum of 50 points and the balance, and values each point at 100đ (BR-40, BR-41 then apply to the code).
  - **Data Storage:** Writes the points spent to loyalty_adjustments and a single-use code to promotions; a receipt is read from payments and invoices.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-11 View Receipts and Redeem Points
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Member
Secondary Actors	None

Field	Detail
Trigger	The member opens Points (/app/points) or a receipt from My schedule.
Description	The member views paid receipts and e-invoices, and turns loyalty points into a single-use discount code.
Preconditions	PRE-1. The member is signed in.
Postconditions	POST-1. Redeem: points are spent and a promo code worth their value is created.
Priority	Should have
Frequency of Use	Weekly
Business Rules	BR-40, BR-41

Normal Flow

11.0 View Receipts and Redeem Points

1. The member opens Points.
2. The system shows the points balance, the tier and the points needed for the next tier.
3. The member enters the points to redeem.
4. The system creates a single-use code worth 100đ per point, valid for 60 days, and lowers the balance. (see 11.0.E1)
5. The member uses the code at checkout (UC-08 or UC-10).

Alternative Flows

11.1 The member opens a receipt instead

1. The system shows the payment and its e-invoice as PDF.
   The use case ends.

Exceptions

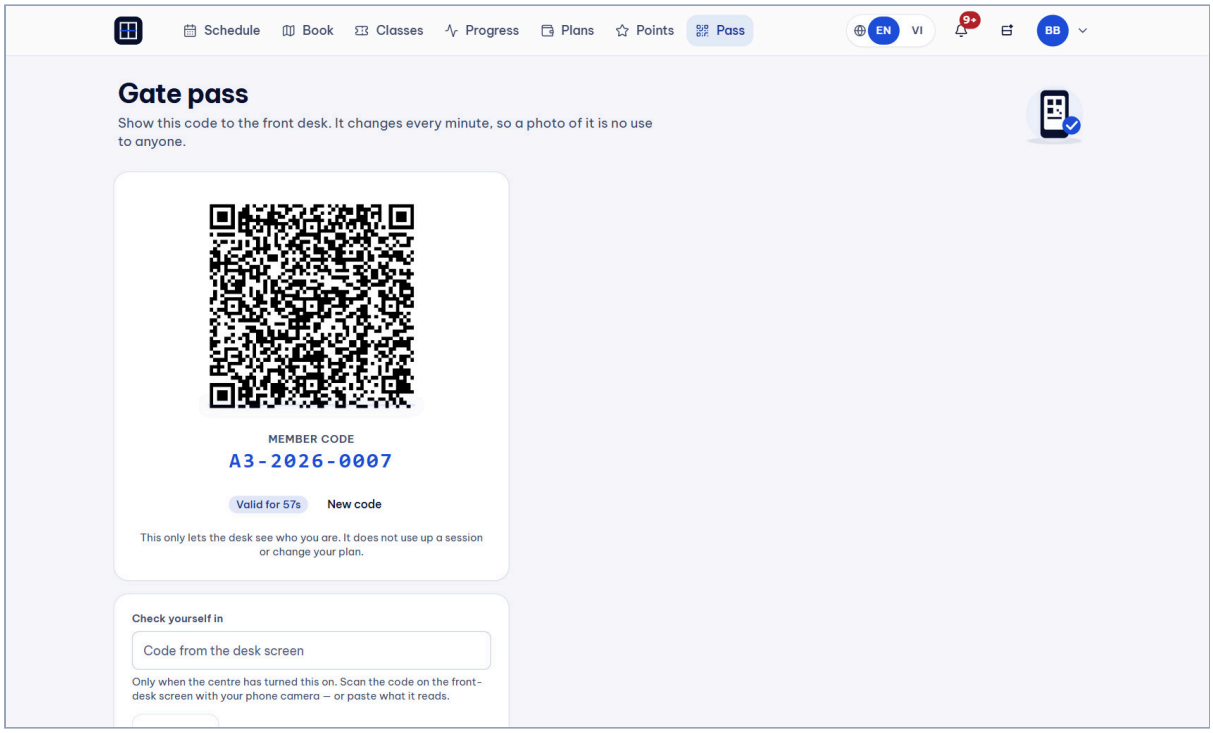
11.0.E1 Fewer than 50 points are entered, or more than the balance

1. The system refuses the request with VALIDATION (5.2 Error Responses).
   The use case ends.

Other information: One point is earned for every 10,000đ paid for bookings, plans and day passes. Spending points never lowers the tier.

3.3.6 Check In by Self-Service

- **Function Trigger:** The member opens Gate pass (/app/pass) at the centre.
- **Function Description:**
  - **Use Case ID:** UC-12
  - **Actors:** Member; Secondary: None
  - **Purpose:** The member shows a short-lived QR pass at the desk, or scans the desk code to check in without staff.
  - **Interface:** Gate pass (/app/pass)
- **Data Processing:**
  - **Data Collect:** The member's request for a pass; with self check-in, the desk code scanned.
  - **Data Computation:** Signs a short-lived token for the member and, with self check-in, checks the desk code (BR-30); spends no session (BR-32).
  - **Data Storage:** Writes the token to checkin_tokens and, with self check-in, the visit to attendance.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-12 Check In by Self-Service
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Member

Field	Detail
Secondary Actors	None
Trigger	The member opens Gate pass (/app/pass) at the centre.
Description	The member shows a short-lived QR pass at the desk, or scans the desk code to check in without staff.
Preconditions	PRE-1. The member is signed in.
Postconditions	POST-1. Self check-in: one visit row exists for the member.
Priority	Should have
Frequency of Use	Daily
Business Rules	BR-30, BR-32

Normal Flow

12.0 Check In by Self-Service

1. The member opens Gate pass.
2. The system shows a short-lived member QR code that refreshes itself.
3. The member shows the code to the receptionist, who continues with UC-26. (see 12.0.E1, 12.0.E2)

Alternative Flows

12.1 Self check-in is switched on

1. The member scans the desk code with the phone, and the system records the visit as in UC-26.

   The use case ends.

Exceptions

12.0.E1 Self check-in is switched off

1. The system refuses the request with BR-30 (ERR_CHECKIN_01), and the member shows the pass at the desk.

   The use case ends.

12.0.E2 The scanned code is not the desk code, or was already used

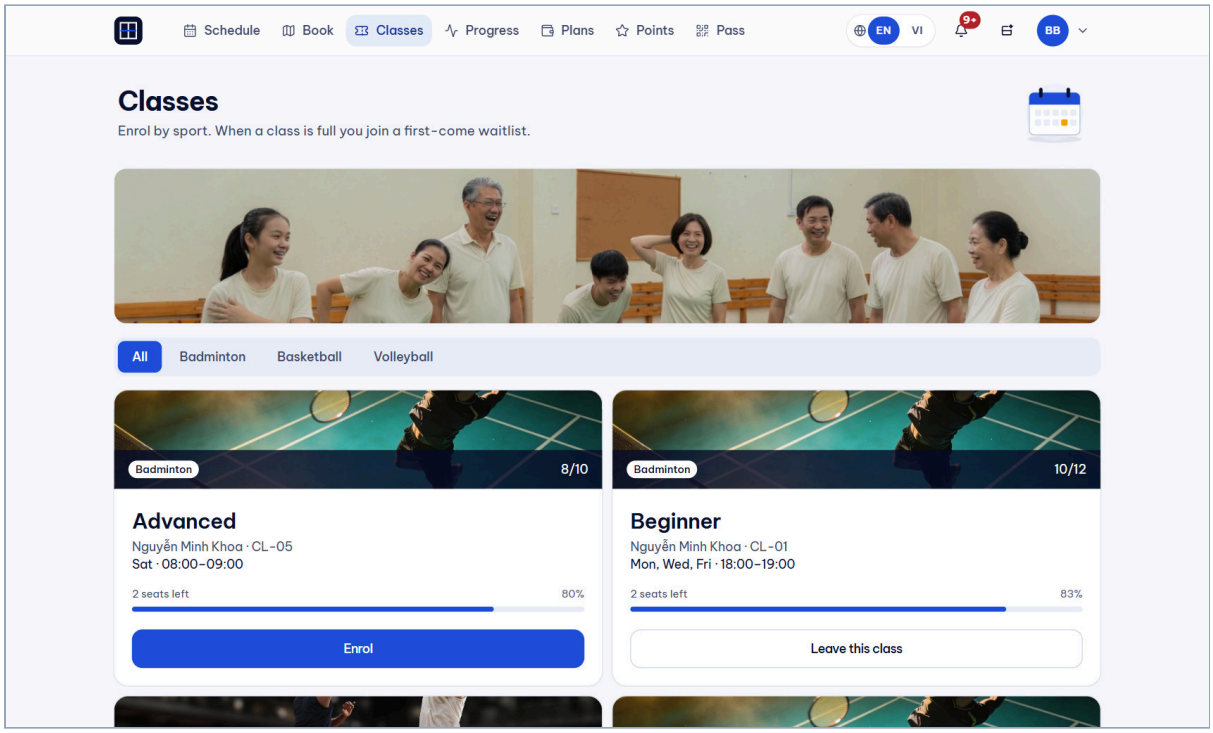
1. The system refuses the request with BR-30 (ERR_CHECKIN_01).

   The use case ends.

3.4 Member Classes and Training

3.4.1 Enrol in a Class

- **Function Trigger:** The member selects Enrol on a class in Classes (/app/classes).
- **Function Description:**
  - **Use Case ID:** UC-13
  - **Actors:** Member; Secondary: Receptionist or Manager, enrolling on the member's behalf
  - **Purpose:** The member takes a seat in an open class, or joins its waitlist when the class is full.
  - **Interface:** Classes (/app/classes); Classes (desk) (/desk/classes)
- **Data Processing:**
  - **Data Collect:** The class chosen.
  - **Data Computation:** Checks the class is open (BR-18), a plan covers the sport and has sessions left (BR-12, BR-19, BR-20), the capacity (BR-21) and clashes (BR-22).
  - **Data Storage:** Writes a confirmed or waitlisted row to enrollments, raises the class's enrolled count in classes and uses a session from subscriptions.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-13 Enrol in a Class
Created By	Arena3 team

Field	Detail
Date Created	2026-10-12
Primary Actor	Member
Secondary Actors	Receptionist or Manager, enrolling on the member's behalf
Trigger	The member selects Enrol on a class in Classes (/app/classes).
Description	The member takes a seat in an open class, or joins its waitlist when the class is full.
Preconditions	PRE-1. The member is signed in and holds an active plan covering the class sport.
Postconditions	POST-1. The member is enrolled and the sessions appear in My schedule, or the member is waitlisted with a position.
Priority	Must have
Frequency of Use	Daily
Business Rules	BR-12, BR-18, BR-19, BR-20, BR-21, BR-22

Normal Flow

13.0 Enrol in a Class

1. The member opens Classes and chooses an open class. (see 13.0.E1)
2. The system checks that a plan covers the sport and has sessions left. (see 13.0.E2, 13.0.E3)
3. The system checks the class does not clash with the member's other classes. (see 13.0.E4)
4. A seat is free: the system enrols the member and adds the class sessions to My schedule.

Alternative Flows

13.1 The class is full

1. The system adds the member to the waitlist and shows the position.
   The use case ends.

13.2 A seat frees up later

1. The next member in line receives an offer valid for two hours (UC-14).
   The use case ends.

Exceptions

13.0.E1 The class is not open for enrolment

1. The system refuses the request with BR-18.
   The use case ends.

13.0.E2 No active plan covers the sport

1. The system refuses the request with BR-19 (ERR_CLASS_01).
   The use case ends.

13.0.E3 The plan has no sessions left

1. The system refuses the request with BR-20.
   The use case ends.

13.0.E4 The class clashes with another class, or the member is already enrolled

1. The system refuses the request with BR-22.

   The use case ends.

A full class never refuses a member outright; it offers a waitlist place instead. Figure 24 shows this use case as an activity diagram.

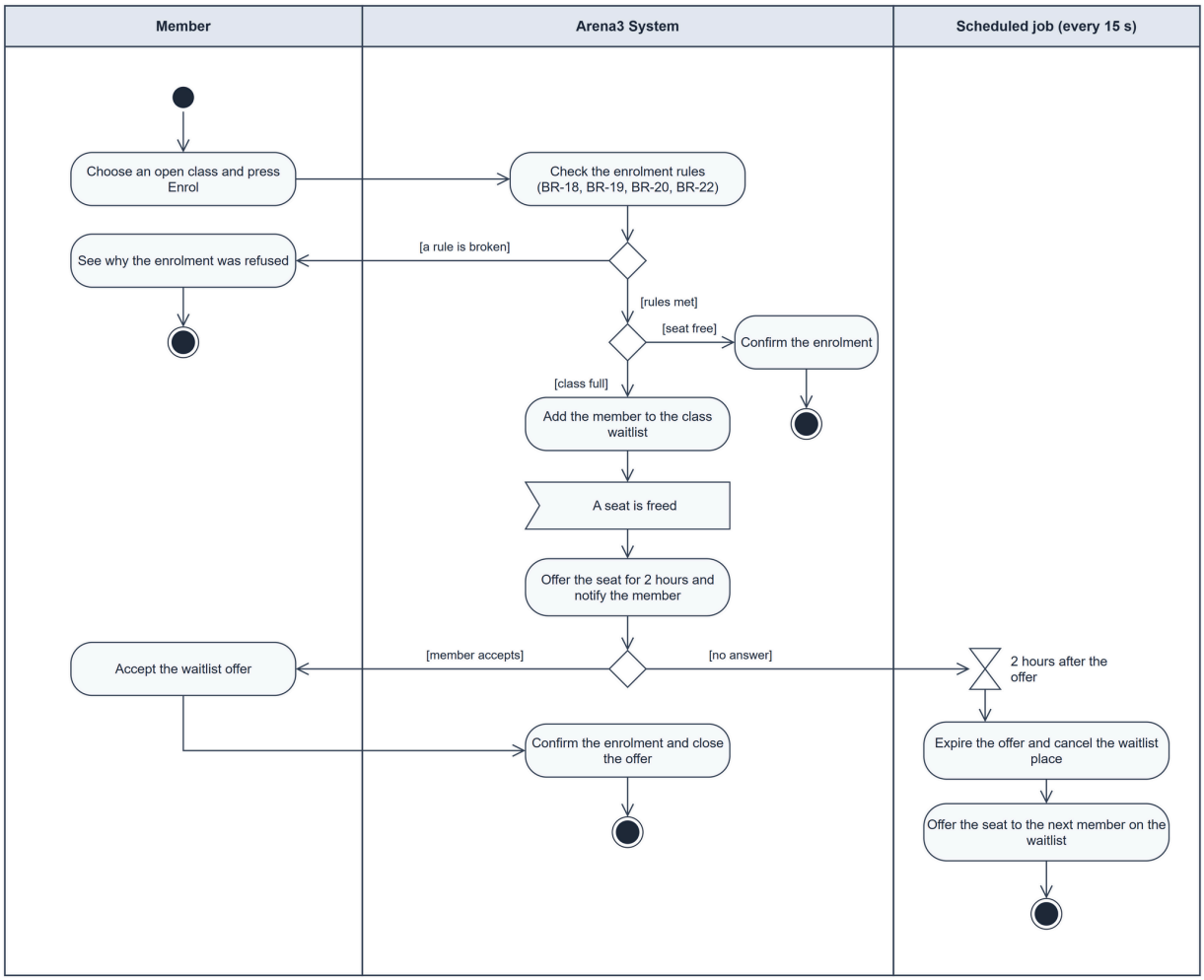
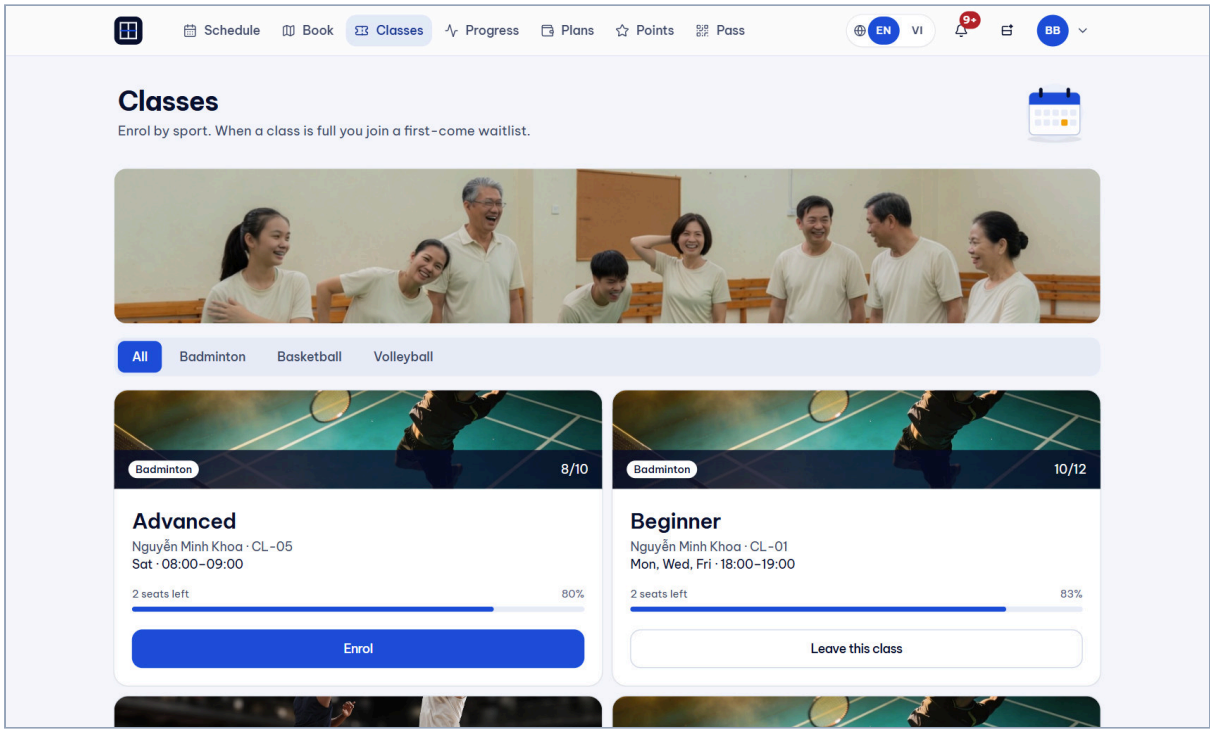


Figure 24. *Activity diagram: Enrol in a Class or Join the Waitlist*

When a seat frees up, the next member gets a 2-hour offer. The scheduled job expires an unanswered offer and moves on to the next member.

3.4.2 Accept a Waitlist Offer

- **Function Trigger:** The member receives a waitlist offer notice.
- **Function Description:**
  - **Use Case ID:** UC-14
  - **Actors:** Member; Secondary: None
  - **Purpose:** A waitlisted member accepts a freed seat before the offer expires.
  - **Interface:** Classes (/app/classes); Notifications (/app/notifications)
- **Data Processing:**
  - **Data Collect:** The Accept choice on the offer.
  - **Data Computation:** Checks the offer has not expired (BR-23), a seat is free (BR-21) and the plan covers the sport (BR-19).
  - **Data Storage:** Closes the offer in waitlist_offers and confirms the enrolment in enrollments.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-14 Accept a Waitlist Offer
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Member
Secondary Actors	None
Trigger	The member receives a waitlist offer notice.

Field	Detail
Description	A waitlisted member accepts a freed seat before the offer expires.
Preconditions	PRE-1. The member holds an offer that has not expired.
Postconditions	POST-1. The member is enrolled and the offer is closed.
Priority	Should have
Frequency of Use	A few times a week
Business Rules	BR-19, BR-21, BR-23

Normal Flow

14.0 Accept a Waitlist Offer

1. The member opens the offer from Notifications or Classes.
2. The member selects Accept.
3. The system checks the offer, the seat and the member's plan. (see 14.0.E1, 14.0.E2, 14.0.E3)
4. The system enrols the member and closes the offer.

Alternative Flows

14.1 The member ignores the offer

1. After two hours the scheduled job expires it and offers the seat to the next member (BR-23, ERR_CLASS_02).

   The use case ends.

Exceptions

14.0.E1 The offer has expired

1. The system refuses the request with BR-23 (ERR_CLASS_02).

   The use case ends.

14.0.E2 The seat was filled in the meantime

1. The system refuses the request with BR-21.

   The use case ends.

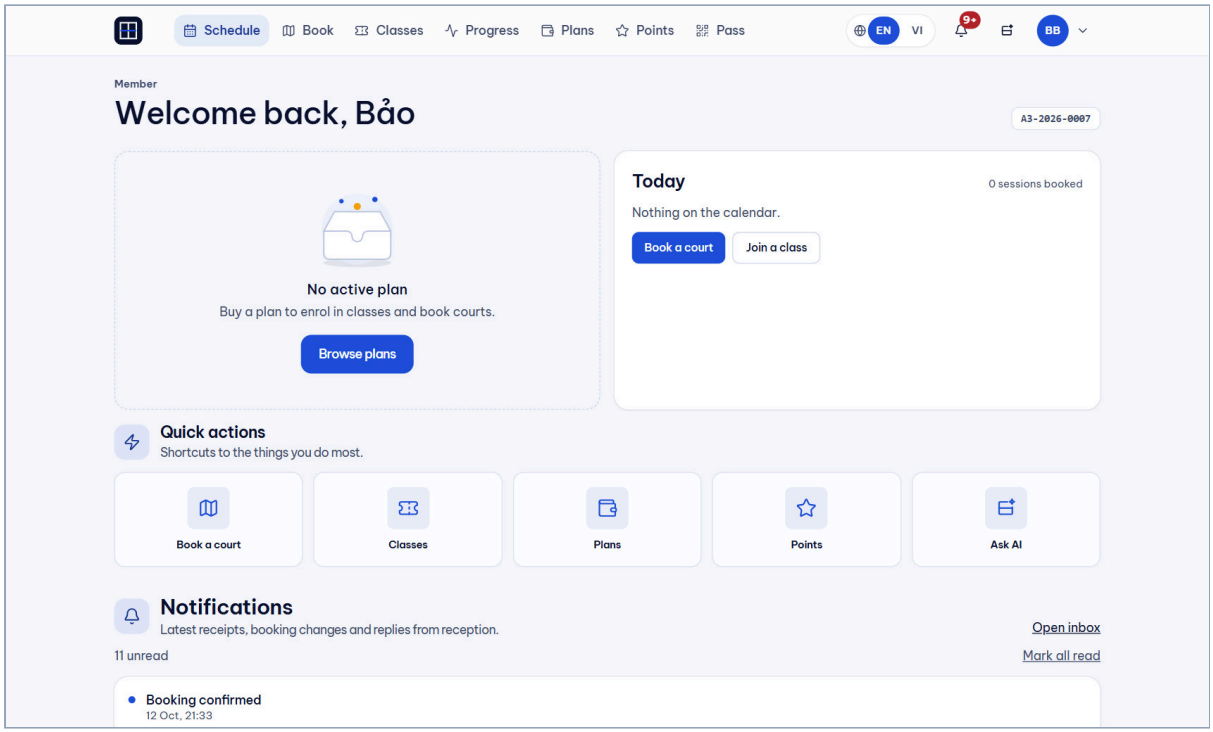
14.0.E3 The plan no longer covers the sport

1. The system refuses the request with BR-19 (ERR_CLASS_01).

   The use case ends.

3.4.3 Leave a Class

- **Function Trigger:** The member selects Leave on an enrolled class.
- **Function Description:**
  - **Use Case ID:** UC-15
  - **Actors:** Member; Secondary: None
  - **Purpose:** The member leaves a class in time, which frees the seat for the waitlist.
  - **Interface:** My schedule (/app)
- **Data Processing:**
  - **Data Collect:** The class to leave.
  - **Data Computation:** Checks the next session starts at least 4 hours later (BR-24) and finds the first waitlisted member.
  - **Data Storage:** Cancels the row in enrollments, lowers the enrolled count in classes, writes an offer to waitlist_offers and queues a notice in outbox.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-15 Leave a Class
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Member
Secondary Actors	None
Trigger	The member selects Leave on an enrolled class.

Field	Detail
Description	The member leaves a class in time, which frees the seat for the waitlist.
Preconditions	PRE-1. The member is enrolled in the class.
Postconditions	POST-1. The enrolment is cancelled, and the next waitlisted member receives an offer.
Priority	Should have
Frequency of Use	A few times a week
Business Rules	BR-24

Normal Flow

15.0 Leave a Class

1. The member opens the class in My schedule and selects Leave.
2. The system checks the time to the next session. (see 15.0.E1)
3. The system cancels the enrolment and removes the future sessions from My schedule.
4. The system offers the seat to the first member on the waitlist.

Exceptions

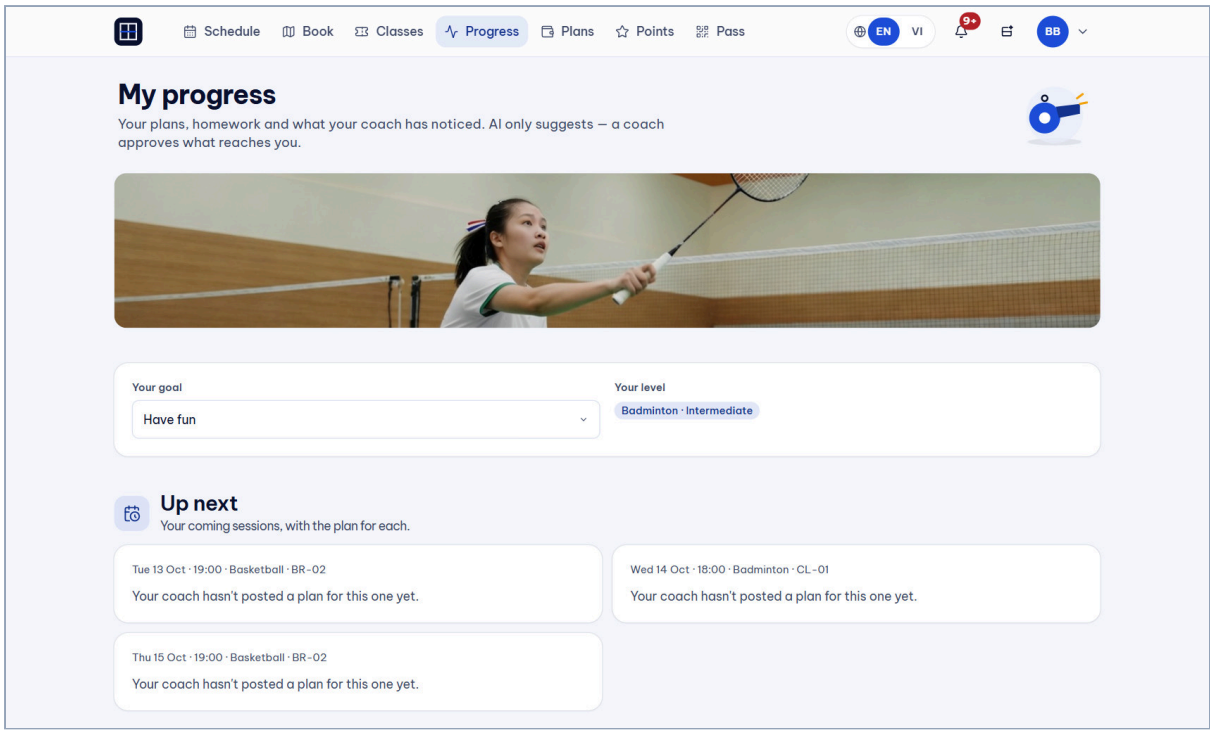
15.0.E1 The next session starts in less than 4 hours

1. The system refuses the request with BR-24, and the member stays enrolled.
   The use case ends.

3.4.4 Follow My Training Progress

- **Function Trigger:** The member opens Training (/app/train).
- **Function Description:**
  - **Use Case ID:** UC-16
  - **Actors:** Member; Secondary: None
  - **Purpose:** The member follows training plans, coach feedback, reviews and homework, and sets a personal goal.
  - **Interface:** My progress (/app/train)
- **Data Processing:**
  - **Data Collect:** Ticked homework items and the personal goal.
  - **Data Computation:** Checks the training module is on (BR-52); homework past its due date stays open; nothing here takes money or changes a plan balance (BR-47).
  - **Data Storage:** Writes homework progress to homework_recipients and the goal to training_profiles.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-16 Follow My Training Progress
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Member
Secondary Actors	None
Trigger	The member opens Training (/app/train).
Description	The member follows training plans, coach feedback, reviews and homework, and sets a personal goal.
Preconditions	PRE-1. The member is signed in and enrolled in at least one class. PRE-2. The training module is switched on.
Postconditions	POST-1. Homework progress and the goal are saved.
Priority	Could have
Frequency of Use	Weekly
Business Rules	BR-47, BR-52

Normal Flow

16.0 Follow My Training Progress

1. The member opens Training. (see 16.0.E1)
2. The system shows the plan of each upcoming session, the latest session results and the coach's reviews.
3. The member opens a homework item and ticks the checklist items done. (see 16.0.E2)

4. The system saves the progress, which the coach sees on the student profile.
5. The member sets or edits a personal goal; the system saves it.

Exceptions

16.0.E1 The training module is switched off

1. The system refuses the request with FORBIDDEN (ERR_AUTH_05) with BR-52.
   The use case ends.

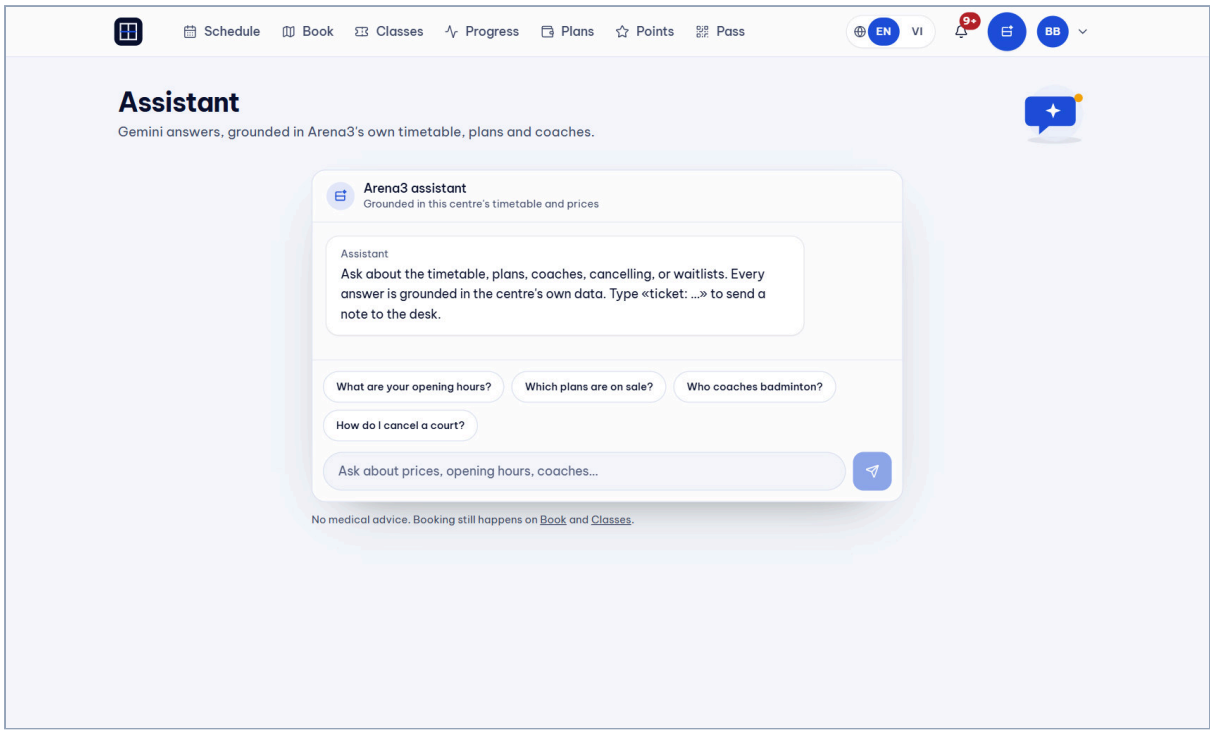
16.0.E2 The homework is past its due date

1. It is shown as overdue but can still be ticked.
   The use case ends.

3.4.5 Get Help from the AI Assistant or Staff

- **Function Trigger:** The member opens Assistant (/app/assistant).
- **Function Description:**
  + **Use Case ID:** UC-17
  + **Actors:** Member; Secondary: Gemini AI
  + **Purpose:** The member asks the AI assistant about the centre, or sends a support request to the front desk.
  + **Interface:** AI assistant (/app/assistant)
- **Data Processing:**
  + **Data Collect:** A question of up to 800 characters, or a support request.
  + **Data Computation:** Builds the prompt from centre facts only, without health notes (BR-46); allows 20 questions a minute; falls back to rule-based answers when the model is unavailable.
  + **Data Storage:** Writes an audit row for each question to audit_logs, or the support request to tickets.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-17 Get Help from the AI Assistant or Staff
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Member
Secondary Actors	Gemini AI
Trigger	The member opens Assistant (/app/assistant).
Description	The member asks the AI assistant about the centre, or sends a support request to the front desk.
Preconditions	PRE-1. The member is signed in.
Postconditions	POST-1. An answer is shown, or a support request is open in the desk queue.
Priority	Could have
Frequency of Use	Daily
Business Rules	BR-46

Normal Flow

17.0 Get Help from the AI Assistant or Staff

1. The member types a question of up to 800 characters.
2. The system sends the question with the centre facts to the Gemini model, without health notes. (see 17.0.E1, 17.0.E2)
3. The system shows the answer.

Alternative Flows

17.1 The member writes a support request instead

1. The system opens a ticket for the desk (UC-29), and the reply arrives as a notice.

   The use case ends.

Exceptions

17.0.E1 More than 20 questions in a minute

1. The system refuses the request with RATE_LIMITED (ERR_AUTH_06).

   The use case ends.

17.0.E2 The model is unavailable

1. The system answers from the centre facts and suggests a support request.

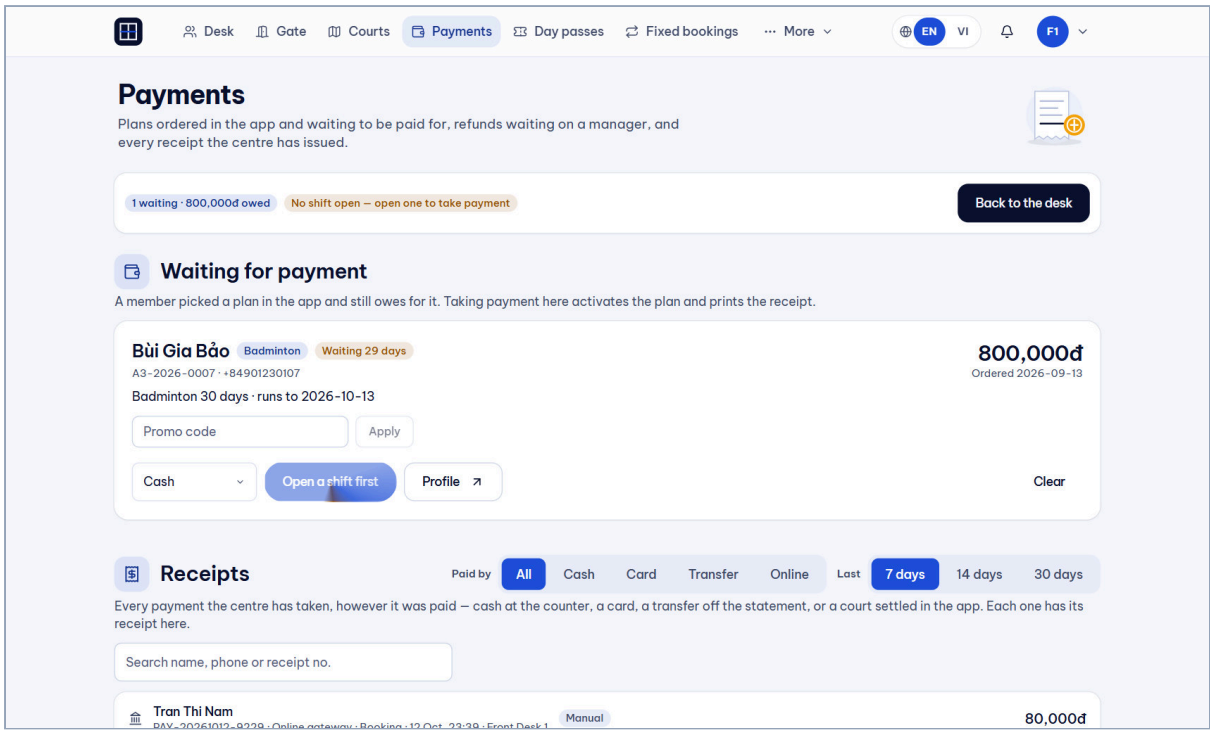
   The use case ends.

3.5 Front Desk Sales

3.5.1 Open and Close a Cashier Shift

- **Function Trigger:** The receptionist starts or ends a working shift on Payments (/desk/payments).
- **Function Description:**
  - **Use Case ID:** UC-18
  - **Actors:** Receptionist; Secondary: None
  - **Purpose:** The receptionist opens a cashier shift with a float and closes it by declaring the cash counted.
  - **Interface:** Payments (/desk/payments)
- **Data Processing:**
  - **Data Collect:** The opening float; at close, the cash counted.
  - **Data Computation:** Allows one open shift per receptionist and works out the expected cash and the difference; desk payments need the open shift (BR-44).
  - **Data Storage:** Writes the shift to cashier_shifts; each desk payment in payments carries the shift.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-18 Open and Close a Cashier Shift
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Receptionist
Secondary Actors	None
Trigger	The receptionist starts or ends a working shift on Payments (/desk/payments).
Description	The receptionist opens a cashier shift with a float and closes it by declaring the cash counted.
Preconditions	PRE-1. The receptionist is signed in.
Postconditions	POST-1. Open: the shift is open and payments are tied to it. POST-2. Close: the shift is closed with the expected and declared cash and the difference.
Priority	Must have
Frequency of Use	Twice a day per receptionist
Business Rules	BR-44

Normal Flow

18.0 Open and Close a Cashier Shift

1. The receptionist selects Open shift and enters the opening float. (see 18.0.E1)
2. The system opens the shift.
3. During the shift, every payment taken at the desk is tied to it.

4. At the end, the receptionist selects Close shift and enters the cash counted. (see 18.0.E2)
5. The system closes the shift and shows expected cash, declared cash and the difference.

Exceptions

18.0.E1 The receptionist already has an open shift

1. The system refuses the request with CONFLICT_STATE (5.2 Error Responses).
   The use case ends.

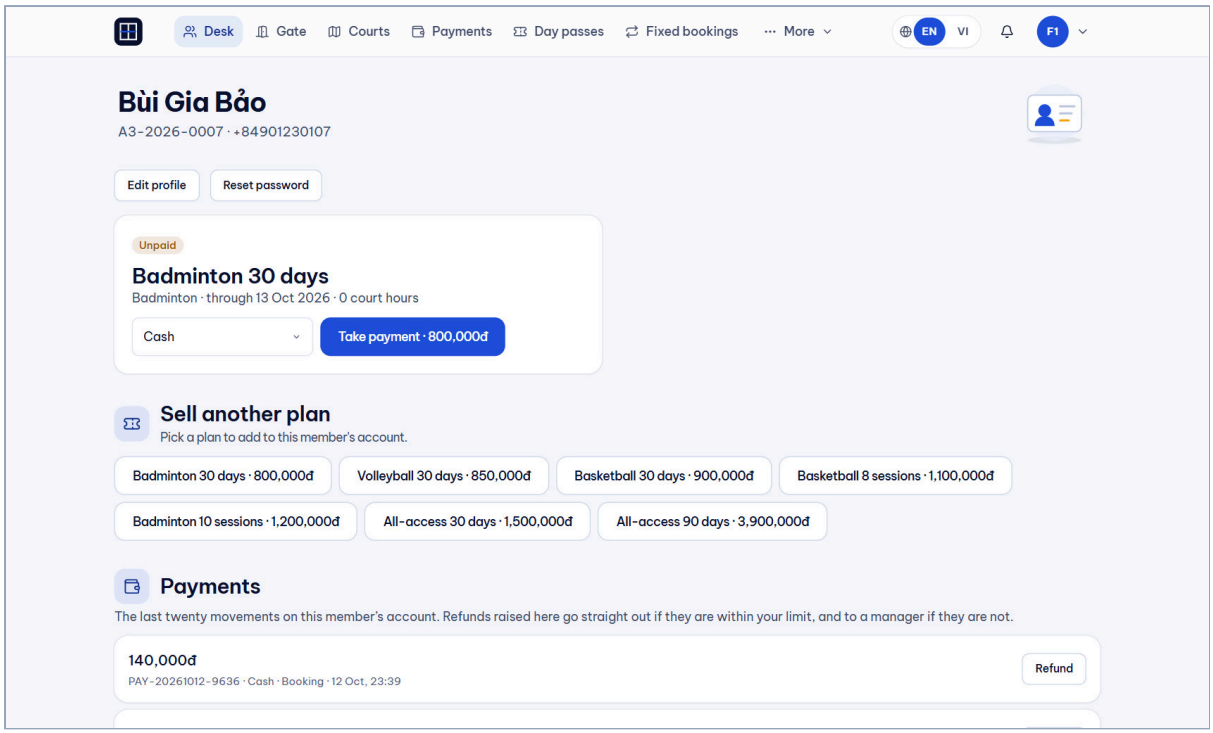
18.0.E2 The counted cash is missing

1. The system refuses the request with VALIDATION (5.2 Error Responses).
   The use case ends.

3.5.2 Manage a Member Record at the Desk

- **Function Trigger:** The receptionist searches for a member on the desk home (/desk) or opens Member profile (/desk/member/$id).
- **Function Description:**
  + **Use Case ID:** UC-19
  + **Actors:** Receptionist; Secondary: Coach (read only, own students)
  + **Purpose:** The receptionist registers a walk-in customer, views and edits a member record, and follows up members at risk of leaving.
  + **Interface:** Member profile (/desk/member/$id); Front desk (/desk); At risk (/desk/at-risk)
- **Data Processing:**
  + **Data Collect:** Search text; for a new member, details, terms consent and guardian details; a follow-up contact.
  + **Data Computation:** Checks that the phone and email are unused (BR-01), the guardian rule (BR-03) and consent (BR-04); lists members at risk by idle days or three absences in a row (BR-36); shows health notes only to those who need them (BR-46).
  + **Data Storage:** Writes the member to users, a follow-up to contact_log and an audit row to audit_logs.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-19 Manage a Member Record at the Desk
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Receptionist
Secondary Actors	Coach (read only, own students)
Trigger	The receptionist searches for a member on the desk home (/desk) or opens Member profile (/desk/member/$id).
Description	The receptionist registers a walk-in customer, views and edits a member record, and follows up members at risk of leaving.
Preconditions	PRE-1. The receptionist is signed in.
Postconditions	POST-1. A member record is created or updated, or a follow-up contact is logged. POST-2. An audit row is written.
Priority	Must have
Frequency of Use	Many times a day
Business Rules	BR-01, BR-03, BR-04, BR-36, BR-46

Normal Flow

19.0 Manage a Member Record at the Desk

1. The receptionist searches by name, phone or member code. (see 19.0.E1)
2. The system shows the matching members.
3. The receptionist opens a member profile.

4. The system shows details, plans, bookings, payments, visits and attendance.
5. The receptionist edits the details and saves; the system checks and stores them. (see 19.0.E2, 19.0.E3)

Alternative Flows

19.1 The person has no account

1. The receptionist registers them at the counter with name, phone, terms consent and, for a minor, guardian details.

   Return to step 3 of the Normal Flow.

19.2 The receptionist resets the member's password

1. The system issues a temporary password.

   The use case ends.

Exceptions

19.0.E1 The terms are not accepted

1. The system refuses the request with BR-04.

   The use case ends.

19.0.E2 The phone already has an account

1. The system refuses the request with BR-01 (ERR_AUTH_01).

   Return to step 5 of the Normal Flow.

19.0.E3 A minor has no guardian details

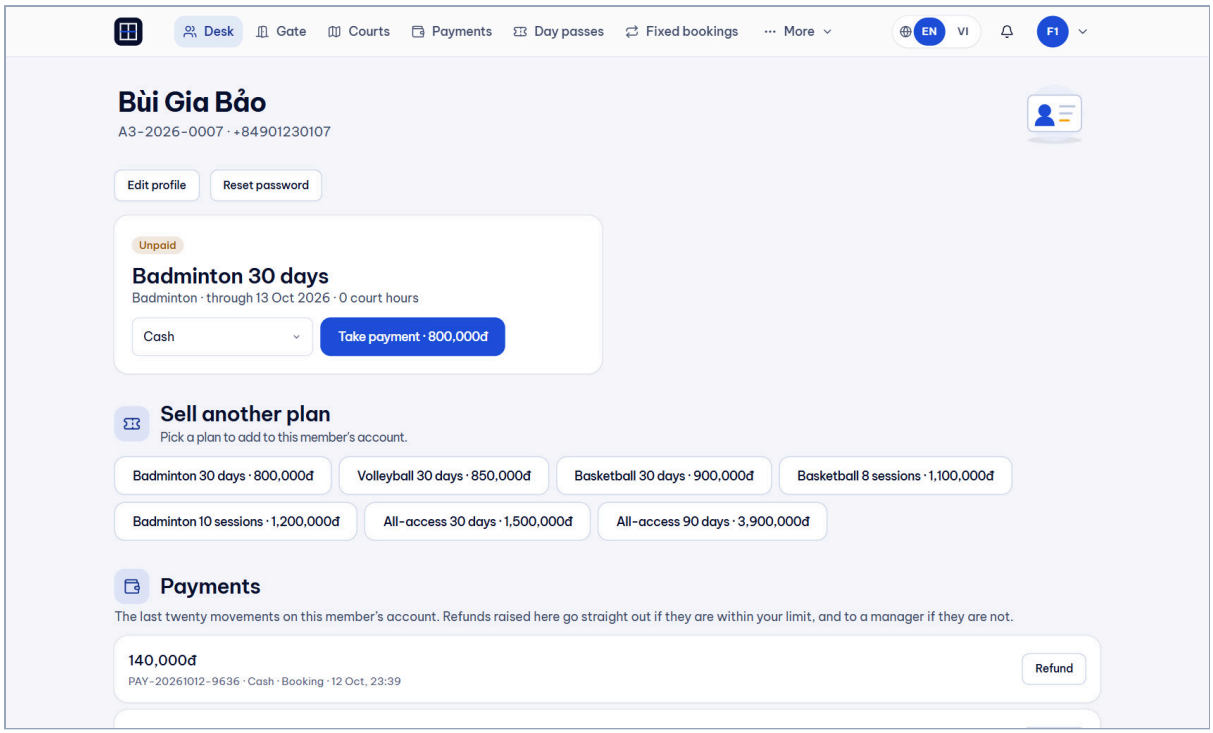
1. The system refuses the request with BR-03.

   Return to step 5 of the Normal Flow.

3.5.3 Manage a Member Plan at the Desk

- **Function Trigger:** The receptionist opens the Plans panel on Member profile.
- **Function Description:**
  + **Use Case ID:** UC-20
  + **Actors:** Receptionist; Secondary: payOS
  + **Purpose:** The receptionist sells, renews, freezes and resumes a member's plan.
  + **Interface:** Member profile (/desk/member/$id)
- **Data Processing:**
  + **Data Collect:** The plan or the member's pending order, a promo code, the payment method, or the freeze days.
  + **Data Computation:** Prices the plan, applies the promo code (BR-40) and rounds once (BR-39); checks the plan is on sale (BR-16), the freeze rules (BR-14, BR-15), the plan dates (BR-12) and the open shift (BR-44).
  + **Data Storage:** Writes subscriptions, payments, invoices and invoice_lines, and an audit row to audit_logs.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-20 Manage a Member Plan at the Desk
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Receptionist
Secondary Actors	payOS
Trigger	The receptionist opens the Plans panel on Member profile.
Description	The receptionist sells, renews, freezes and resumes a member's plan.
Preconditions	PRE-1. The receptionist is signed in with an open shift. PRE-2. The member exists.
Postconditions	POST-1. Sell or renew: a posted payment, an e-invoice and an active plan. POST-2. Freeze or resume: the plan status and end date change. POST-3. An audit row is written.
Priority	Must have
Frequency of Use	Many times a day
Business Rules	BR-12, BR-14, BR-15, BR-16, BR-39, BR-40, BR-44

Normal Flow

20.0 Manage a Member Plan at the Desk

1. The receptionist selects a plan, or a pending order the member placed (UC-10). (see 20.0.E1, 20.0.E2, 20.0.E3)
2. The system prices it, applies any promo code and shows the amount due.

3. The receptionist takes cash, card or transfer and confirms. (see 20.0.E4)
4. The system posts the payment, issues the e-invoice and activates the plan.

Alternative Flows

20.1 Renew

1. The new plan starts the day after the current one ends.
   Return to step 2 of the Normal Flow.

20.2 Resume

1. The system sets a frozen plan active again.
   The use case ends.

20.3 The member pays by VietQR

1. The receptionist raises a payOS link for the order.
   Return to step 4 of the Normal Flow.

Exceptions

20.0.E1 The plan has been withdrawn

1. The system refuses the request with BR-16.
   The use case ends.

20.0.E2 The plan cannot be frozen

1. The system refuses the request with BR-14. Over the yearly allowance: BR-15.
   The use case ends.

20.0.E3 Resume on a plan that is not frozen

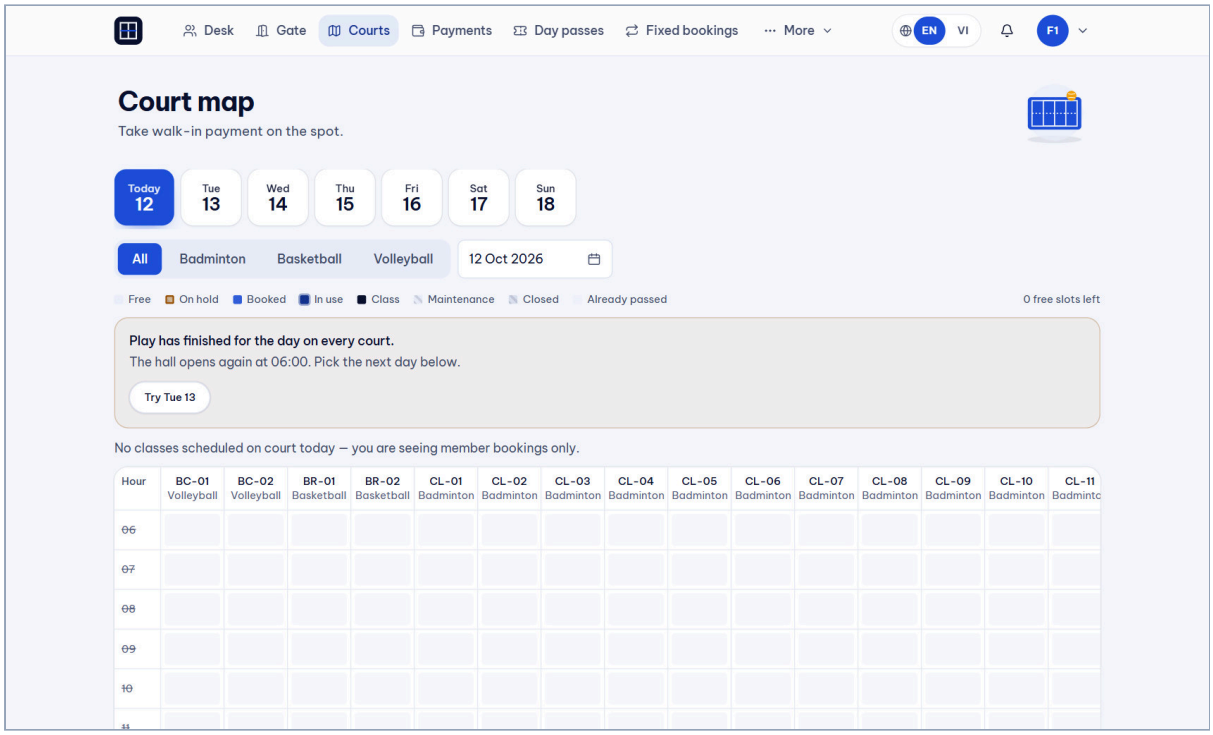
1. The system refuses the request with CONFLICT_STATE (5.2 Error Responses).
   The use case ends.

20.0.E4 No open shift

1. The system refuses the request with BR-44 (ERR_PAY_02).
   The use case ends.

3.5.4 Book a Court at the Desk

- **Function Trigger:** The receptionist selects a slot on Courts (/desk/courts) or opens Fixed series (/desk/series).
- **Function Description:**
  - **Use Case ID:** UC-21
  - **Actors:** Receptionist; Secondary: None
  - **Purpose:** The receptionist books a court for a walk-in customer, or a fixed weekly series of 4 to 12 weeks at 10% off.
  - **Interface:** Court map (/desk/courts); Fixed bookings (/desk/series)
- **Data Processing:**
  - **Data Collect:** The slot, the customer's name and phone; for a series, the court, weekday, time and number of weeks.
  - **Data Computation:** Checks the slot (BR-05 to BR-08), prices it from the price rules (BR-37), with 10% off for a series, and checks the open shift (BR-44).
  - **Data Storage:** Writes court_bookings and occupancies, booking_series for a series, and payments and invoices.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-21 Book a Court at the Desk
Created By	Arena3 team
Date Created	2026-10-12

Field	Detail
Primary Actor	Receptionist
Secondary Actors	None
Trigger	The receptionist selects a slot on Courts (/desk/courts) or opens Fixed series (/desk/series).
Description	The receptionist books a court for a walk-in customer, or a fixed weekly series of 4 to 12 weeks at 10% off.
Preconditions	PRE-1. The receptionist is signed in with an open shift.
Postconditions	POST-1. Walk-in: a confirmed, paid booking. POST-2. Series: one booking per week, all confirmed, with one payment.
Priority	Must have
Frequency of Use	Many times a day
Business Rules	BR-05, BR-06, BR-07, BR-08, BR-37, BR-44

Normal Flow

21.0 Book a Court at the Desk

1. The receptionist selects a free slot on the court map.
2. The receptionist enters the customer's name and phone and selects Walk-in. (see 21.0.E1)
3. The system checks the slot and prices it. (see 21.0.E2, 21.0.E3, 21.0.E4, 21.0.E5)
4. The receptionist takes payment. (see 21.0.E6)
5. The system confirms the booking and posts the payment.

Alternative Flows

21.1 Book a fixed weekly series

1. The receptionist picks the court, weekday, time and 4 to 12 weeks. The system previews every week, marks any week that is taken, and books the free weeks at 10% off after payment.

   The use case ends.

Exceptions

21.0.E1 The name, phone or court is missing

1. The system refuses the request with VALIDATION (5.2 Error Responses).

   Return to step 2 of the Normal Flow.

21.0.E2 The slot has less than 20 minutes left

1. The system refuses the request with BR-07.

   The use case ends.

21.0.E3 The slot is outside opening hours or the court is not ready

1. The system refuses the request with BR-06.

   The use case ends.

21.0.E4 No price rule matches

1. The system refuses the request with BR-37 (ERR_BOOK_04).

   The use case ends.

21.0.E5 Another booking took the slot first

1. The system refuses the request with CONFLICT_SLOT (ERR_BOOK_02).

   The use case ends.

21.0.E6 No open shift

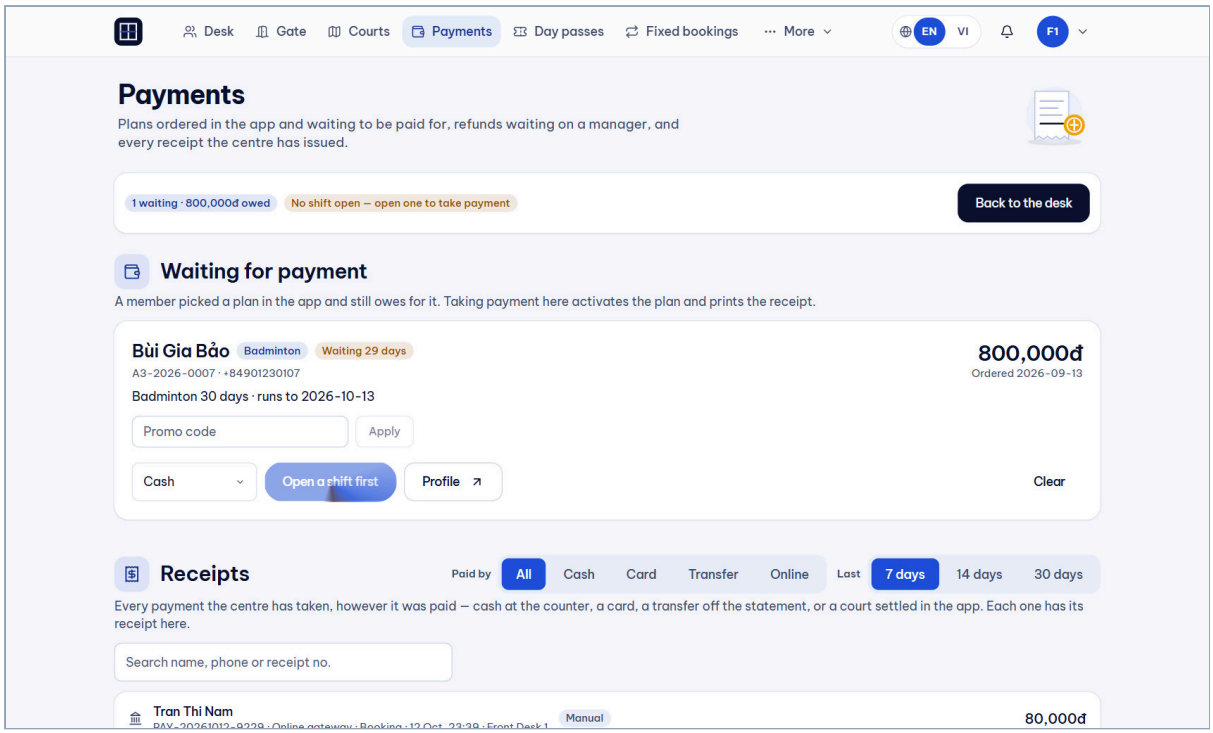
1. The system refuses the request with BR-44 (ERR_PAY_02).

   The use case ends.

3.5.5 Settle Pending Payments

- **Function Trigger:** The receptionist opens Payments (/desk/payments) or a booking on Courts.
- **Function Description:**
  + **Use Case ID:** UC-22
  + **Actors:** Receptionist; Secondary: payOS
  + **Purpose:** The receptionist settles money that is still open: a booking balance, a bank transfer to confirm, or a payment for an order.
  + **Interface:** Payments (/desk/payments)
- **Data Processing:**
  + **Data Collect:** The pending item, the amount and the method; Confirm or Reject for a transfer.
  + **Data Computation:** Checks the booking state and the balance left, rounds once (BR-39) and checks the open shift (BR-44).
  + **Data Storage:** Writes payments and invoices and confirms the court_bookings or subscriptions row, or releases the slot and queues a notice in outbox.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-22 Settle Pending Payments
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Receptionist
Secondary Actors	payOS
Trigger	The receptionist opens Payments (/desk/payments) or a booking on Courts.
Description	The receptionist settles money that is still open: a booking balance, a bank transfer to confirm, or a payment for an order.
Preconditions	PRE-1. The receptionist is signed in with an open shift.
Postconditions	POST-1. The payment is posted with a receipt and e-invoice, and the booking or order it pays for is confirmed.
Priority	Must have
Frequency of Use	Many times a day
Business Rules	BR-39, BR-44

Normal Flow

22.0 Settle Pending Payments

1. The receptionist opens the list of pending items.
2. The receptionist selects a booking with a balance left after its deposit. (see 22.0.E1, 22.0.E2)
3. The receptionist takes the balance in cash, card or transfer. (see 22.0.E3)

4. The system posts the payment and marks the booking as paid.

Alternative Flows

22.1 A bank transfer is waiting

1. The receptionist checks the bank statement and selects Confirm; the system confirms the booking.

   The use case ends.

22.2 The transfer has not arrived

1. The receptionist selects Reject; the system releases the slot and notifies the member.

   The use case ends.

22.3 An order has no money taken yet

1. The receptionist may cancel it.

   The use case ends.

Exceptions

22.0.E1 The booking is not confirmed or has nothing left to pay

1. The system refuses the request with CONFLICT_STATE (5.2 Error Responses).

   The use case ends.

22.0.E2 The transfer is not confirmed within 120 minutes

1. The scheduled job releases the slot.

   The use case ends.

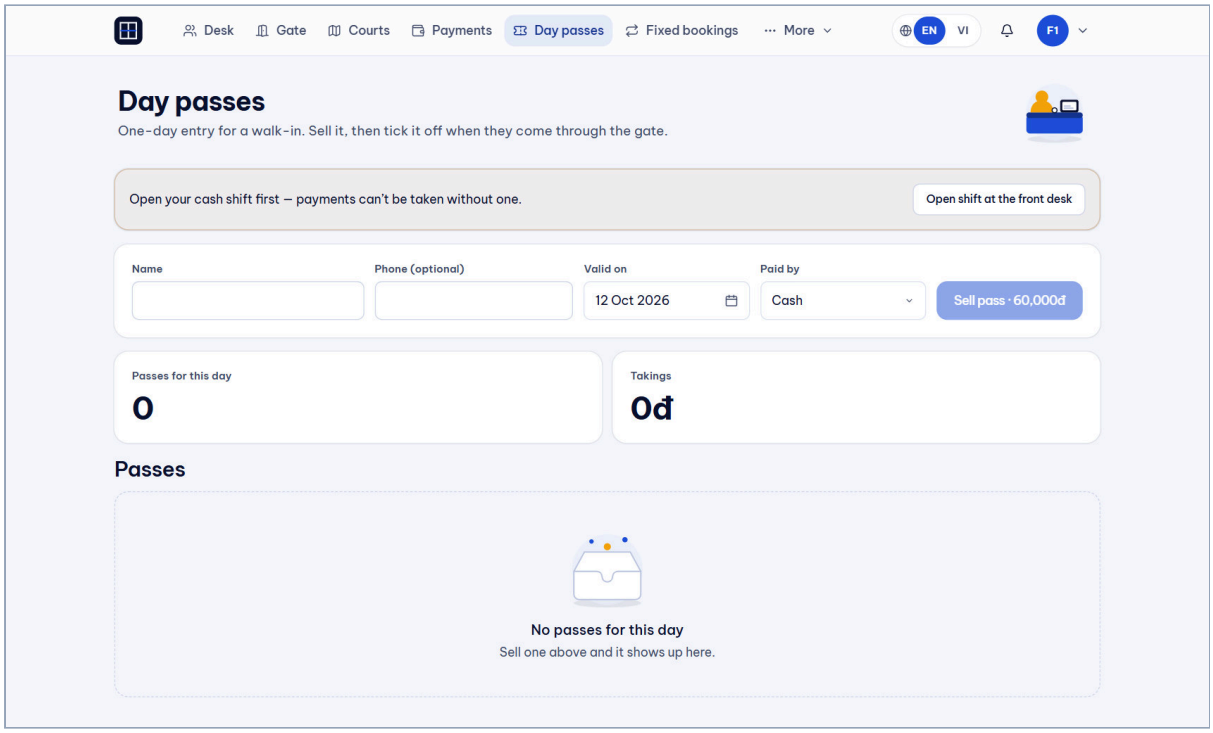
22.0.E3 No open shift

1. The system refuses the request with BR-44 (ERR_PAY_02).

   The use case ends.

3.5.6 Sell and Redeem a Day Pass

- **Function Trigger:** The receptionist opens Day passes (/desk/day-passes).
- **Function Description:**
  - **Use Case ID:** UC-23
  - **Actors:** Receptionist; Secondary: None
  - **Purpose:** The receptionist sells a day pass for entry on a chosen day and marks it used when the customer arrives.
  - **Interface:** Day passes (/desk/day-passes)
- **Data Processing:**
  - **Data Collect:** The customer's name or a member's phone, the day and the payment method; Use or Void on the day.
  - **Data Computation:** Prices the pass at the day-pass setting, checks the day is not past and checks the open shift (BR-44).
  - **Data Storage:** Writes day_passes, payments and invoices; a void writes a negative row to payments.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-23 Sell and Redeem a Day Pass
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Receptionist

Field	Detail
Secondary Actors	None
Trigger	The receptionist opens Day passes (/desk/day-passes).
Description	The receptionist sells a day pass for entry on a chosen day and marks it used when the customer arrives.
Preconditions	PRE-1. The receptionist is signed in with an open shift.
Postconditions	POST-1. Sell: an issued pass with a posted payment and e-invoice. POST-2. Redeem: the pass is used.
Priority	Should have
Frequency of Use	Daily
Business Rules	BR-44

Normal Flow

23.0 Sell and Redeem a Day Pass

1. The receptionist enters the customer's name, or the phone of a member, and the day. (see 23.0.E1)
2. The system prices the pass at the centre day-pass price (60,000đ by default).
3. The receptionist takes payment. (see 23.0.E2)
4. The system issues the pass with its own code, posts the payment and issues the e-invoice.
5. On the day, the receptionist finds the pass and selects Use. (see 23.0.E3)

Alternative Flows

23.1 The pass was sold in error and not used

1. The receptionist voids it and the payment is refunded.

   The use case ends.

Exceptions

23.0.E1 The day is in the past

1. The system refuses the request with VALIDATION (5.2 Error Responses).

   The use case ends.

23.0.E2 No open shift

1. The system refuses the request with BR-44 (ERR_PAY_02).

   The use case ends.

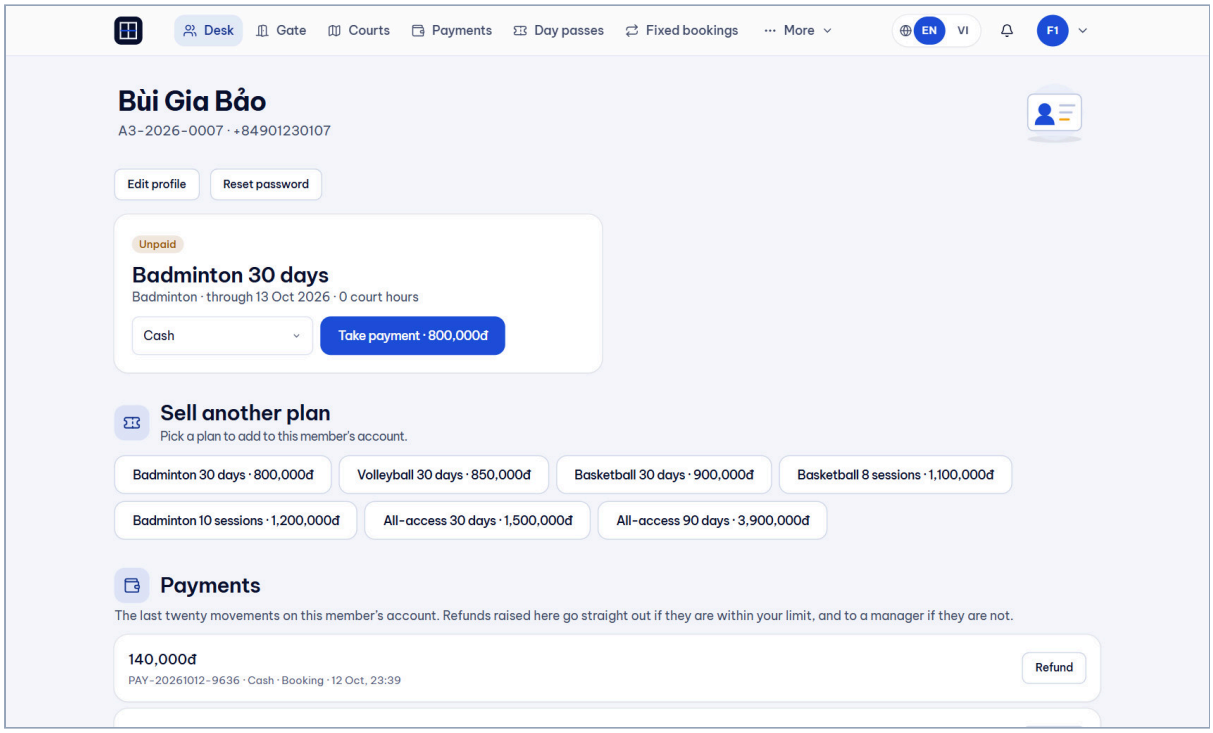
23.0.E3 The pass is used on another day, or was already used or voided

1. The system refuses the request with CONFLICT_STATE (5.2 Error Responses).

   The use case ends.

3.5.7 Raise a Refund

- **Function Trigger:** The receptionist chooses Raise a refund on a payment in Member profile.
- **Function Description:**
  - **Use Case ID:** UC-24
  - **Actors:** Receptionist; Secondary: Manager, who can raise a refund directly (flow 4b)
  - **Purpose:** The receptionist gives back all or part of a posted payment, and a large refund waits for a manager's approval.
  - **Interface:** Member profile (/desk/member/$id)
- **Data Processing:**
  - **Data Collect:** The payment, the refund amount and the reason.
  - **Data Computation:** Works out what is still refundable, compares the amount with the refund limit (1,000,000đ) and checks the open shift for a cash refund (BR-44).
  - **Data Storage:** Writes a negative row to payments, posted or pending approval; restores a promo code in promotion_redemptions after a full refund; writes audit_logs (BR-53).
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-24 Raise a Refund
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Receptionist

Field	Detail
Secondary Actors	Manager, who can raise a refund directly (flow 4b)
Trigger	The receptionist chooses Raise a refund on a payment in Member profile.
Description	The receptionist gives back all or part of a posted payment, and a large refund waits for a manager's approval.
Preconditions	PRE-1. The receptionist is signed in. PRE-2. The payment is posted and has money left to refund.
Postconditions	POST-1. A negative payment row exists with its own receipt code, either posted or pending approval. POST-2. An audit row is written.
Priority	Must have
Frequency of Use	A few times a week
Business Rules	BR-44, BR-53

Normal Flow

24.0 Raise a Refund

1. The receptionist opens the member profile, selects a posted payment and chooses Raise a refund.
2. The receptionist enters the amount and a reason.
3. The system works out what is still refundable: the payment, less refunds already posted or pending. (see 24.0.E1)
4. The amount is under the refund limit (1,000,000đ): the system posts the refund with its own receipt code. (see 24.0.E2)
5. The system writes an audit row and shows the refund receipt.
6. If the order is now refunded in full, a used promo code is given back to the member.

Alternative Flows

24.1 The amount is at or above the refund limit

1. The refund is saved as pending approval and goes to a manager (UC-42).

   The use case ends.

24.2 A manager raises the refund

1. It is posted directly, whatever the amount.

   Return to step 5 of the Normal Flow.

Exceptions

24.0.E1 The payment is already refunded in full, or the amount is larger than what remains

1. The system refuses the request with VALIDATION (ERR_PAY_03).

   The use case ends.

24.0.E2 A cash refund without an open shift

1. The system refuses the request with BR-44 (ERR_PAY_02).

The use case ends.

A large refund needs a second person, so no receptionist can refund their own sale above the limit. Figure 25 shows this use case as an activity diagram.

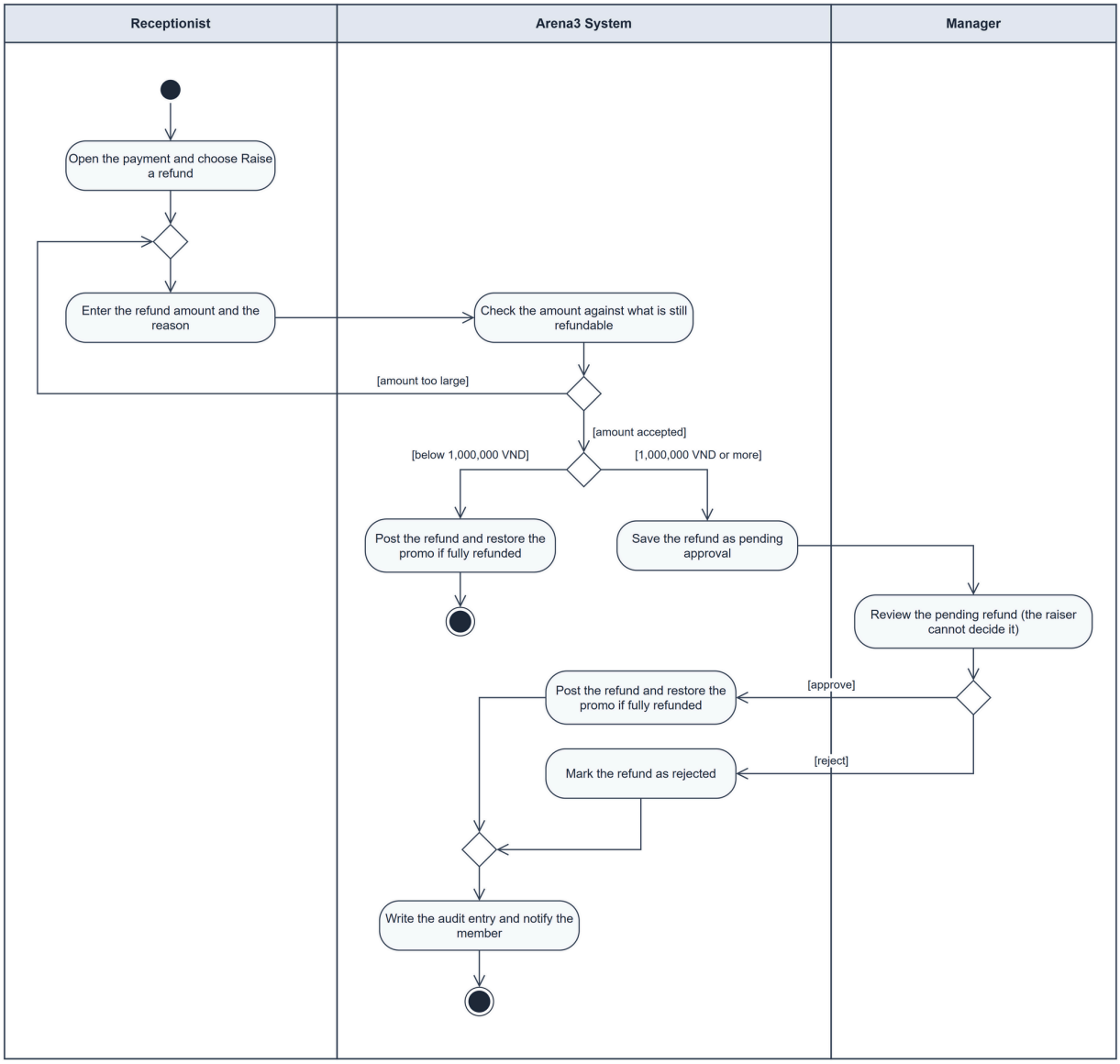


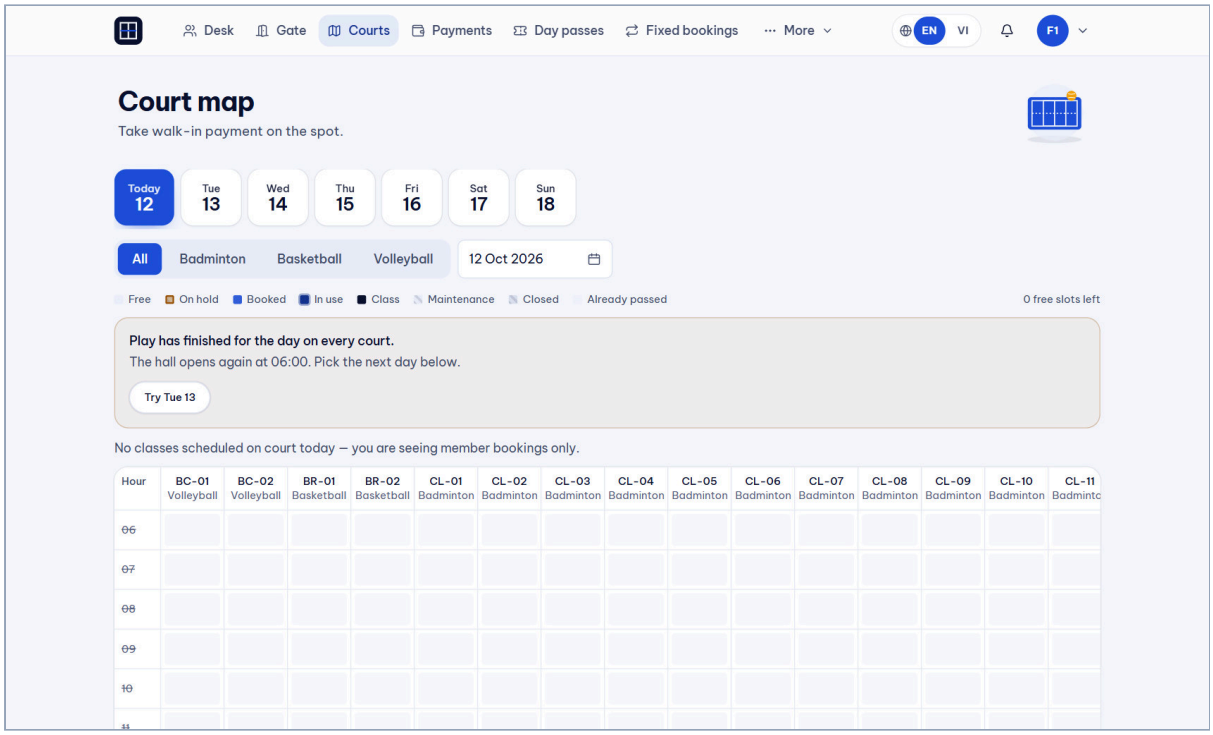
Figure 25. *Activity diagram: Raise and Approve a Refund*

Refunds up to 1,000,000đ post at once. Larger refunds wait for a manager, who approves or rejects them. Both outcomes are audited and the member is notified.

3.6 Front Desk Operations

3.6.1 Check In a Court Booking

- **Function Trigger:** The customer arrives for a booking and the receptionist opens Courts (/desk/courts).
- **Function Description:**
  - **Use Case ID:** UC-25
  - **Actors:** Receptionist; Secondary: Member, who shows the booking QR code (flow 3b)
  - **Purpose:** The receptionist checks a court booking in on the live court map, inside the check-in window.
  - **Interface:** Court map (/desk/courts)
- **Data Processing:**
  - **Data Collect:** The booking selected, or its QR code.
  - **Data Computation:** Checks the booking is confirmed and the time is inside the check-in window (BR-11).
  - **Data Storage:** Sets the booking's status in court_bookings to in use.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-25 Check In a Court Booking
Created By	Arena3 team
Date Created	2026-10-12

Field	Detail
Primary Actor	Receptionist
Secondary Actors	Member, who shows the booking QR code (flow 3b)
Trigger	The customer arrives for a booking and the receptionist opens Courts (/desk/courts).
Description	The receptionist checks a court booking in on the live court map, inside the check-in window.
Preconditions	PRE-1. The receptionist is signed in. PRE-2. The booking is confirmed.
Postconditions	POST-1. The booking is checked in; a booking missed past the window becomes a no-show.
Priority	Must have
Frequency of Use	Many times a day
Business Rules	BR-11

Normal Flow

25.0 Check In a Court Booking

1. The receptionist opens the live court map for today.
2. The system shows each court with its current and next bookings and their payment status.
3. The receptionist selects the booking and chooses Check in. (see 25.0.E1)
4. The system checks the window and records the check-in time. (see 25.0.E2)

Alternative Flows

25.1 The booking still has a balance

1. The receptionist collects it first (UC-22).
   Return to step 3 of the Normal Flow.

25.2 The member shows a booking QR code

1. The receptionist scans it to open the booking.
   Return to step 3 of the Normal Flow.

Exceptions

25.0.E1 The booking is not confirmed

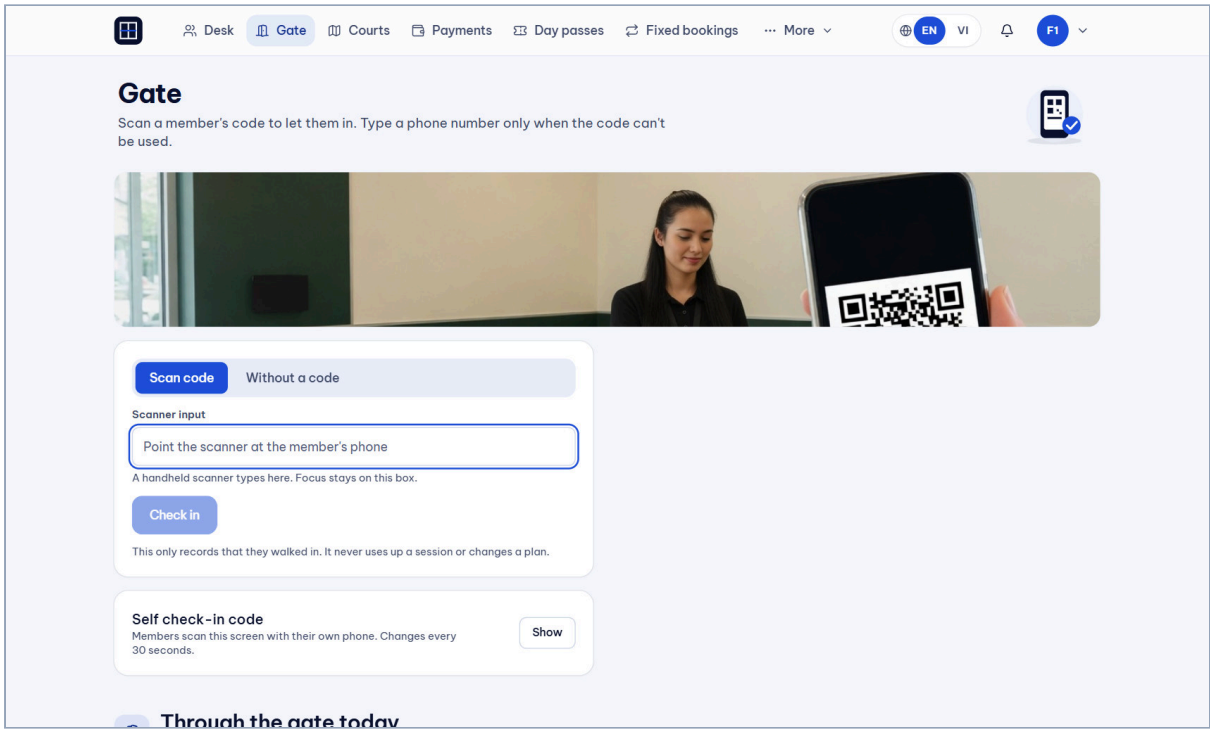
1. The system refuses the request with CONFLICT_STATE (5.2 Error Responses).
   The use case ends.

25.0.E2 The time is outside the window of 15 minutes before to 10 minutes after the start

1. The system refuses the request with BR-11.
   The use case ends.

3.6.2 Check a Member In at the Gate

- **Function Trigger:** The member shows the gate pass QR at the front desk.
- **Function Description:**
  + **Use Case ID:** UC-26
  + **Actors:** Receptionist; Secondary: Member, who shows the gate pass
  + **Purpose:** The receptionist admits a member and the system records the visit, flagging any override.
  + **Interface:** Gate (/desk/gate)
- **Data Processing:**
  + **Data Collect:** The scanned member code or the member looked up, and an override reason when needed.
  + **Data Computation:** Checks the code (BR-30), today's entitlement and the repeat-scan window (BR-31); spends no session (BR-32) and fills no register (BR-33).
  + **Data Storage:** Marks the token used in checkin_tokens, writes the visit to attendance and audits an override in audit_logs.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-26 Check a Member In at the Gate
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Receptionist

Field	Detail
Secondary Actors	Member, who shows the gate pass
Trigger	The member shows the gate pass QR at the front desk.
Description	The receptionist admits a member and the system records the visit, flagging any override.
Preconditions	PRE-1. The receptionist is signed in on Gate (/desk/gate). PRE-2. The member opens Gate pass (/app/pass).
Postconditions	POST-1. One visit row exists for the member within the 120-minute window. POST-2. Any override is flagged and audited.
Priority	Must have
Frequency of Use	Hundreds of times a day
Business Rules	BR-30, BR-31, BR-32, BR-33

Normal Flow

26.0 Check a Member In at the Gate

1. The member opens Gate pass, which shows a short-lived QR code.
2. The receptionist scans the code on the Gate screen.
3. The system checks the code is a valid member code that has not been used. (see 26.0.E1, 26.0.E2)
4. The system checks today's entitlement: a usable plan, a court booking or a class session today.
5. The system records the check-in time, returns any warnings and shows a green pass.

Alternative Flows

26.1 The code cannot be scanned

1. The receptionist looks the member up by name, phone or member code and picks why: no phone, dead battery, QR failed, forgot, or other.

   Return to step 4 of the Normal Flow.

26.2 The member is not entitled today

1. The system lists the reasons and asks for an override.
2. The receptionist picks a reason: expired but allowed, manager approval, guest pass, renewing, or other. The visit is flagged and audited, and the screen shows an amber pass (BR-31).

   The use case ends.

26.3 The member already checked in within 120 minutes

1. The system returns that visit instead of writing a second (BR-31).

   The use case ends.

Exceptions

26.0.E1 The code is invalid or already used

1. The system refuses the request with BR-30 (ERR_CHECKIN_01), and the member refreshes the code.

   Return to step 2 of the Normal Flow.

26.0.E2 The desk's own self check-in code is scanned at the desk

1. The system refuses the request with BR-30 (ERR_CHECKIN_01).

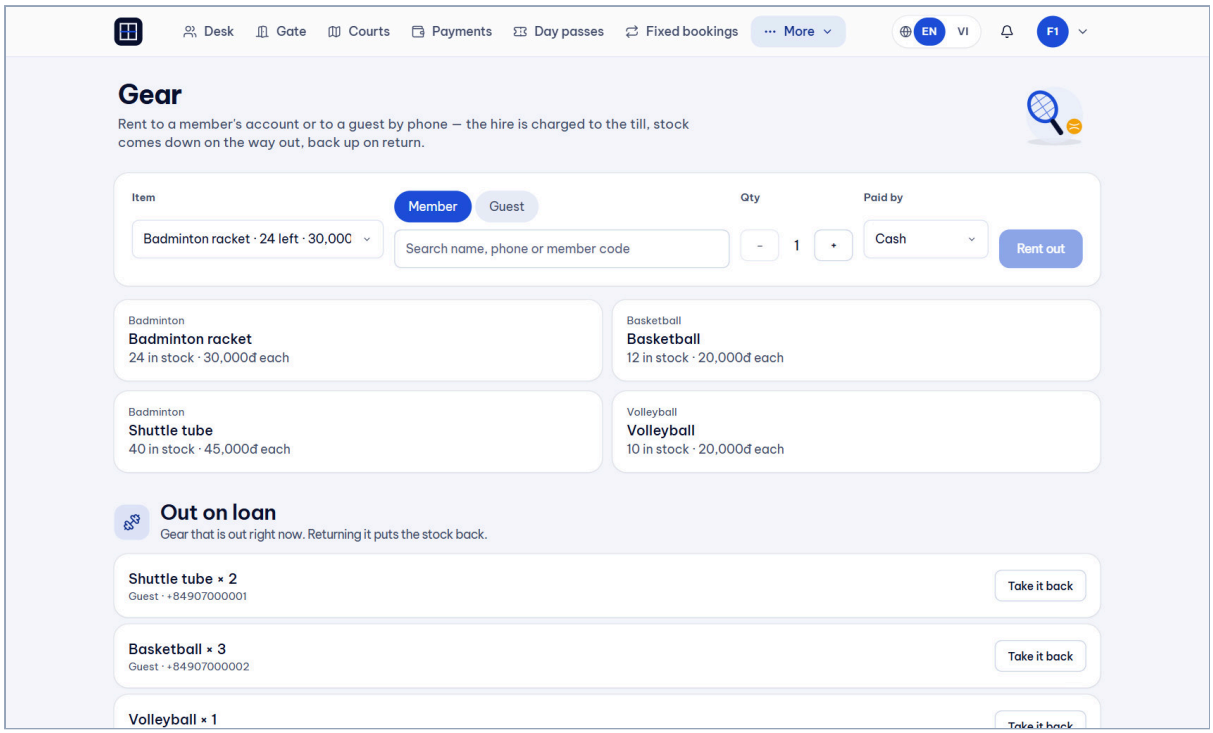
   The use case ends.

Other information: Warnings cover a frozen plan, an expired plan, a plan ending within seven days and a plan with no sessions left. Walking through the gate never spends a session (BR-32) and never fills a class register (BR-33).

3.6.3 Rent Out Gear and Record Its Return

- **Function Trigger:** The customer asks for gear at the desk, and the receptionist opens Gear (/desk/gear).
- **Function Description:**
  - **Use Case ID:** UC-27
  - **Actors:** Receptionist; Secondary: None
  - **Purpose:** The receptionist rents out gear, charges the hire to the till with an e-invoice, and records the return.
  - **Interface:** Gear (/desk/gear)
- **Data Processing:**
  - **Data Collect:** The gear item, quantity, member or guest phone and payment method; Return later.
  - **Data Computation:** Checks the stock (BR-50) and the open shift (BR-44); charges rent × quantity with an e-invoice (BR-51).
  - **Data Storage:** Lowers the stock in equipment_items and writes equipment_loans, payments and invoices; a return closes the loan and restores the stock.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-27 Rent Out Gear and Record Its Return
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Receptionist
Secondary Actors	None
Trigger	The customer asks for gear at the desk, and the receptionist opens Gear (/desk/gear).
Description	The receptionist rents out gear, charges the hire to the till with an e-invoice, and records the return.
Preconditions	PRE-1. The receptionist is signed in with an open shift. PRE-2. The gear item exists.
Postconditions	POST-1. Rent: stock is lowered, a loan due in three hours is open, the hire is posted to the shift and an e-invoice is issued. POST-2. Return: the loan is closed and stock is restored.
Priority	Should have
Frequency of Use	Daily
Business Rules	BR-44, BR-50, BR-51

Normal Flow

27.0 Rent Out Gear and Record Its Return

1. The receptionist selects the gear item and the quantity (1 to 99).

2. The receptionist picks the member, or enters a guest's phone, and the payment method. (see 27.0.E1)
3. The system checks the stock and the open shift. (see 27.0.E2, 27.0.E3)
4. The system lowers the stock and opens the loan, due back in three hours.
5. The system posts the hire (rent × quantity) to the shift, issues the e-invoice and sends the receipt.
6. When the gear comes back, the receptionist selects Return; the system closes the loan and restores the stock. (see 27.0.E4)

Alternative Flows

27.1 The item's rent is 0đ

1. The loan opens without a payment or invoice.
   Return to step 6 of the Normal Flow.

Exceptions

27.0.E1 The member account is not active, or the guest phone is invalid

1. The system refuses the request with VALIDATION (5.2 Error Responses).
   The use case ends.

27.0.E2 The quantity is more than the stock

1. The system refuses the request with BR-50 (ERR_GEAR_01).
   The use case ends.

27.0.E3 No open shift

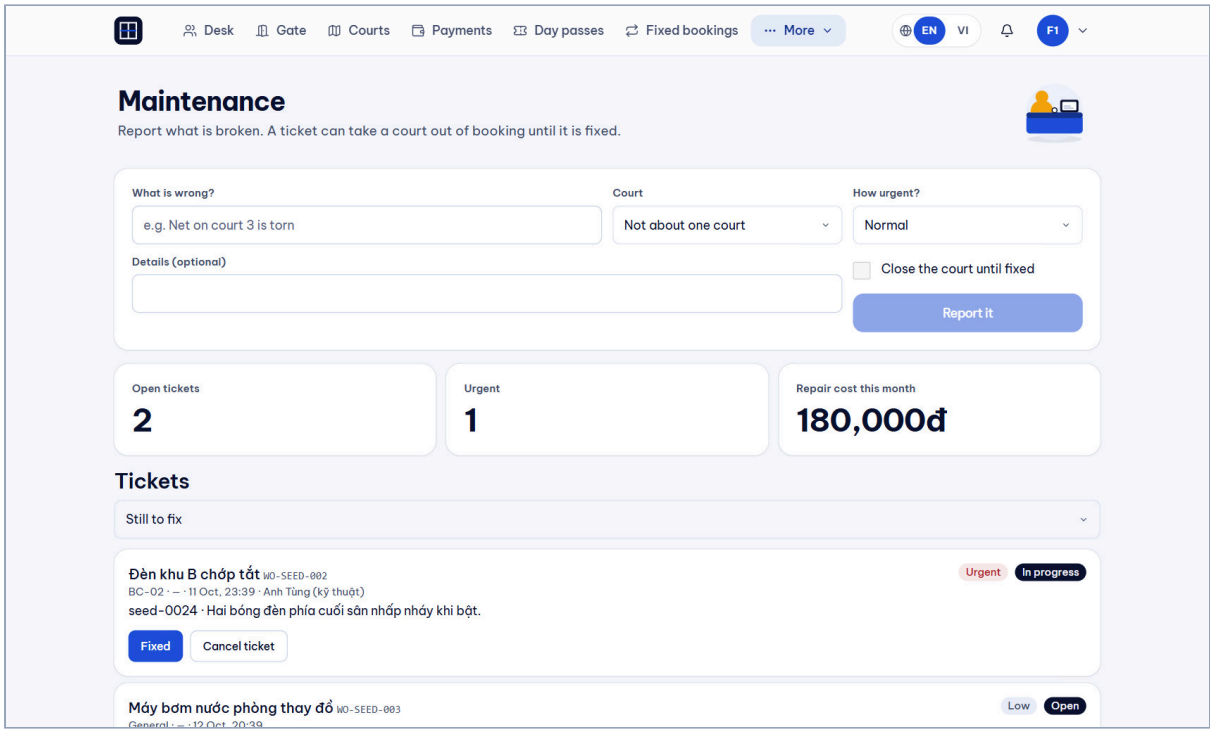
1. The system refuses the request with BR-44 (ERR_PAY_02).
   The use case ends.

27.0.E4 The loan is already returned

1. The system refuses the request with CONFLICT_STATE (5.2 Error Responses).
   The use case ends.

3.6.4 Handle Maintenance Work Orders

- **Function Trigger:** A receptionist or a manager notices a fault and opens Maintenance (/desk/maintenance).
- **Function Description:**
  + **Use Case ID:** UC-28
  + **Actors:** Receptionist, Manager; Secondary: None
  + **Purpose:** A receptionist reports a fault as a work order, and a manager tracks it to done with its cost.
  + **Interface:** Maintenance (/desk/maintenance)
- **Data Processing:**
  + **Data Collect:** Title, court or area and priority; later the status and the cost.
  + **Data Computation:** Checks the title, the priority and a cost that is not negative; a blocking order takes the court out of use (BR-08).
  + **Data Storage:** Writes work_orders, sets the court's status in courts to maintenance when needed, and writes audit_logs (BR-53).
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-28 Handle Maintenance Work Orders
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Receptionist, Manager

Field	Detail
Secondary Actors	None
Trigger	A receptionist or a manager notices a fault and opens Maintenance (/desk/maintenance).
Description	A receptionist reports a fault as a work order, and a manager tracks it to done with its cost.
Preconditions	PRE-1. The receptionist or the manager is signed in.
Postconditions	POST-1. A work order exists with its priority, status and, when done, its cost. POST-2. An audit row is written.
Priority	Should have
Frequency of Use	Weekly
Business Rules	BR-08, BR-53

Normal Flow

28.0 Handle Maintenance Work Orders

1. The receptionist enters a title, the court or area, and a priority: low, normal or urgent. (see 28.0.E1)
2. The system opens the work order.
3. The manager opens the list, filtered by status.
4. The manager moves the order to in progress, then to done, and records the cost. (see 28.0.E2)
5. The system stores each change.

Alternative Flows

28.1 The fault takes a court out of use

1. The manager sets the court to maintenance, so no booking can use it (BR-08).

   Return to step 2 of the Normal Flow.

Exceptions

28.0.E1 The title is empty or the priority is unknown

1. The system refuses the request with VALIDATION (5.2 Error Responses).

   The use case ends.

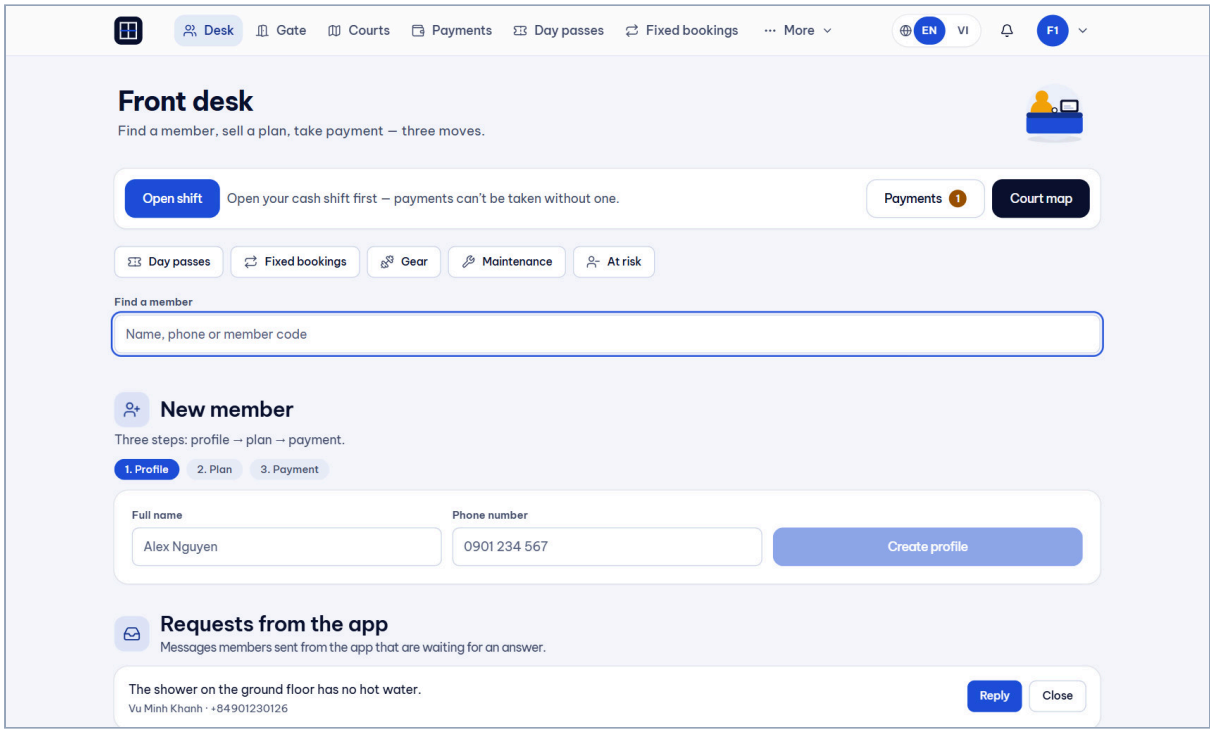
28.0.E2 The cost is negative

1. The system refuses the request with VALIDATION (5.2 Error Responses).

   The use case ends.

3.6.5 Reply to a Support Request

- **Function Trigger:** A support request appears in the desk queue on the desk home (/desk).
- **Function Description:**
  - **Use Case ID:** UC-29
  - **Actors:** Receptionist; Secondary: None
  - **Purpose:** The receptionist answers a member's support request, which closes it and notifies the member.
  - **Interface:** Front desk (/desk)
- **Data Processing:**
  - **Data Collect:** The reply text, or Close without a reply.
  - **Data Computation:** Lists open requests oldest first and checks the reply is not empty.
  - **Data Storage:** Writes the reply and the status to tickets, queues a notice in outbox and writes audit_logs (BR-53).
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-29 Reply to a Support Request
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Receptionist
Secondary Actors	None
Trigger	A support request appears in the desk queue on the desk home (/desk).

Field	Detail
Description	The receptionist answers a member's support request, which closes it and notifies the member.
Preconditions	PRE-1. The receptionist is signed in. PRE-2. An open request exists.
Postconditions	POST-1. The request is closed with the reply, the member is notified, and an audit row is written.
Priority	Should have
Frequency of Use	Daily
Business Rules	BR-53

Normal Flow

29.0 Reply to a Support Request

1. The receptionist opens the queue of open requests, oldest first.
2. The receptionist reads a request and writes a reply.
3. The system saves the reply, closes the request and notifies the member. (see 29.0.E1)

Alternative Flows

29.1 Show all requests

1. The receptionist shows all requests to read back an earlier answer.
   Return to step 2 of the Normal Flow.

29.2 The request needs no reply

1. The receptionist closes it.
   The use case ends.

Exceptions

29.0.E1 The reply is empty

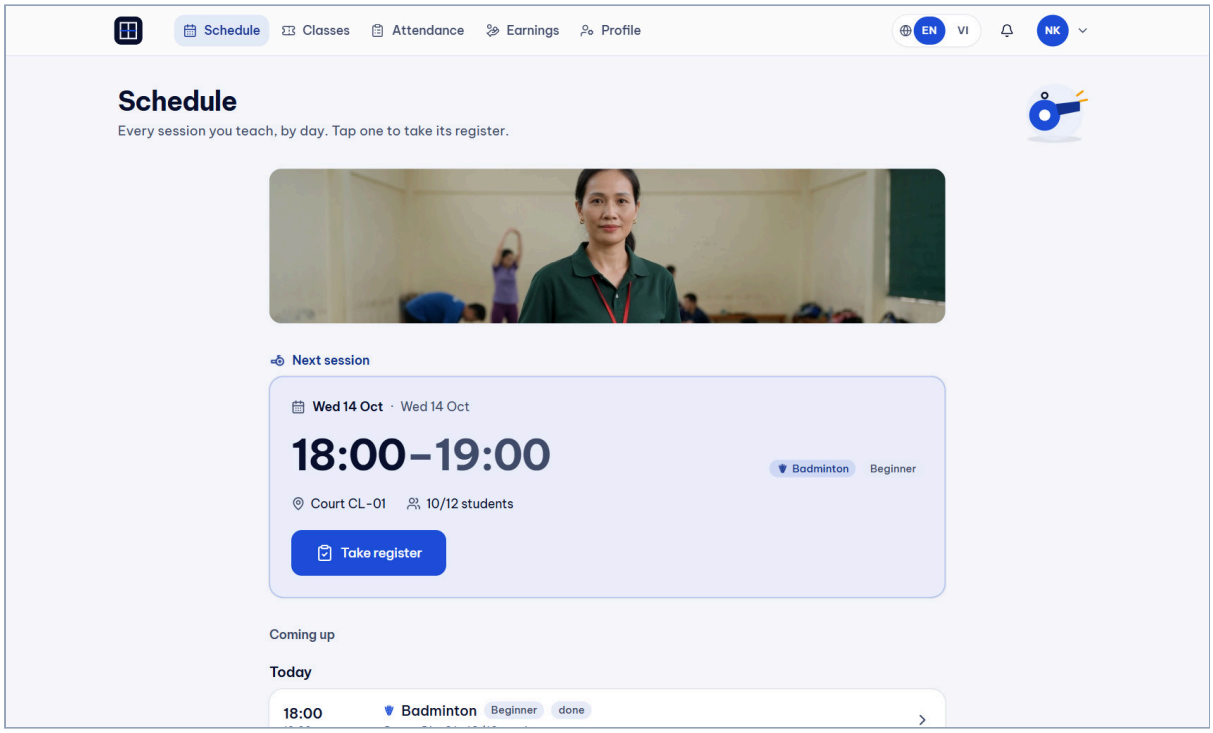
1. The system refuses the request with VALIDATION (5.2 Error Responses).
   The use case ends.

Other information: A member who writes again opens a new request, so an answered request never stays in the open queue.

3.7 Coach Teaching and Earnings

3.7.1 View the Teaching Schedule and Rosters

- **Function Trigger:** The coach opens the coach home (/coach) or My classes (/coach/classes).
- **Function Description:**
  - **Use Case ID:** UC-30
  - **Actors:** Coach; Secondary: None
  - **Purpose:** The coach sees the week's sessions and the roster of each class taught.
  - **Interface:** Schedule (/coach); Classes (coach) (/coach/classes)
- **Data Processing:**
  - **Data Collect:** The week and the class opened.
  - **Data Computation:** Limits the classes to those the coach leads or assists (BR-45) and health notes to those who need them (BR-46).
  - **Data Binding:** Reads sessions, classes, enrollments, member_levels and attendance, and shows the schedule and the rosters.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-30 View the Teaching Schedule and Rosters
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Coach

Field	Detail
Secondary Actors	None
Trigger	The coach opens the coach home (/coach) or My classes (/coach/classes).
Description	The coach sees the week's sessions and the roster of each class taught.
Preconditions	PRE-1. The coach is signed in.
Postconditions	POST-1. No data changes.
Priority	Must have
Frequency of Use	Daily
Business Rules	BR-45, BR-46

Normal Flow

30.0 View the Teaching Schedule and Rosters

1. The coach opens the coach home.
2. The system lists the coach's sessions for the week, with court and time.
3. The coach opens a class. (see 30.0.E1)
4. The system shows the roster with each student's level and attendance.
5. The coach opens a student to see the profile (UC-34).

Exceptions

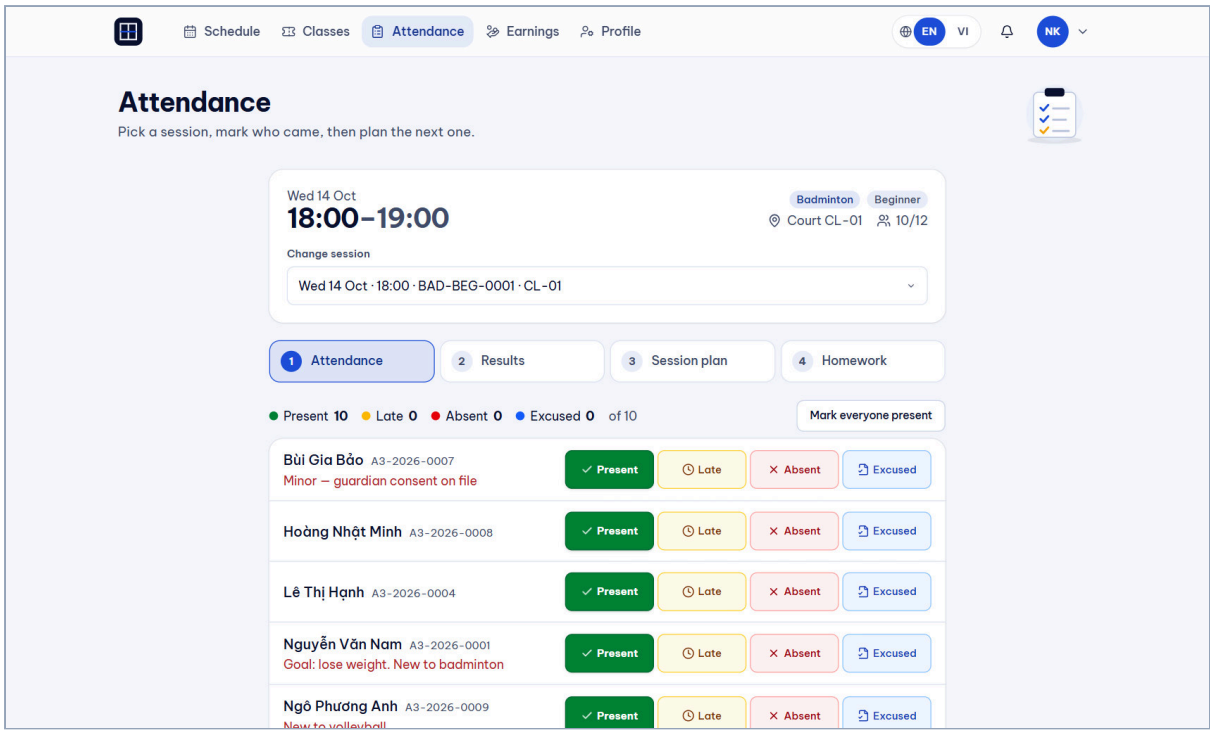
30.0.E1 The coach opens a class the coach neither leads nor assists

1. The system refuses the request with FORBIDDEN (ERR_AUTH_05, BR-45).
   The use case ends.

3.7.2 Take Session Attendance

- **Function Trigger:** The session starts and the coach opens Attendance (/coach/attendance).
- **Function Description:**
  - **Use Case ID:** UC-31
  - **Actors:** Coach; Secondary: Manager, who corrects closed registers
  - **Purpose:** The coach marks each student present, late, absent or excused for a session.
  - **Interface:** Attendance (/coach/attendance)
- **Data Processing:**
  - **Data Collect:** The session and each student's mark; a reason for a manager's correction.
  - **Data Computation:** Checks the coach leads or assists the class (BR-45), the lock 2 hours after the session (BR-35), the marks a coach may set (BR-33, BR-34) and absence streaks (BR-36).
  - **Data Storage:** Writes attendance, queues streak alerts in outbox and audits a correction in audit_logs.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-31 Take Session Attendance
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Coach
Secondary Actors	Manager, who corrects closed registers
Trigger	The session starts and the coach opens Attendance (/coach/attendance).
Description	The coach marks each student present, late, absent or excused for a session.
Preconditions	PRE-1. The coach leads or assists the class. PRE-2. The register is not locked.
Postconditions	POST-1. The register is saved. POST-2. Three absences in a row raise an alert.
Priority	Must have
Frequency of Use	Every session
Business Rules	BR-33, BR-34, BR-35, BR-36, BR-45

Normal Flow

31.0 Take Session Attendance

1. The coach selects the session. (see 31.0.E1, 31.0.E2)
2. The system lists the enrolled students.
3. The coach marks each one present, late, absent or excused. (see 31.0.E3)

4. The system saves the register and checks for absence streaks.

Alternative Flows

31.1 A manager corrects a locked register

1. The manager enters a reason, and the change is audited (BR-35, ERR_COACH_01).
   Return to step 3 of the Normal Flow.

31.2 A student reaches three absences in a row

1. The system alerts the coach and the managers (BR-36).
   The use case ends.

Exceptions

31.0.E1 More than 2 hours have passed since the session

1. The system refuses the request with BR-35 (ERR_COACH_01).
   The use case ends.

31.0.E2 The session belongs to another coach's class

1. The system refuses the request with FORBIDDEN (ERR_AUTH_05, BR-45).
   The use case ends.

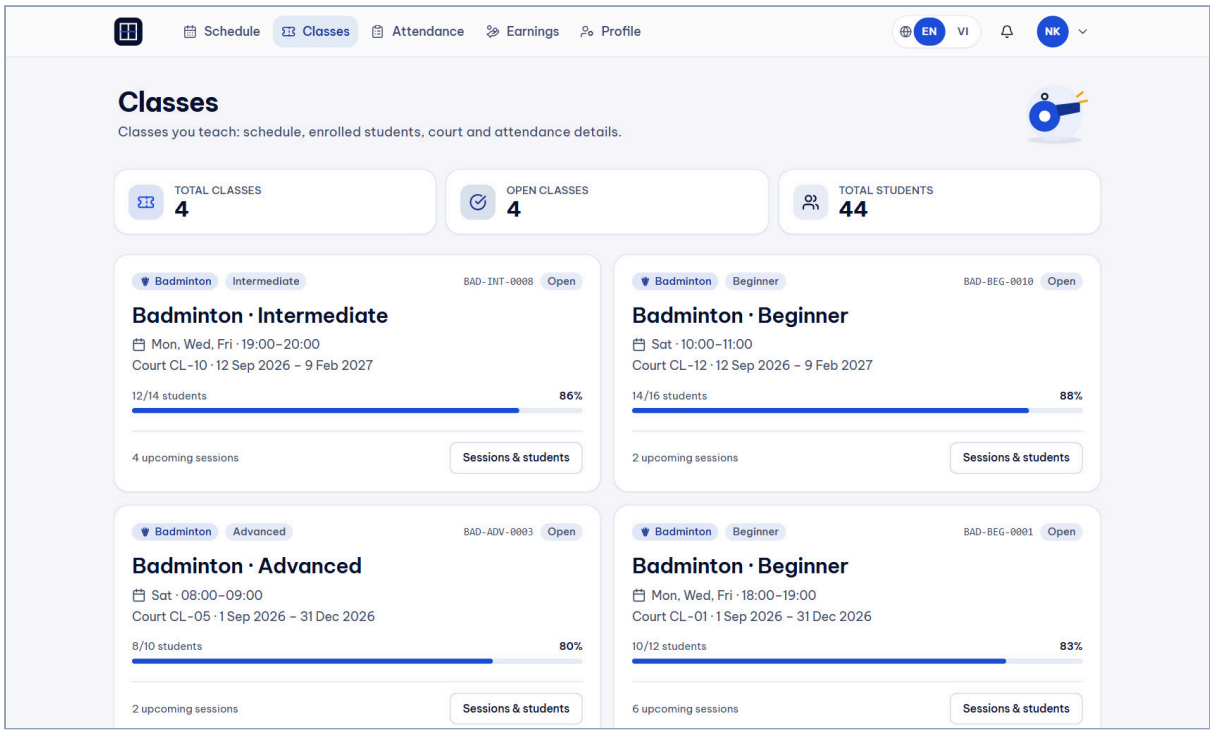
31.0.E3 A result other than present, late, absent or excused

1. The system refuses the request with VALIDATION (5.2 Error Responses).
   The use case ends.

3.7.3 Plan a Training Session

- **Function Trigger:** The coach opens a session in Training plans (/coach/classes).
- **Function Description:**
  + **Use Case ID:** UC-32
  + **Actors:** Coach; Secondary: None
  + **Purpose:** The coach builds the plan of a session from blocks, a saved template or an AI suggestion.
  + **Interface:** Classes (coach) (/coach/classes)
- **Data Processing:**
  + **Data Collect:** Plan blocks with title, type, intensity, minutes and description; a template or a suggestion.
  + **Data Computation:** Checks the module is on (BR-52), the coach's class (BR-45), 1 to 12 valid blocks (BR-49) and that the session has no results (BR-48); a suggestion comes from a built-in drill list for the sport and level. Nothing here takes money (BR-47).
  + **Data Storage:** Writes training_plans and a new row in training_plan_versions.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-32 Plan a Training Session
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Coach
Secondary Actors	None
Trigger	The coach opens a session in Training plans (/coach/classes).
Description	The coach builds the plan of a session from blocks, a saved template or an AI suggestion.
Preconditions	PRE-1. The coach teaches the class. PRE-2. The training module is switched on.
Postconditions	POST-1. The session plan is saved as a new version.
Priority	Should have
Frequency of Use	Weekly
Business Rules	BR-45, BR-47, BR-48, BR-49, BR-52

Normal Flow

32.0 Plan a Training Session

1. The coach opens the session and selects Plan. (see 32.0.E1)
2. The coach adds blocks, each with a title, type, intensity, minutes and description.
3. The system checks the blocks and saves the plan. (see 32.0.E2, 32.0.E3)

Alternative Flows

32.1 Start from a saved template

1. The coach starts from a saved template, then edits the blocks.
   Return to step 3 of the Normal Flow.

32.2 The coach asks for an AI suggestion

1. The system sends the class sport and level to Gemini and fills the blocks, which the coach reviews.
   Return to step 3 of the Normal Flow.

32.3 Save as a template or copy a week

1. The coach saves the plan as a template, or copies a week of plans to the next week.
   The use case ends.

Exceptions

32.0.E1 The training module is switched off

1. The system refuses the request with BR-52.
   The use case ends.

32.0.E2 A block breaks BR-49

1. The system refuses the request with VALIDATION (5.2 Error Responses), and the block is marked.
   Return to step 2 of the Normal Flow.

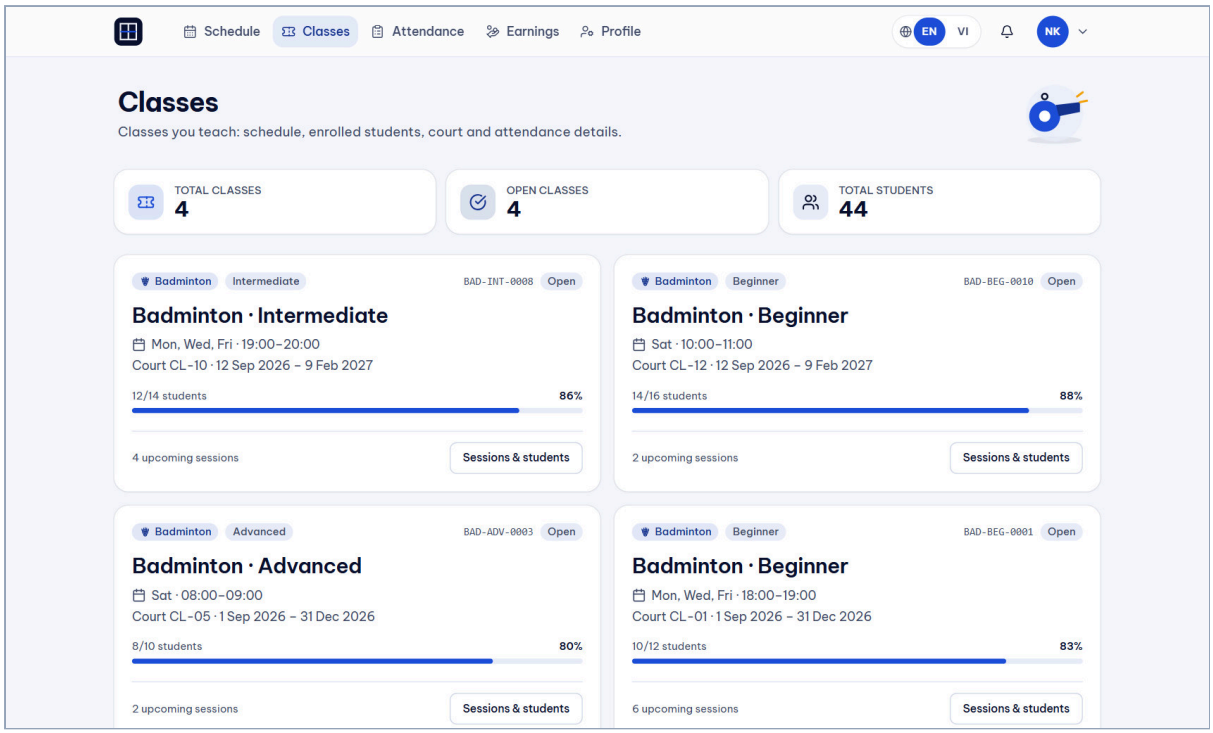
32.0.E3 The session already has results

1. The system refuses the request with BR-48, and the coach copies the plan instead.
   The use case ends.

3.7.4 Record Session Results and Homework

- **Function Trigger:** The session ends and the coach opens it in Training plans.
- **Function Description:**
  - **Use Case ID:** UC-33
  - **Actors:** Coach; Secondary: None
  - **Purpose:** The coach records how each student did in the session and sets homework.
  - **Interface:** Classes (coach) (/coach/classes)
- **Data Processing:**
  - **Data Collect:** Each student's score and note; the homework title, checklist and due date.
  - **Data Computation:** Checks the coach's class (BR-45), the homework limits and that the session plan is no longer edited (BR-48); records never take money (BR-47).
  - **Data Storage:** Writes session_results, homework and homework_recipients, and queues notices in outbox.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-33 Record Session Results and Homework
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Coach
Secondary Actors	None
Trigger	The session ends and the coach opens it in Training plans.
Description	The coach records how each student did in the session and sets homework.
Preconditions	PRE-1. The coach teaches the class. PRE-2. The training module is switched on.
Postconditions	POST-1. Results and homework are saved; students see them in Training.
Priority	Should have
Frequency of Use	Every session
Business Rules	BR-45, BR-47, BR-48

Normal Flow

33.0 Record Session Results and Homework

1. The coach opens the session results.
2. The coach records a score and a note for each student.
3. The system saves the results. (see 33.0.E1)

4. The coach sets homework with a title, a checklist and a due date, for the class or one student. (see 33.0.E2)
5. The system saves the homework and notifies the students.

Exceptions

33.0.E1 The session belongs to another coach's class

1. The system refuses the request with FORBIDDEN (ERR_AUTH_05, BR-45).
   The use case ends.

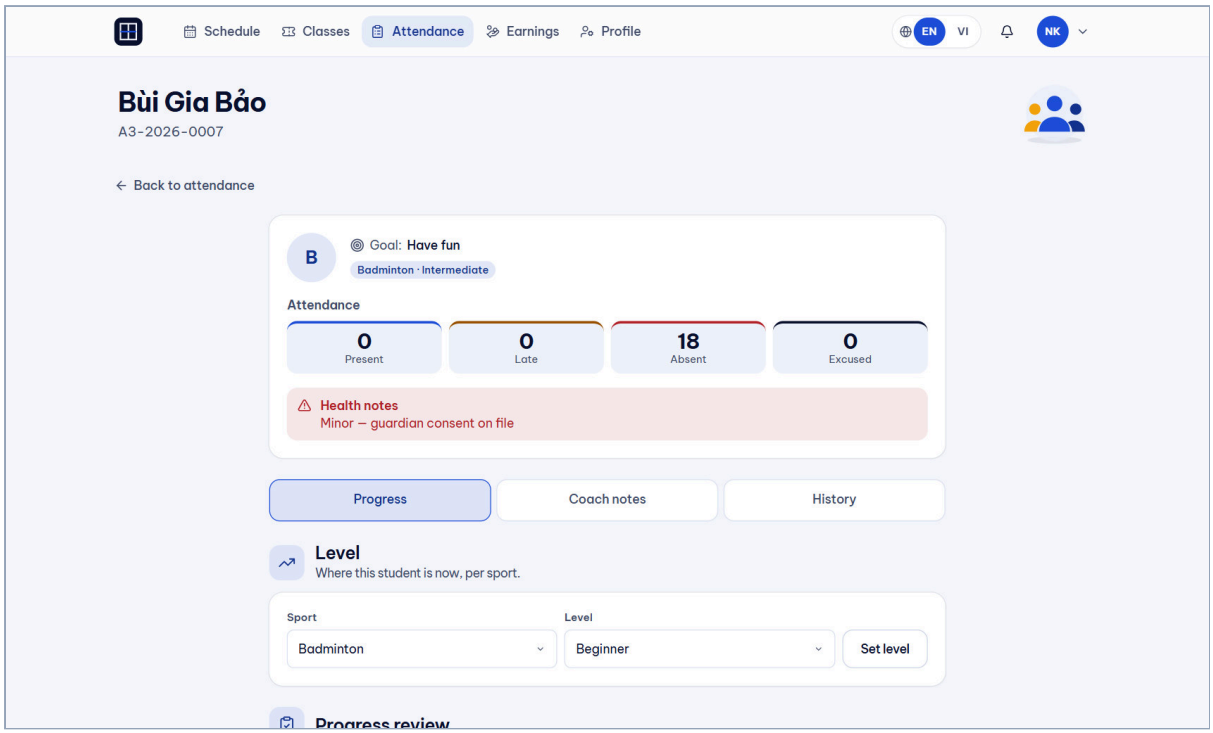
33.0.E2 The title is longer than 120 characters, the checklist has more than 20 items, or the due date is past

1. The system refuses the request with VALIDATION (5.2 Error Responses).
   Return to step 4 of the Normal Flow.

3.7.5 Assess a Student's Progress

- **Function Trigger:** The coach opens a student profile (/coach/student/$id).
- **Function Description:**
  - **Use Case ID:** UC-34
  - **Actors:** Coach; Secondary: None
  - **Purpose:** The coach sets a student's level, writes a 2- or 4-week review and keeps private notes.
  - **Interface:** Student progress (/coach/student/$id)
- **Data Processing:**
  - **Data Collect:** The review period and scores, a level, or a private note.
  - **Data Computation:** Checks the student is in the coach's classes (BR-45), scores from 1 to 5 and a 2- or 4-week period; shows health notes only in scope (BR-46).
  - **Data Storage:** Writes progress_reviews, member_levels and coach_notes.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-34 Assess a Student's Progress
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Coach
Secondary Actors	None
Trigger	The coach opens a student profile (/coach/student/$id).
Description	The coach sets a student's level, writes a 2- or 4-week review and keeps private notes.
Preconditions	PRE-1. The student is in a class the coach teaches.
Postconditions	POST-1. The level, review or note is saved and the student sees the review.
Priority	Should have
Frequency of Use	Weekly
Business Rules	BR-45, BR-46

Normal Flow

34.0 Assess a Student's Progress

1. The coach opens the student profile. (see 34.0.E1)
2. The system shows level, attendance, results, homework progress and health notes in scope.
3. The coach writes a review covering 2 or 4 weeks with scores from 1 to 5. (see 34.0.E2)
4. The system saves the review and shows it to the student.

Alternative Flows

34.1 Change the student's level

1. The coach changes the student's level.
   Return to step 4 of the Normal Flow.

34.2 Add a private note

1. The coach adds a private note of under 2,000 characters.
   Return to step 4 of the Normal Flow.

Exceptions

34.0.E1 The student is not in the coach's classes

1. The system refuses the request with FORBIDDEN (ERR_AUTH_05, BR-45).
   The use case ends.

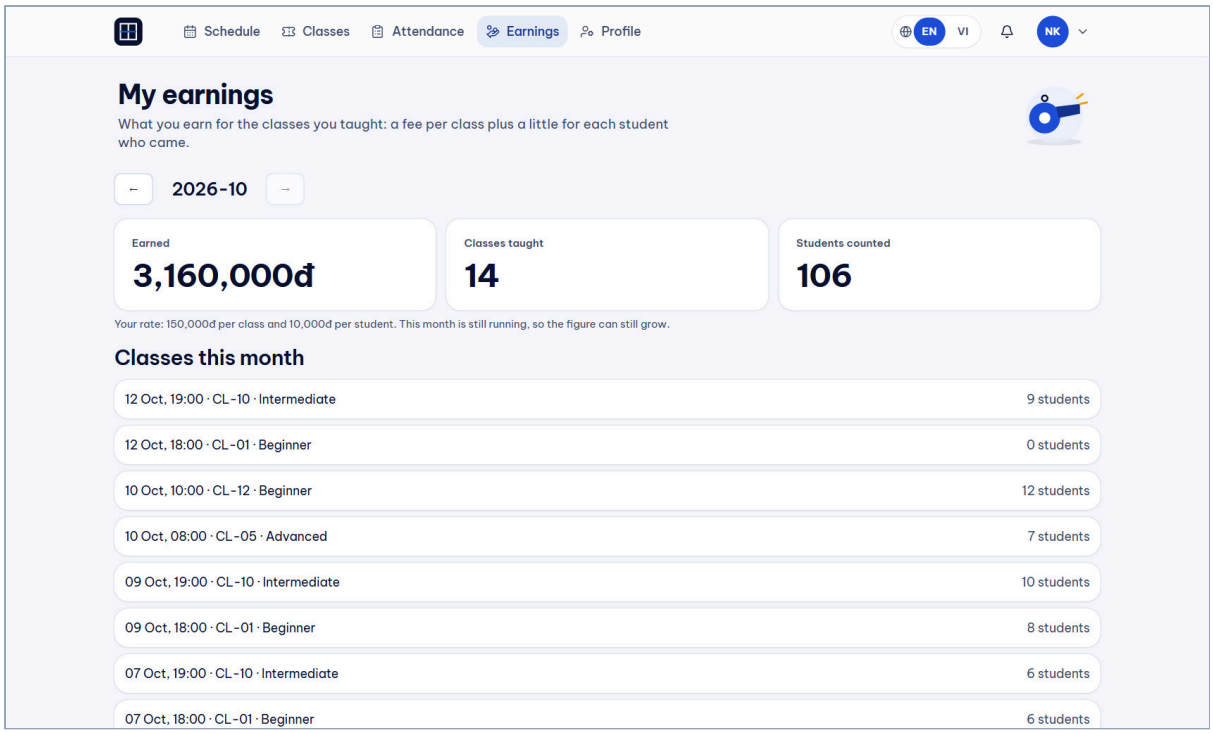
34.0.E2 A score outside 1 to 5, or a period other than 2 or 4 weeks

1. The system refuses the request with VALIDATION (5.2 Error Responses).
   Return to step 3 of the Normal Flow.

3.7.6 View Monthly Earnings

- **Function Trigger:** The coach opens Earnings (/coach/earnings).
- **Function Description:**
  + **Use Case ID:** UC-35
  + **Actors:** Coach; Secondary: None
  + **Purpose:** The coach sees the sessions taught and the commission earned for a month, and whether it is paid.
  + **Interface:** My earnings (/coach/earnings)
- **Data Processing:**
  + **Data Collect:** The month chosen.
  + **Data Computation:** Counts the sessions taught and the students present, and applies the coach's rates.
  + **Data Binding:** Reads coach_rates, coach_payouts, sessions and attendance, and shows the month's earnings.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-35 View Monthly Earnings
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Coach
Secondary Actors	None
Trigger	The coach opens Earnings (/coach/earnings).
Description	The coach sees the sessions taught and the commission earned for a month, and whether it is paid.
Preconditions	PRE-1. The coach is signed in.
Postconditions	POST-1. No data changes.
Priority	Could have
Frequency of Use	Monthly
Business Rules	None

Normal Flow

35.0 View Monthly Earnings

1. The coach opens Earnings and picks a month. (see 35.0.E1)
2. The system shows the sessions taught, the students taught, the rate and the amount.
3. The system shows whether the month is open, closed or paid.

Exceptions

35.0.E1 The month has no sessions

1. The system shows zero.

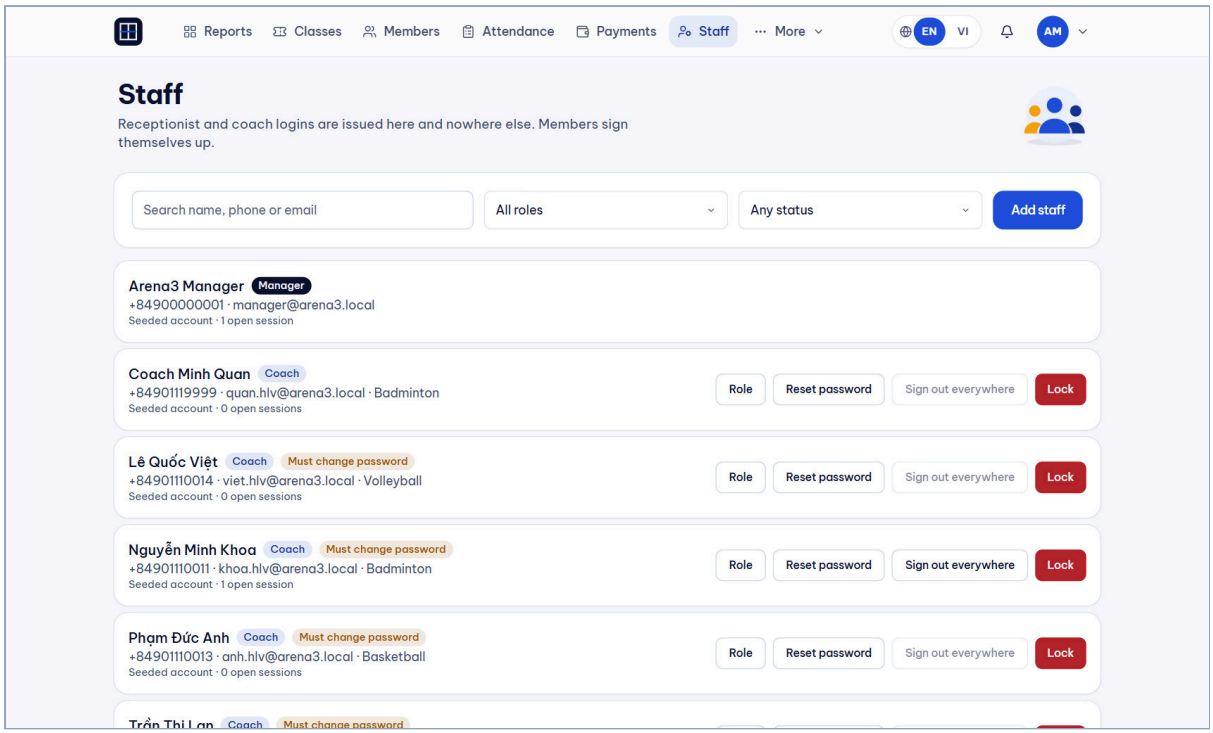
   The use case ends.

3.8 Manager Setup

3.8.1 Manage User Accounts

- **Function Trigger:** The manager opens Staff (/manager/staff) or Members (/manager/members).
- **Function Description:**
  - **Use Case ID:** UC-36
  - **Actors:** Manager; Secondary: None
  - **Purpose:** The manager creates and maintains staff accounts and looks after member accounts.
  - **Interface:** Staff (/manager/staff); Members (/manager/members)
- **Data Processing:**
  - **Data Collect:** Name, phone, email, role and, for a coach, the sports; lock, reset and sign-out actions; member filters.
  - **Data Computation:** Checks that the phone and email are unused (BR-01) and a coach's sports (BR-25), and refuses changes to the manager's own account.
  - **Data Storage:** Writes users and coach_sports, revokes rows in sessions_auth and writes audit_logs (BR-53).

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-36 Manage User Accounts
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Manager
Secondary Actors	None
Trigger	The manager opens Staff (/manager/staff) or Members (/manager/members).
Description	The manager creates and maintains staff accounts and looks after member accounts.
Preconditions	PRE-1. The manager is signed in.
Postconditions	POST-1. An account is created or changed, sessions may be revoked, and an audit row is written.
Priority	Must have
Frequency of Use	Weekly
Business Rules	BR-01, BR-25, BR-53

Normal Flow

36.0 Manage User Accounts

1. The manager selects Add staff. (see 36.0.E1, 36.0.E2)
2. The manager enters name, phone, email and role: receptionist or coach, with the sports for a coach. (see 36.0.E3, 36.0.E4)
3. The system creates the account with a temporary password.

4. The manager passes the temporary password to the new staff member, who changes it on first sign-in.

Alternative Flows

36.1 Lock, unlock, reset or sign out an account

1. The manager locks or unlocks an account, resets its password, or signs it out everywhere.
   The use case ends.

Exceptions

36.0.E1 The manager tries to change their own account here

1. The system refuses the request with FORBIDDEN (ERR_AUTH_05).
   The use case ends.

36.0.E2 A coach with running classes is given another role

1. The system refuses the request with CONFLICT_STATE (5.2 Error Responses).
   The use case ends.

36.0.E3 The phone or email already has an account

1. The system refuses the request with BR-01 (ERR_AUTH_01).
   Return to step 2 of the Normal Flow.

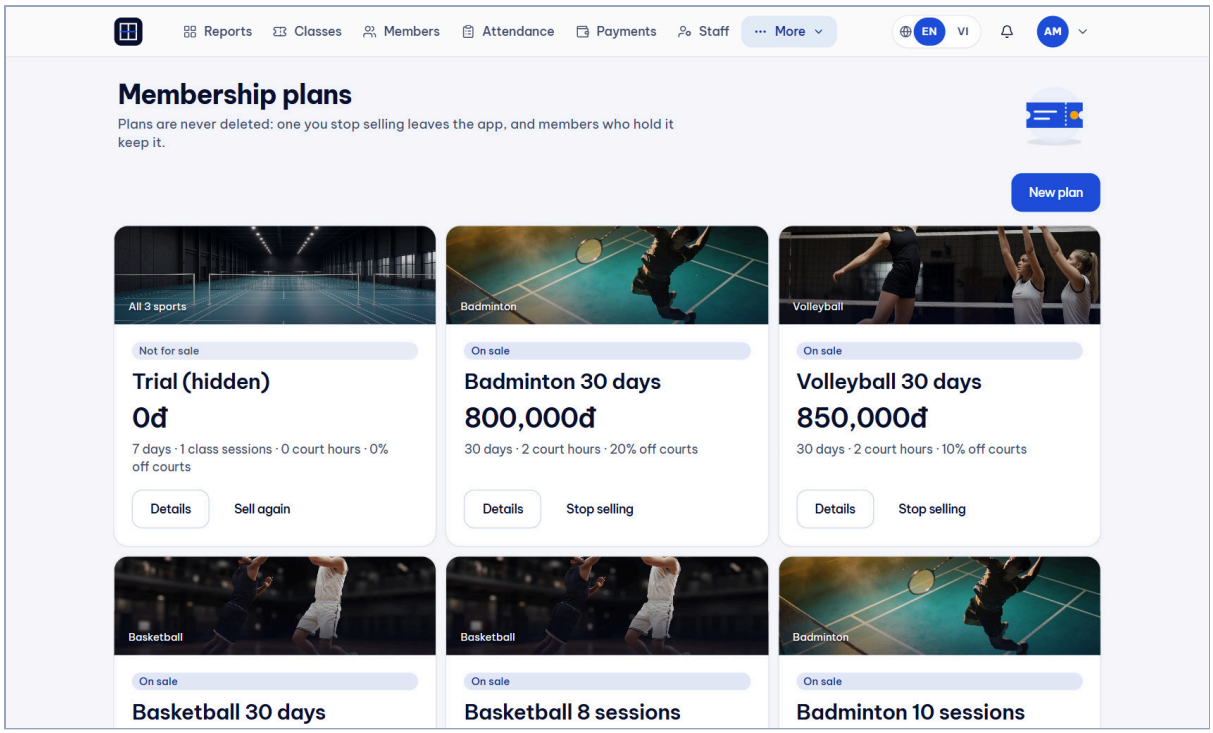
36.0.E4 A coach has no sport

1. The system refuses the request with VALIDATION (5.2 Error Responses).
   Return to step 2 of the Normal Flow.

3.8.2 Manage Membership Plans and Prices

- **Function Trigger:** The manager opens Plans (/manager/plans) or Prices (/manager/prices).
- **Function Description:**
  + **Use Case ID:** UC-37
  + **Actors:** Manager; Secondary: None
  + **Purpose:** The manager sets up the plans on sale and the court price rules.
  + **Interface:** Membership plans (/manager/plans); Pricing (/manager/prices)
- **Data Processing:**
  + **Data Collect:** Plan fields; price rules by sport, court, weekday and hour band.
  + **Data Computation:** Checks the discount is a whole percentage (BR-17), the plan is on sale or withdrawn (BR-16) and each rule has a price and no overlap (BR-37); prices include VAT (BR-38).
  + **Data Storage:** Writes membership_plans and price_rules, and writes audit_logs (BR-53).

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-37 Manage Membership Plans and Prices
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Manager
Secondary Actors	None
Trigger	The manager opens Plans (/manager/plans) or Prices (/manager/prices).
Description	The manager sets up the plans on sale and the court price rules.
Preconditions	PRE-1. The manager is signed in.
Postconditions	POST-1. Plans or price rules are saved and apply to new orders and bookings. POST-2. An audit row is written.
Priority	Must have
Frequency of Use	Monthly
Business Rules	BR-16, BR-17, BR-37, BR-38, BR-53

Normal Flow

37.0 Manage Membership Plans and Prices

1. The manager creates or edits a plan: name, sport, length, sessions, court hours, discount and price.
2. The system checks and saves the plan. (see 37.0.E1)
3. The manager opens Prices and edits the price rules by sport, court, weekday and hour.

4. The system saves the rules. (see 37.0.E2)

Alternative Flows

37.1 Withdraw a plan

1. It stays on existing members but can no longer be ordered (BR-16).
   Return to step 3 of the Normal Flow.

Exceptions

37.0.E1 The discount is not a whole number from 0 to 100

1. The system refuses the request with BR-17.
   Return to step 1 of the Normal Flow.

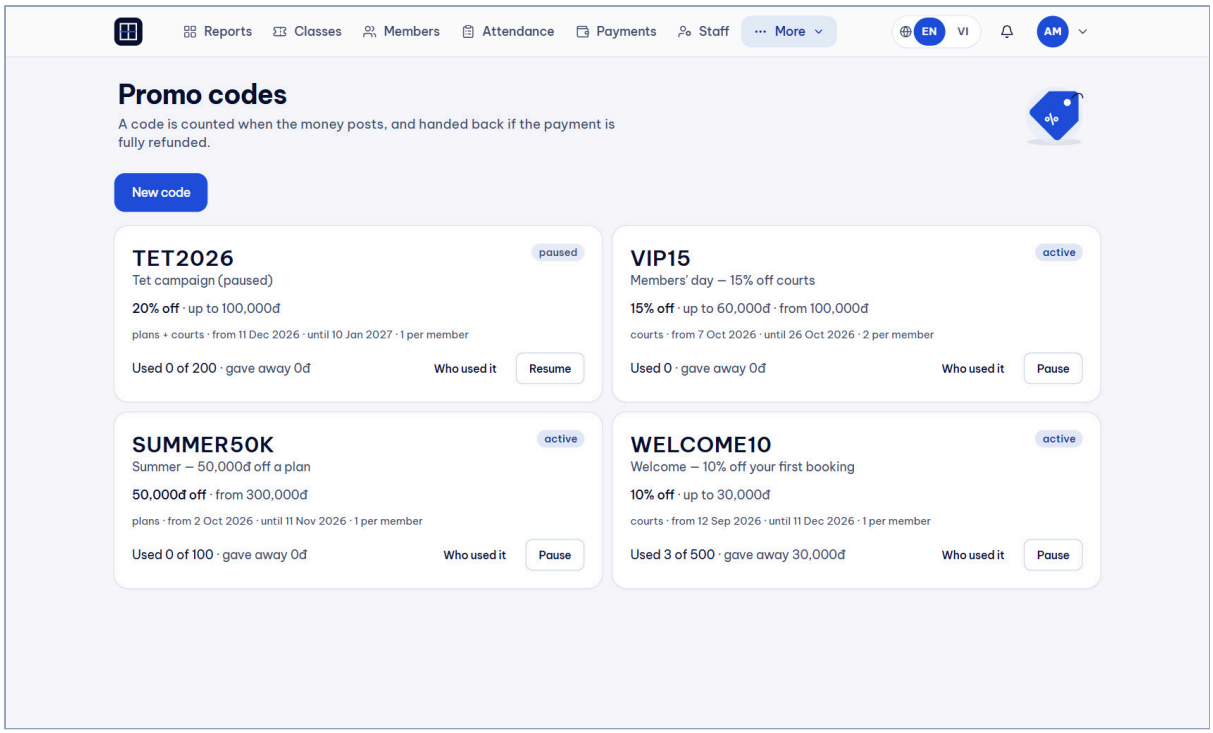
37.0.E2 A rule has no price, or overlaps another rule

1. The system refuses the request with VALIDATION (5.2 Error Responses).
   Return to step 3 of the Normal Flow.

3.8.3 Manage Promo Codes

- **Function Trigger:** The manager opens Promo codes (/manager/promos).
- **Function Description:**
  - **Use Case ID:** UC-38
  - **Actors:** Manager; Secondary: None
  - **Purpose:** The manager creates promo codes with their scope, limits and dates, and pauses them.
  - **Interface:** Promo codes (/manager/promos)
- **Data Processing:**
  - **Data Collect:** Code, kind and value, scope, dates and limits; pause or resume.
  - **Data Computation:** Checks the code length and the value ranges, the scope and limits applied at checkout (BR-40, BR-41, BR-42), and that a used code keeps its code (BR-43).
  - **Data Storage:** Writes promotions and audit_logs; the use counts come from promotion_redemptions.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-38 Manage Promo Codes
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Manager
Secondary Actors	None
Trigger	The manager opens Promo codes (/manager/promos).
Description	The manager creates promo codes with their scope, limits and dates, and pauses them.
Preconditions	PRE-1. The manager is signed in.
Postconditions	POST-1. The code is saved or paused. POST-2. An audit row is written.
Priority	Should have
Frequency of Use	Monthly
Business Rules	BR-40, BR-41, BR-42, BR-43

Normal Flow

38.0 Manage Promo Codes

1. The manager enters the code, a percent or fixed amount, the scope, the dates and the limits.
2. The system checks and saves the code. (see 38.0.E1, 38.0.E2)
3. The manager views how many times each code has been used.

Alternative Flows

38.1 Pause or resume a code

1. The manager pauses or resumes a code.
   The use case ends.

Exceptions

38.0.E1 The code is not 3 to 24 characters, a percent is over 100, or a fixed amount is not a multiple of 1,000đ

1. The system refuses the request with VALIDATION (5.2 Error Responses).
   Return to step 1 of the Normal Flow.

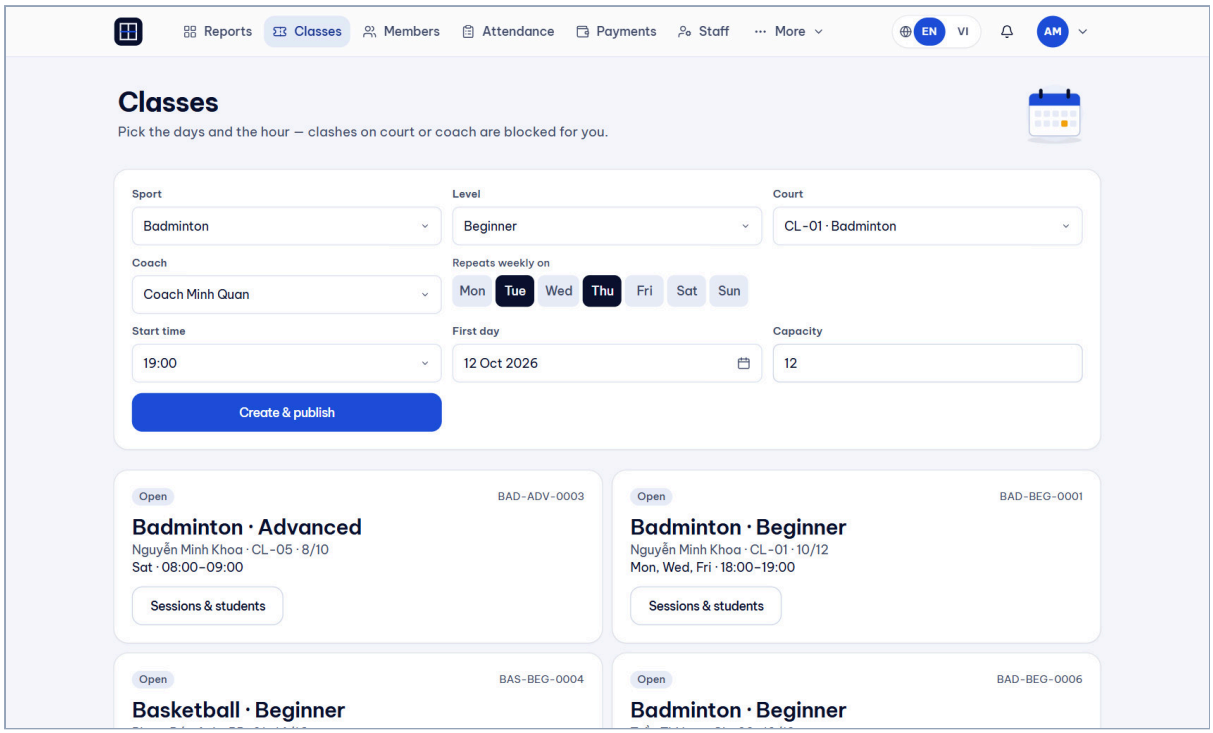
38.0.E2 The manager renames a code that has been used

1. The system refuses the request with BR-43.
   The use case ends.

3.8.4 Schedule Classes and Sessions

- **Function Trigger:** The manager opens Classes (/manager/classes).
- **Function Description:**
  + **Use Case ID:** UC-39
  + **Actors:** Manager; Secondary: None
  + **Purpose:** The manager creates a class with its coach, court, schedule and capacity, opens it, and changes or cancels sessions.
  + **Interface:** Classes (manager) (/manager/classes)
- **Data Processing:**
  + **Data Collect:** Sport, coach, court, weekdays, time, length, capacity and dates; later moves, cancellations, substitutes and capacity.
  + **Data Computation:** Checks the coach (BR-25), the court and coach times (BR-26, BR-08), the capacity (BR-21), late moves (BR-27), substitutes (BR-28) and sessions given back (BR-29).
  + **Data Storage:** Writes classes, sessions, occupancies and coach_occupancies, and queues notices in outbox.

- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-39 Schedule Classes and Sessions
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Manager
Secondary Actors	None
Trigger	The manager opens Classes (/manager/classes).
Description	The manager creates a class with its coach, court, schedule and capacity, opens it, and changes or cancels sessions.
Preconditions	PRE-1. The manager is signed in.
Postconditions	POST-1. The class and its sessions exist and hold their court times. POST-2. Enrolled members are notified of changes.
Priority	Must have
Frequency of Use	Weekly
Business Rules	BR-08, BR-21, BR-25, BR-26, BR-27, BR-28, BR-29

Normal Flow

39.0 Schedule Classes and Sessions

1. The manager enters the sport, coach, court, weekdays, time, length (60, 120 or 180 minutes), capacity and dates.
2. The system checks the coach and the court times and creates the sessions. (see 39.0.E1, 39.0.E2)

3. The manager publishes the class, which opens it for enrolment. (see 39.0.E3)

Alternative Flows

39.1 Move a session

1. The manager moves a session to a future time, with a reason when it is under 12 hours away (BR-27).
   The use case ends.

39.2 Cancel a session

1. The manager cancels a session with a reason; session-based plans get the session back once (BR-29).
   The use case ends.

39.3 Hand the class to a substitute coach

1. The manager hands the class to a substitute coach from a date onward (BR-28).
   The use case ends.

39.4 Change the capacity

1. The manager changes the capacity.
   The use case ends.

Exceptions

39.0.E1 The coach is inactive or not qualified for the sport

1. The system refuses the request with BR-25.
   Return to step 1 of the Normal Flow.

39.0.E2 The court or the coach is busy at a session time

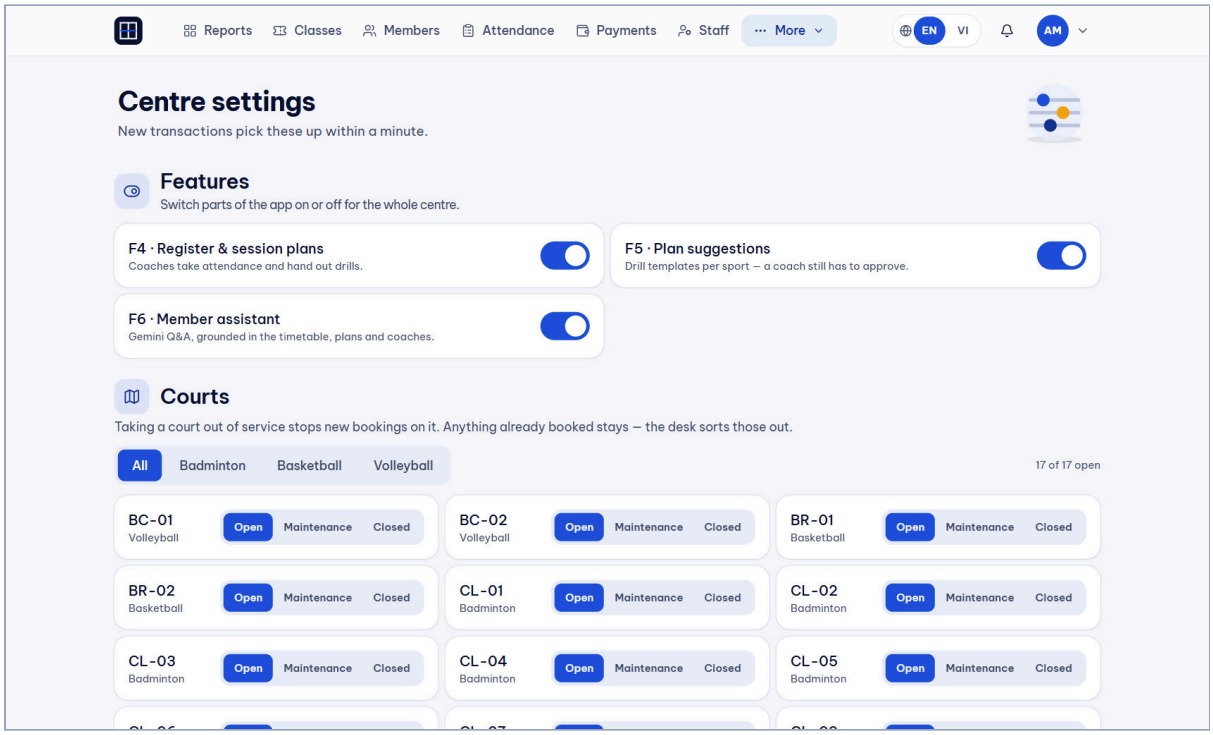
1. The system refuses the request with BR-26 or CONFLICT_SLOT (ERR_BOOK_02).
   Return to step 1 of the Normal Flow.

39.0.E3 The capacity is outside 1 to 200, or below the number enrolled

1. The system refuses the request with BR-21.
   The use case ends.

3.8.5 Configure the Centre

- **Function Trigger:** The manager opens Centre settings (/manager/settings).
- **Function Description:**
  - **Use Case ID:** UC-40
  - **Actors:** Manager; Secondary: None
  - **Purpose:** The manager sets the centre's hours, booking limits, refund tiers, check-in and other settings, and switches modules on or off.
  - **Interface:** Centre settings (/manager/settings)
- **Data Processing:**
  - **Data Collect:** Setting values and module switches.
  - **Data Computation:** Checks each value's range; a module switched off answers 403 with BR-52.
  - **Data Storage:** Writes center_settings and feature_flags, and writes audit_logs (BR-53).
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-40 Configure the Centre
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Manager
Secondary Actors	None

Field	Detail
Trigger	The manager opens Centre settings (/manager/settings).
Description	The manager sets the centre's hours, booking limits, refund tiers, check-in and other settings, and switches modules on or off.
Preconditions	PRE-1. The manager is signed in.
Postconditions	POST-1. The settings apply to every later request. POST-2. An audit row is written.
Priority	Must have
Frequency of Use	Rarely
Business Rules	BR-52, BR-53

Normal Flow

40.0 Configure the Centre

1. The manager opens Centre settings.
2. The system shows the settings in groups with their current values.
3. The manager edits values and saves.
4. The system checks each value's range and stores them. (see 40.0.E1)

Alternative Flows

40.1 The manager switches a module off

1. Every route of that module answers 403 with BR-52 until it is switched on.
   Return to step 4 of the Normal Flow.

Exceptions

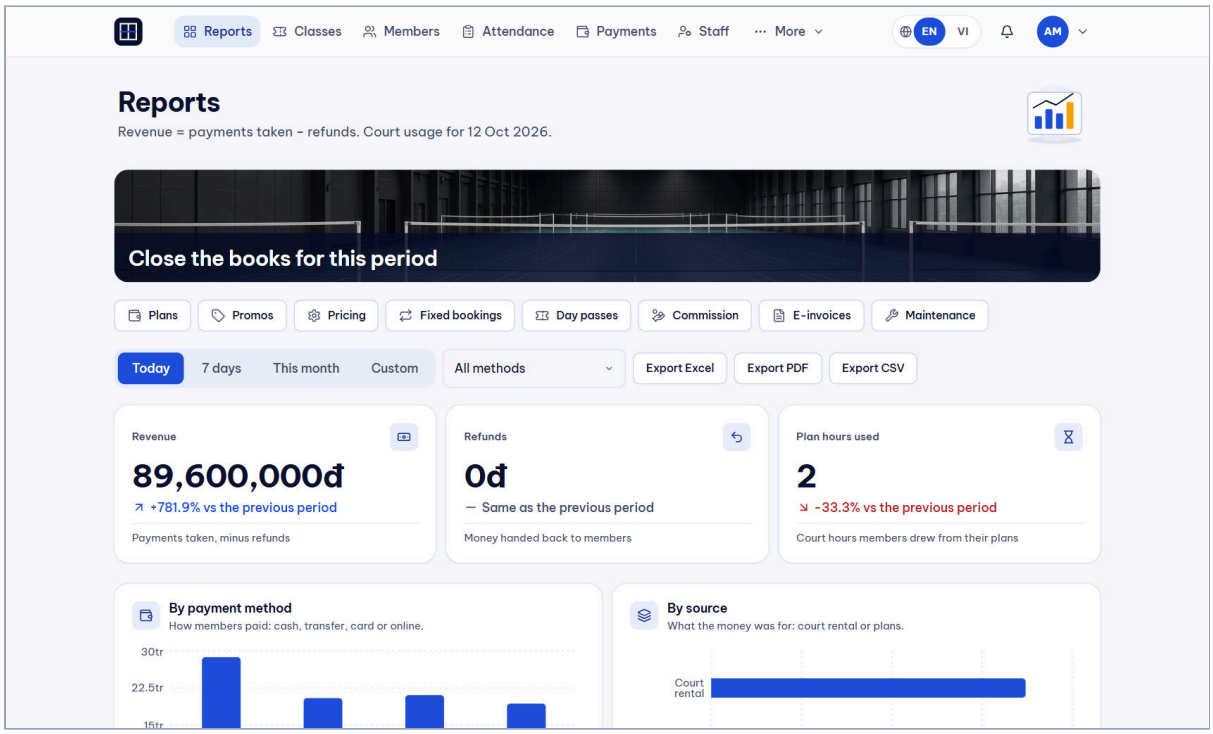
40.0.E1 A value is out of range

1. The system refuses the request with VALIDATION (5.2 Error Responses), naming the setting.
   Return to step 3 of the Normal Flow.

3.9 Manager Oversight

3.9.1 View Reports and Export Data

- **Function Trigger:** The manager opens Reports (/manager), Attendance (/manager/attendance) or Invoices (/manager/invoices).
- **Function Description:**
  - **Use Case ID:** UC-41
  - **Actors:** Manager; Secondary: None
  - **Purpose:** The manager reviews revenue, occupancy, capacity, members and attendance, and exports reports and e-invoices.
  - **Interface:** Reports (/manager); Attendance report (/manager/attendance); E-invoice export (/manager/invoices)
- **Data Processing:**
  - **Data Collect:** The report, the date range and the export format.
  - **Data Computation:** Builds each report from the same queries as the screens; prices include VAT (BR-38) and attendance follows the locked registers (BR-35).
  - **Data Binding:** Reads payments, court_bookings, subscriptions, attendance and invoices; an export is built on request and not stored.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-41 View Reports and Export Data

Field	Detail
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Manager
Secondary Actors	None
Trigger	The manager opens Reports (/manager), Attendance (/manager/attendance) or Invoices (/manager/invoices).
Description	The manager reviews revenue, occupancy, capacity, members and attendance, and exports reports and e-invoices.
Preconditions	PRE-1. The manager is signed in.
Postconditions	POST-1. No data changes; an export file is downloaded.
Priority	Must have
Frequency of Use	Daily
Business Rules	BR-35, BR-38

Normal Flow

41.0 View Reports and Export Data

1. The manager picks a report and a date range.
2. The system builds the report from the same queries as the screens. (see 41.0.E1)
3. The manager selects Export and a format: xlsx or pdf.
4. The system builds the file on request and the browser downloads it.

Alternative Flows

41.1 Export the e-invoices

1. The manager opens Invoices and exports the e-invoices of a period as XML.
   The use case ends.

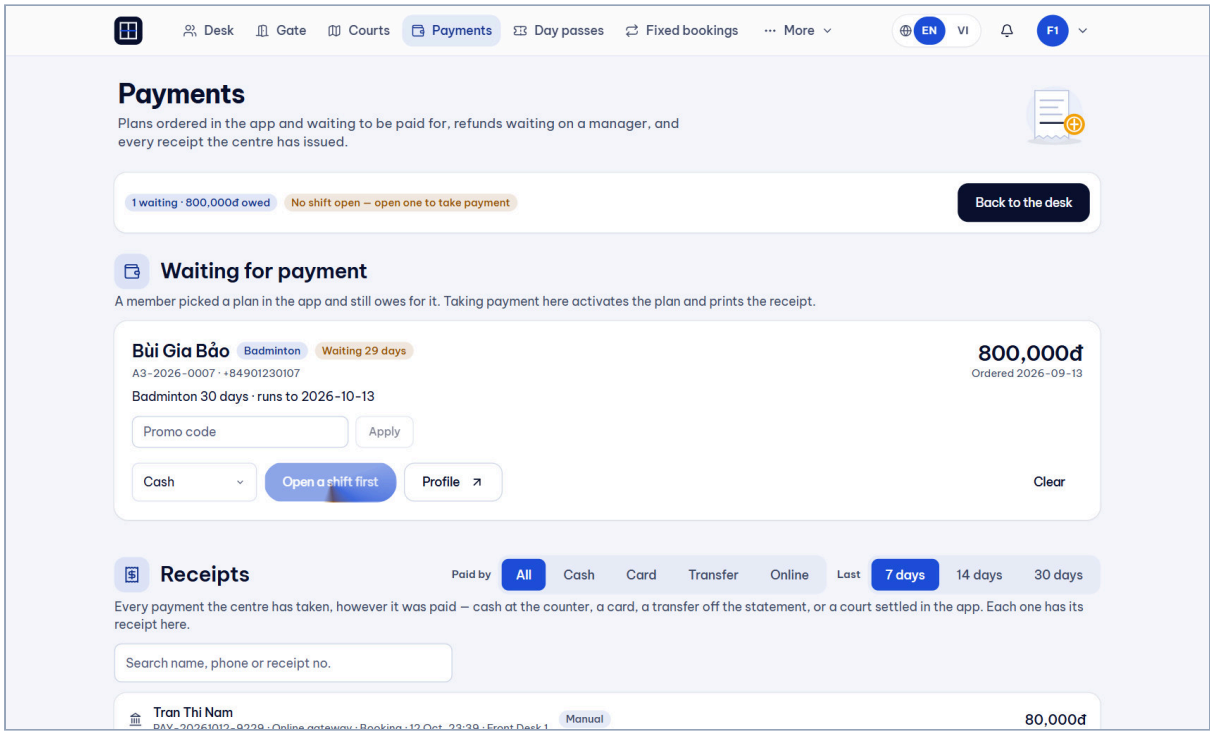
Exceptions

41.0.E1 The date range is invalid

1. The system refuses the request with VALIDATION (5.2 Error Responses).
   Return to step 1 of the Normal Flow.

3.9.2 Approve or Reject a Refund

- **Function Trigger:** A refund pending approval appears in Refund sign-off (/desk/payments).
- **Function Description:**
  - **Use Case ID:** UC-42
  - **Actors:** Manager; Secondary: None
  - **Purpose:** A manager other than the person who raised a large refund approves or rejects it.
  - **Interface:** Payments (/desk/payments)
- **Data Processing:**
  - **Data Collect:** Approve, or Reject with a reason.
  - **Data Computation:** Refuses the person who raised the refund, and a refund that was already decided.
  - **Data Storage:** Posts or closes the refund in payments, queues a notice in outbox and writes audit_logs (BR-53).
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-42 Approve or Reject a Refund
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Manager
Secondary Actors	None

Field	Detail
Trigger	A refund pending approval appears in Refund sign-off (/desk/payments).
Description	A manager other than the person who raised a large refund approves or rejects it.
Preconditions	PRE-1. The manager is signed in. PRE-2. The refund is pending approval.
Postconditions	POST-1. Approve: the refund is posted and the member notified. POST-2. Reject: it is closed with a reason. POST-3. An audit row is written.
Priority	Must have
Frequency of Use	A few times a week
Business Rules	BR-53

Normal Flow

42.0 Approve or Reject a Refund

1. The manager opens the pending refunds.
2. The manager reviews the payment, the amount and the reason.
3. The manager selects Approve. (see 42.0.E1, 42.0.E2)
4. The system posts the refund and notifies the member.

Alternative Flows

42.1 The manager selects Reject and gives a reason; the system closes the refund and notifies the member

1. The manager selects Reject and gives a reason; the system closes the refund and notifies the member.

   The use case ends.

Exceptions

42.0.E1 The manager raised this refund

1. The system refuses the request with FORBIDDEN (ERR_PAY_04), and another manager must sign it off.

   The use case ends.

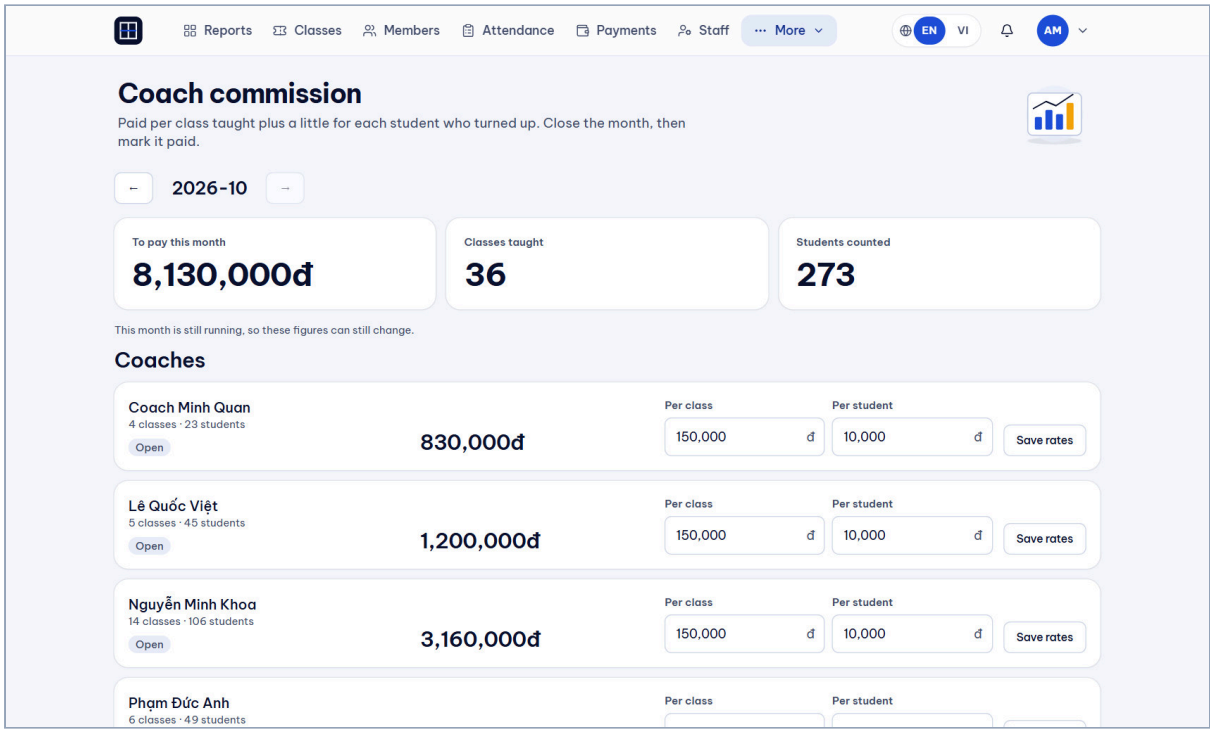
42.0.E2 Another manager decided it first

1. The system refuses the request with CONFLICT_STATE (ERR_PAY_05).

   The use case ends.

3.9.3 Close and Pay Coach Commission

- **Function Trigger:** A month ends and the manager opens Commission (/manager/commission).
- **Function Description:**
  + **Use Case ID:** UC-43
  + **Actors:** Manager; Secondary: None
  + **Purpose:** The manager sets coach rates, closes a finished month so its figures are frozen, and marks each coach paid.
  + **Interface:** Coach commission (/manager/commission)
- **Data Processing:**
  + **Data Collect:** The month, coach rates, Close month and Mark paid.
  + **Data Computation:** Counts each coach's sessions and students and applies the rates; a month closes only after it ends.
  + **Data Storage:** Writes coach_rates, one row per coach to coach_payouts and audit_logs (BR-53).
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-43 Close and Pay Coach Commission
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Manager

Field	Detail
Secondary Actors	None
Trigger	A month ends and the manager opens Commission (/manager/commission).
Description	The manager sets coach rates, closes a finished month so its figures are frozen, and marks each coach paid.
Preconditions	PRE-1. The manager is signed in.
Postconditions	POST-1. Close: one payout row per coach with sessions. POST-2. Pay: the payout is paid. POST-3. An audit row is written.
Priority	Should have
Frequency of Use	Monthly
Business Rules	BR-53

Normal Flow

43.0 Close and Pay Coach Commission

1. The manager picks a month and reviews each coach's sessions, students and amount.
2. The manager selects Close month; the system copies the figures into payout rows. (see 43.0.E1)
3. The manager pays a coach outside the system and selects Mark paid. (see 43.0.E2)

Alternative Flows

43.1 Change a coach's rate

1. The manager changes a coach's rate; closed months keep their figures.

   Return to step 2 of the Normal Flow.

Exceptions

43.0.E1 The month has not ended

1. The system refuses the request with CONFLICT_STATE (5.2 Error Responses).

   The use case ends.

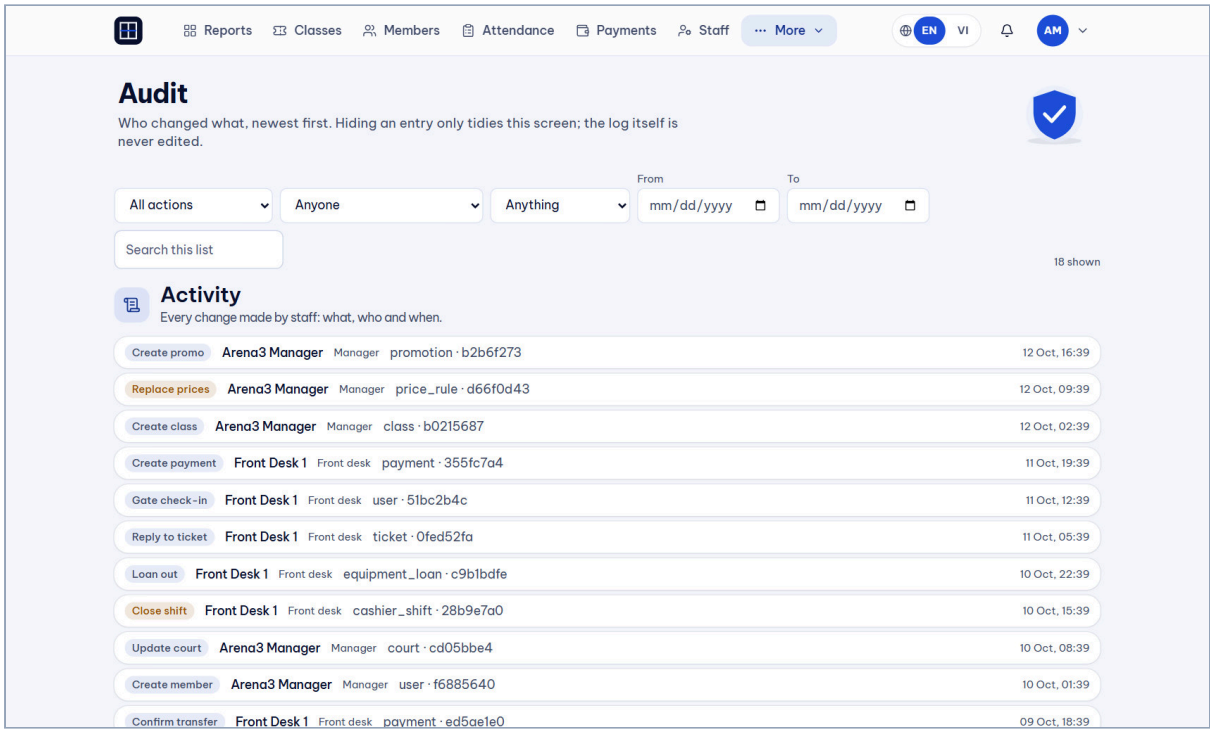
43.0.E2 The month is not closed, or already paid

1. The system refuses the request with CONFLICT_STATE (5.2 Error Responses).

   The use case ends.

3.9.4 Review the Audit Log

- **Function Trigger:** The manager opens Audit log (/manager/audit).
- **Function Description:**
  - **Use Case ID:** UC-44
  - **Actors:** Manager; Secondary: None
  - **Purpose:** The manager reviews who changed what and when, to investigate a dispute or a mistake.
  - **Interface:** Audit log (/manager/audit)
- **Data Processing:**
  - **Data Collect:** Filters by actor, action, record and date.
  - **Data Computation:** Orders the rows newest first; no row can be changed or deleted (BR-53).
  - **Data Binding:** Reads audit_logs and shows the values before and after each change.
- **Screen Layout:**



- **Function Details:**

Field	Detail
ID and Name	UC-44 Review the Audit Log
Created By	Arena3 team
Date Created	2026-10-12
Primary Actor	Manager
Secondary Actors	None
Trigger	The manager opens Audit log (/manager/audit).

Field	Detail
Description	The manager reviews who changed what and when, to investigate a dispute or a mistake.
Preconditions	PRE-1. The manager is signed in.
Postconditions	POST-1. No data changes.
Priority	Must have
Frequency of Use	Weekly
Business Rules	BR-53

Normal Flow

44.0 Review the Audit Log

1. The manager opens Audit log.
2. The system lists audit rows, newest first, with actor, action, record and time.
3. The manager filters by actor, action or date. (see 44.0.E1)
4. The manager opens a row to see the values before and after.

Exceptions

44.0.E1 No rows match

1. The system shows an empty list.

   The use case ends.

4. Non-Functional Requirements

Arena 3 must keep money and schedules correct under concurrency, respond quickly on a phone, and protect member data. This section names the external interfaces and the quality attributes that make this so.

4.1 External Interfaces

4.1.1 User Interface

Arena 3 is one responsive web application with four portals, and every screen works from a 375 px phone width.

- The interface is available in English and Vietnamese, and the user switches language in the header.
- Running text uses Be Vietnam Pro, which renders Vietnamese diacritics correctly. Headings use Newsreader and figures use Anton.
- Amounts are written in VND with a lowercase đ, for example 1,500,000đ.
- A refusal is shown as a sentence that says what to do instead. The error code travels with it for support.
- Motion is decorative only, and no action is discoverable only through animation.

4.1.2 Communications Interfaces

The application talks to five outside systems, and each has a defined fallback.

Interface	Purpose	Protocol and data	Failure behaviour
Arena 3 API (`/v1`)	Every screen reads and writes through it	JSON over HTTPS with a bearer token; served inside the web app	Errors return `{ code, message }`, plus `field` or `br` where one applies
payOS	Payment links, VietQR codes, link status and payment webhooks	HTTPS REST; the webhook is signed with the checksum key	A payment posts only when payOS confirms it, and the amount must equal the amount due
Google Gemini (`gemini-2.5-flash`)	Answers in the AI assistant	HTTPS; the prompt carries centre data only, never member data	12-second timeout, then xAI Grok, then rule-based answers
Email (SMTP through nodemailer)	One-time codes and notifications	SMTP with the centre's sender address	Codes are sent at once; notifications are queued and retried up to five times
PostgreSQL 16	All records	Neon in production; embedded PGLite when no `DATABASE_URL` is set	Schema changes are forward-only numbered migrations, each run in one transaction

The application is hosted on Vercel. Background jobs start with the first API request and run in a loop every 15 seconds.

4.2 Quality Attributes

4.2.1 Usability

Every screen must be usable on a phone and must explain a refusal in words, in English or Vietnamese.

ID	Requirement
NFR-11	Every screen works at a width of 375 px.
NFR-12	A refusal tells the user what to do instead, as a sentence rather than a code.
NFR-13	Text renders Vietnamese diacritics correctly, and currency is never upper-cased (1,500,000đ, never 1,500,000Đ).
NFR-14	Motion is decorative and never carries meaning on its own.

4.2.2 Reliability

Two bookings can never own the same court time, and money never disagrees with the schedule.

ID	Requirement
NFR-01	Two bookings must never own the same court time, under any concurrency. Both booking paths lock the court row, and a database trigger rejects any overlap.
NFR-02	A coach must never be scheduled into two overlapping sessions.
NFR-03	A confirmed booking always has its payment and invoice, written in the same transaction.
NFR-08	Any request that takes money or changes a schedule accepts an `Idempotency-Key`. A replay returns the first result and never charges twice.
NFR-09	With no `DATABASE_URL`, the application migrates and seeds an embedded PGLite database, so a fresh copy runs offline.
NFR-10	Losing the assistant's model degrades only the assistant, which falls back to rule-based answers.

4.2.3 Performance

The screens staff and members open most often must cost as few database round trips as possible.

ID	Requirement
NFR-04	`GET /v1/me` issues its independent reads concurrently, so a sign-in costs one round trip rather than seven.
NFR-05	Read-only endpoints open no transaction, which saves two round trips to a remote database.
NFR-06	The front desk badge asks for a count only (`?brief=1`), which runs one aggregate query.
NFR-07	The job loop first checks, in one query, which jobs are due. A pass with nothing due costs one query.

4.2.4 Security

Authorisation is decided on the server for every request, and member data stays inside the centre.

ID	Requirement
NFR-17	Passwords are stored only as salted hashes and are never written to logs.

ID	Requirement
NFR-18	Session tokens are stored as SHA-256 hashes. Member sessions last seven days and staff sessions twelve hours.
NFR-19	Every request is authorised on the server. Hiding a button is not a control.
NFR-20	A member can read only their own bookings, receipts and profile. A read of another member's data returns `403`.
NFR-21	Health notes and guardian details are shown only to those who need them and are never sent to the assistant's model.
NFR-22	In production, one-tap demo sign-in and on-screen codes are off unless `VITE_DEMO_LOGINS` or `OTP_ECHO` is set explicitly.
NFR-23	A switched-off module (F4 training) answers `403` with `BR-52` on every route.
NFR-24	Report exports are built on request from the same queries as the screens, and are not stored.
NFR-25	An online payment is trusted only when read from payOS or carried by a verified webhook signature.
NFR-26	Sign-in locks for 15 minutes after five wrong passwords, and a one-time code after five wrong attempts.

4.2.5 Auditability

Every staff change must leave a trace that names who did what and when, and the codes on money must never skip a number.

ID	Requirement
NFR-15	Every staff-only change writes an audit row naming the actor, action, record and time. This covers refunds, reconciliation, shift closes, settings, sessions, classes, staff accounts, attendance corrections and payment links. No endpoint deletes an audit row.
NFR-16	Member, payment and invoice codes come from a counter table and have no gaps within each kind.

4.2.6 Maintainability

The requirements in this document are checked by executable tests, so a change that breaks one fails before release.

- `npm test` runs the unit tests for pricing, refunds, check-in windows and other rules.
- `npm run check:api` drives the real API against a real database and checks the money and schedule rules end to end.
- `npm run typecheck` checks the TypeScript types across the application.

5. Requirement Appendix

This appendix lists every business rule, configurable setting, shared requirement and user message that the sections above refer to. Each entry matches the running application.

5.1 Business Rules

A refused request returns 422 BR_VIOLATION with the rule code in br, so any refusal on screen traces to one row below. Every rule is cited in the Business Rules row of at least one use case.

ID	Rule Definition
BR-01	Each user account must have a phone number and an email address that no other account uses.
BR-02	A password must contain at least 8 characters, including at least one letter and one digit.
BR-03	A member under 16 years of age must have a guardian's name and phone number on record.
BR-04	A member account must be created only after the person accepts the terms of use and the data-privacy notice.
BR-05	A member must not book a court more than 7 days ahead or hold more than 2 court slots on the same day.
BR-06	A court booking must lie within opening hours, on a court whose status is ready.
BR-07	A court booking must not start in the past. A walk-in booking must have at least 20 minutes left in its slot.
BR-08	Each court time must have exactly one owner: a booking, a class session or a maintenance block.
BR-09	A slot that overlaps one of the member's classes must be held only after the member confirms the clash.
BR-10	A replacement hold must pass every booking check before the hold it replaces is released.
BR-11	Court check-in is allowed from 15 minutes before the start until 10 minutes after it. A booking not checked in by then is a no-show.
BR-12	A plan grants its benefits only between its start date and its end date.
BR-13	A court booking paid from plan hours requires enough court hours left on that plan.
BR-14	A frozen plan must be resumed before it is frozen again or paid for.
BR-15	A plan must not be frozen for longer than the yearly freeze allowance (30 days by default).
BR-16	A withdrawn plan must not be ordered or sold.
BR-17	A plan discount is a whole-number percentage from 0 to 100.
BR-18	A member must enrol only in a class whose status is open.
BR-19	Class enrolment requires an active plan that covers the class sport.
BR-20	Enrolment through a session-based plan requires sessions left on that plan.
BR-21	Enrolments must not exceed class capacity, and capacity must not be set below the number already enrolled.
BR-22	A member must not hold a duplicate enrolment or two enrolments whose sessions overlap.
BR-23	A waitlist offer expires 2 hours after it is sent; the seat then passes to the next member in line.
BR-24	A member must leave a class no later than 4 hours before its next session.

ID	Rule Definition
BR-25	A class coach must hold an active account and be qualified for the class sport.
BR-26	A session must not be moved, or its coach changed, to a time when the court or the coach is already busy.
BR-27	Moving a session less than 12 hours before its original start requires a reason.
BR-28	A substitute coach takes over a class from the date of the change onward; past sessions keep their coach.
BR-29	When the centre cancels a session, a member on a session-based plan receives that session back once.
BR-30	A gate code must be a valid member code that has not been used before.
BR-31	A member with no entitlement today is admitted only with an override reason, and the visit is flagged. A repeat scan within 120 minutes returns the first visit.
BR-32	A gate check-in never spends a session or changes any balance.
BR-33	A gate check-in never marks class attendance; only the session register does.
BR-34	A student is marked late only by the coach in the session register.
BR-35	A session register locks 2 hours after the session; after that, a correction must come from a manager and must give a reason.
BR-36	Three absences in a row alert the coach and the managers; an excused absence breaks the streak.
BR-37	A price is quoted only from a matching price rule; the system never invents a price.
BR-38	Every price shown includes VAT.
BR-39	Money is rounded once, half up, to the nearest 1,000₫ after every discount is applied.
BR-40	A promo code must be active, within its dates, and match the order's scope, sport, plan and minimum spend.
BR-41	A promo code must not be used beyond its total limit or its per-member limit.
BR-42	A promo code that does not stack never combines with a plan discount; the larger of the two applies.
BR-43	A promo code that has been used must keep its code; staff pause it and create a new one instead.
BR-44	Money taken at the desk must belong to the receptionist's open cashier shift.
BR-45	A coach must act only on classes that the coach leads or assists.
BR-46	A student's health notes are shown only to staff whose role or class gives them a need to know.
BR-47	Training records never take money or change a plan balance.
BR-48	A training session that has results cannot change or add a plan; the coach copies the plan instead.
BR-49	A training plan has 1 to 12 blocks, and each block carries a type, an intensity, a description and a length of 1 to 180 minutes.
BR-50	Gear must not be rented out beyond the stock on hand.
BR-51	Gear hire is charged to the open cashier shift and issues an e-invoice; a return puts the gear back in stock.
BR-52	A module that a manager has switched off answers every request with 403 and BR-52.
BR-53	Every change to money, plans, prices, accounts or settings writes an audit row, and no audit row is ever deleted.

Version 2.1 renumbered the rules by category, from 75 codes to 53. Codes that only described a screen, or repeated another rule, were folded into the rule they belong to, and the API returns the

new codes. BR-49, BR-51 and BR-53 state rules the application already enforced without a code.

5.2 Common Requirements

- **User Login:** every user signs in with a phone number or email and a password (3.2.3). Member sessions last seven days and staff sessions twelve hours, and five wrong passwords lock sign-in for 15 minutes.
- **User Authorization:** each portal checks the signed-in role before it renders (3.1.3), and the API checks every request again on the server. The actions below are narrower than the screens that show them.

Action	Allowed roles	Refused with
Approve or reject a refund above the limit	Manager	403 FORBIDDEN
Mark a bank transfer as received or rejected	Receptionist, Manager	403 FORBIDDEN
Raise a payOS link for a plan order	Receptionist, Manager	403 FORBIDDEN
Raise a payOS link for a held court	Member (own hold only), Receptionist, Manager	403 FORBIDDEN
Write a register for a session	Coach (own classes only), Manager	403 FORBIDDEN
Correct a closed register	Manager, with a reason	422 BR-35
Read another member's bookings or receipts	Receptionist, Manager	403 FORBIDDEN

These requirements apply to every function, so the use cases do not repeat them.

Area	Requirement
Idempotency	Requests that take money or change a schedule accept an `Idempotency-Key`; a replay returns the first result.
Audit	Every staff-only change writes an audit row, and no endpoint deletes one.
Money	Amounts are whole VND, rounded once to the nearest 1,000đ after discounts. Payments are never deleted; a refund is a new negative payment.
Codes	Member, payment and invoice codes have no gaps within each kind.
Time	All times are stored with a time zone and shown in Asia/Ho_Chi_Minh time.
Sessions	Member sessions last seven days and staff sessions twelve hours.
Lockout	Five wrong passwords, or five wrong one-time codes, lock that path for 15 minutes.
Rate limits	Assistant questions: 800 characters each, 20 a minute. Plan orders: 3 a minute, 10 an hour.
Notifications	Notifications are queued and sent in batches of 50, with up to five attempts each. Plan reminders go 7, 3 and 0 days before the end date.
Background jobs	The jobs listed in 3.1.4 run in one loop every 15 seconds.
Language	Every screen is available in English and Vietnamese, and Vietnamese diacritics render correctly.
Layout	Every screen works at a width of 375 px.

Error Responses

Every API error uses one of these codes, so a screen can react to the code rather than parse the text.

HTTP	Code	Meaning
400	`VALIDATION`	A value is malformed; `field` names the input to correct
401	`UNAUTHENTICATED`	No session, or the session has expired
403	`FORBIDDEN`	Wrong role, or another member's data. With `br: "BR-52"`, the module is switched off
404	`NOT_FOUND`	The record does not exist
409	`CONFLICT_SLOT`	Another person took that court time first
409	`HOLD_EXPIRED`	The five-minute hold ran out
409	`CONFLICT_STATE`	The record is in the wrong state for this action
422	`BR_VIOLATION`	A named business rule refused the request; `br` carries the rule code
422	`VALIDATION`	The request is well formed but not acceptable in the current state
429	`RATE_LIMITED`	Too many attempts in a short time

Configurable Settings

Managers change these values on Centre settings (`/manager/settings`), and the rules above read them on every request.

Group	Setting	Default
Locale	Time zone; currency	Asia/Ho_Chi_Minh; VND (both fixed)
Opening hours	Open; close; slot length	06:00; 22:00; 60 minutes
Booking	Hold time; days ahead; slots per member per day	5 minutes; 7 days; 2
Cancellation	Court booking notice; class notice	2 hours; 4 hours
Court refunds by notice	24 hours or more; 4 to 24 hours; under 4 hours	100%; 50%; 0%
Check-in	Court check-in window; no-show grace	15 minutes before; 10 minutes after
Deposit	Deposit share; peak slots only	30%; yes
Fixed weekly bookings	Length; discount	4 to 12 weeks; 10%
Money	Refund limit without a manager; VAT rate; rounding	1,000,000đ; 0%; nearest 1,000đ
Members	Freeze days a year; minor age; waitlist offer	30 days; under 16; 2 hours
Transfers	Bank transfer hold	120 minutes
Loyalty and day pass	Spend per point; value per point; day pass price	10,000đ; 100đ; 60,000đ
Gate	Self check-in; repeat-scan window	Off; 120 minutes

5.3 Application Messages List

Each message below is the text the application shows today. The Context/Trigger column names the screen and the action, then the v2.2 code and the business rule it reports.

Message ID	Message Text	Message Type	Context/Trigger
ERR_AUTH_01	That phone or email already has an account.	Error	Create an account (submit) (was MSG01; BR-01)
ERR_AUTH_02	Password needs at least 8 characters, with letters and numbers.	Validation	Create an account (submit) (was MSG02; BR-02)
ERR_AUTH_03	Too many OTP attempts — locked for 15 minutes.	Error	Create an account (verify the code) (was MSG03)
ERR_AUTH_04	5 wrong passwords — locked for 15 minutes.	Error	Sign in (sign in) (was MSG04)
ERR_AUTH_05	You do not have permission.	Error	Any screen (an action outside the role) (was MSG19)
ERR_AUTH_06	Too many attempts — try again later.	Error	AI assistant (send a question) (was MSG20)
ERR_BOOK_01	You can book at most 7 days ahead.	Error	Book a court (hold a slot) (was MSG05; BR-05)
WARN_BOOK_01	This slot clashes with a class you are in. Confirm to hold it anyway.	Warning	Book a court (hold a slot) (was MSG06; BR-09)
ERR_BOOK_02	Someone just took that slot.	Error	Book a court (hold a slot) (was MSG07)
ERR_BOOK_03	Your hold has expired.	Error	Book a court (pay) (was MSG08)
ERR_BOOK_04	There is no price for that slot yet — ask the desk.	Error	Book a court (hold a slot) (was MSG09; BR-37)
ERR_PAY_01	That code has expired.	Error	Book a court (apply a promo code) (was MSG10; BR-40)
ERR_PAY_02	Open a till shift before taking payment.	Error	Member profile (take a payment) (was MSG13; BR-44)
ERR_PAY_03	This payment has already been refunded in full.	Validation	Member profile (raise a refund) (was MSG15)
ERR_PAY_04	You raised this refund, so someone else has to sign it off.	Error	Payments (approve a refund) (was MSG16)
ERR_PAY_05	That refund is no longer waiting for a decision.	Error	Payments (approve or reject a refund) (was MSG17)
ERR_CLASS_01	You need an active plan covering this sport.	Error	Classes (enrol) (was MSG11; BR-19)
ERR_CLASS_02	That offer has expired.	Error	Classes (accept a waitlist offer) (was MSG12; BR-23)
ERR_CHECKIN_01	That code was already used — ask the member to refresh it.	Error	Gate (scan a pass) (was MSG14; BR-30)

Message ID	Message Text	Message Type	Context/Trigger
ERR_COACH_01	This register closed 2 hours after the session. Ask a manager to correct it.	Error	Attendance (save the register) (was MSG18; BR-35)
ERR_GEAR_01	Only {stock} {item} left — you asked for {quantity}.	Error	Gear (rent out gear) (was MSG21; BR-50)

II. Software Design Description

1. System Design

1.1 System Architecture

Figure 26 shows how Arena3 is deployed. The browser runs the five portals as one React client. One TanStack Start server renders the pages and serves the JSON API under /v1; the API router dispatches each request to a feature handler that applies the business rules. The handlers and the background job loop read and write PostgreSQL (Neon in production, embedded PGLite locally). The server calls payOS for payment links and receives its webhook, asks Gemini AI for assistant answers, and sends mail through Mail delivery (SMTP).

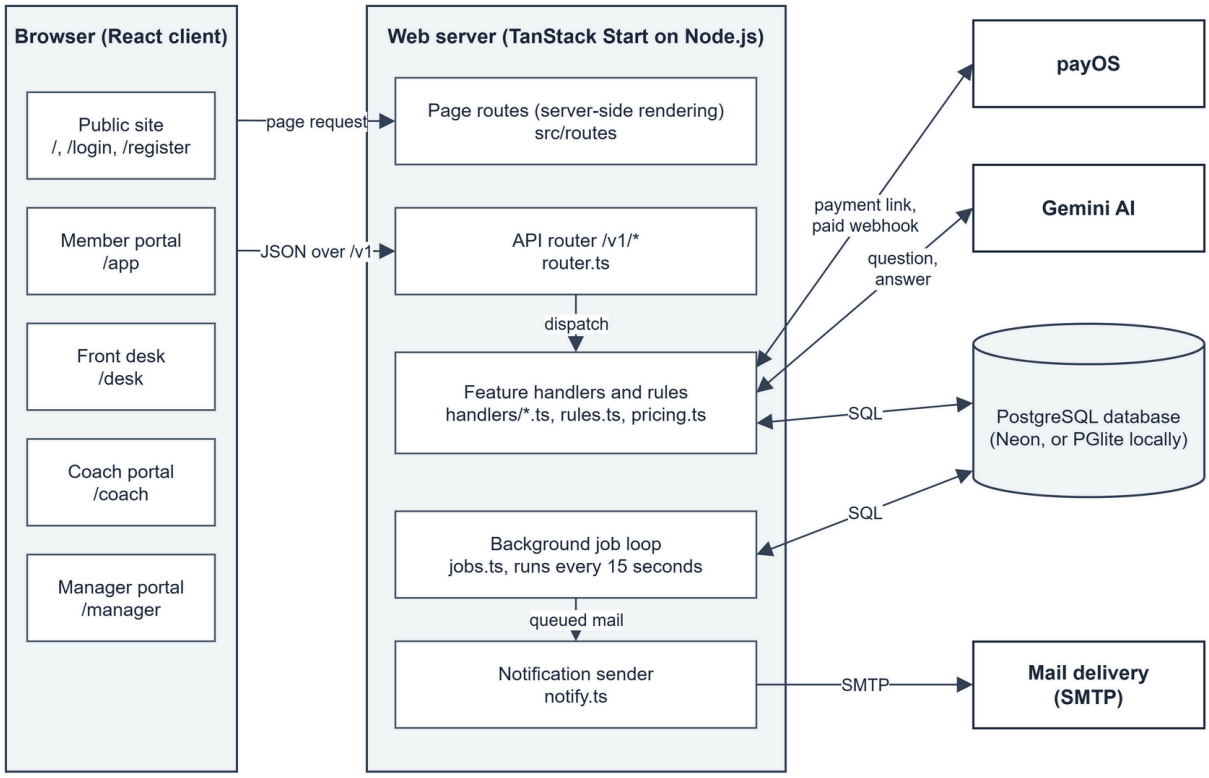


Figure 26. *System architecture*

1.2 Package Diagram

Figure 27 shows the packages of the source tree and the dependencies between them.

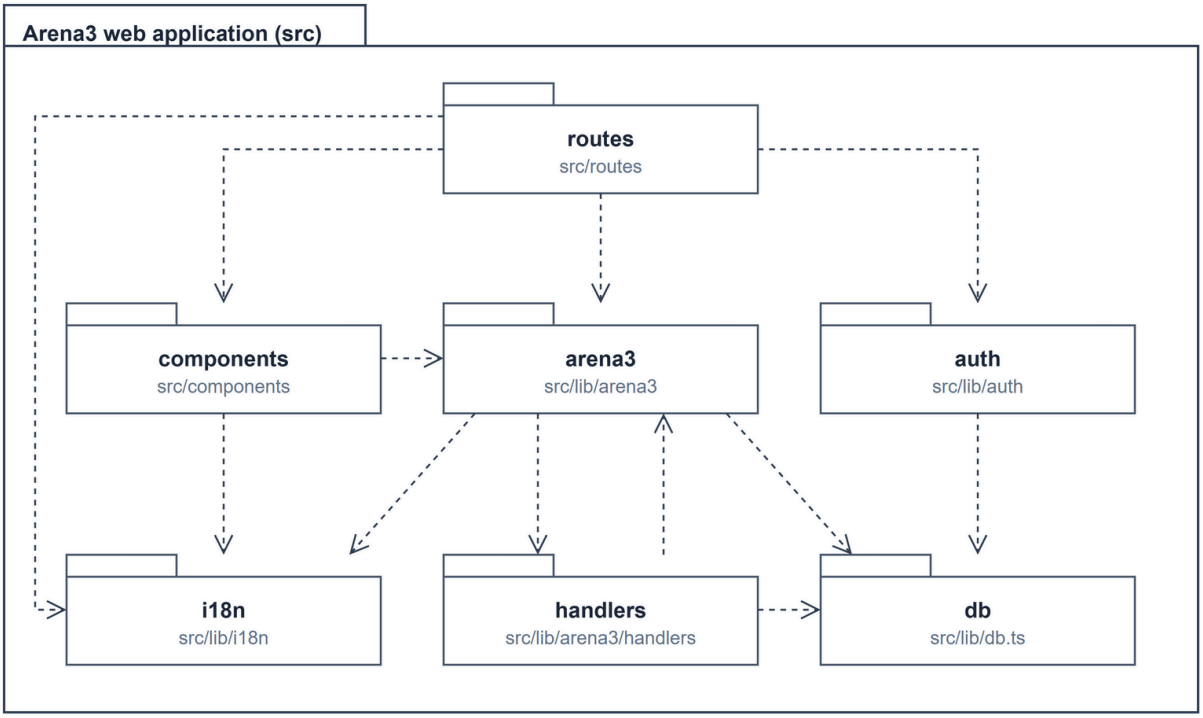


Figure 27. *Package diagram*

Package Descriptions

No	Package Name	Description
1	routes (src/routes)	File-based page routes for the member, desk, coach and manager portals, and the /v1 API entry point
2	components (src/components)	Shared React components: shell, court grid, dialogs, panels and the UI kit
3	auth (src/lib/auth)	Sign-in gate, session cookie handling and sign-out helpers
4	arena3 (src/lib/arena3)	Core domain: API router, business rules, pricing, jobs, notifications, payOS and Gemini clients
5	i18n (src/lib/i18n)	Vietnamese and English text dictionaries and the language provider
6	handlers (src/lib/arena3/handlers)	One request handler module per feature: bookings, classes, desk, plans, reports and more
7	db (src/lib/db.ts)	Database access: Neon PostgreSQL, or embedded PGlite when no DATABASE_URL is set

2. Database Design

The database has 49 tables, created by the numbered migrations in migrations/*.sql. The logical ERD in 3.1.5.2 draws them; the table below describes each one with its keys.

Table Descriptions

#	Entity	Description
1	Attendance	Store one member's mark for one session (table `attendance`). - Primary Key: id - Foreign Key: user_id → users, session_id → sessions
2	AuditLog	Store every audited change with its actor, action, record and the values before and after (table `audit_logs`). - Primary Key: id - Foreign Key: None
3	BookingSeries	Store fixed weekly bookings of 4 to 12 weeks for one court, weekday and time (table `booking_series`). - Primary Key: id - Foreign Key: user_id → users, court_id → courts, created_by → users
4	CashierShift	Store one receptionist's till session, from opening float to counted close (table `cashier_shifts`). - Primary Key: id - Foreign Key: receptionist_id → users
5	CenterSettings	Store the centre's single row of settings: hours, limits, windows, money and loyalty values (table `center_settings`). - Primary Key: id - Foreign Key: None
6	CheckinToken	Store the short-lived signed codes behind each gate pass and when each was used (table `checkin_tokens`). - Primary Key: jti - Foreign Key: user_id → users
7	TrainingClass	Store a coached class with a coach, a court, a weekly timetable and a capacity (table `classes`). - Primary Key: id - Foreign Key: coach_id → users, assistant_id → users, court_id → courts
8	CoachNote	Store a coach's private notes about a student (table `coach_notes`). - Primary Key: id - Foreign Key: user_id → users, coach_id → users
9	CoachOccupancy	Store the times a coach is busy with a session, so no coach is booked twice (table `coach_occupancies`). - Primary Key: id - Foreign Key: coach_id → users
10	CoachPayout	Store each coach's closed monthly commission and whether it is paid (table `coach_payouts`). - Primary Key: id - Foreign Key: coach_id → users, closed_by → users

#	Entity	Description
11	CoachRate	Store each coach's fee per session and per student present (table `coach_rates`). - Primary Key: coach_id - Foreign Key: coach_id → users
12	CoachSport	Store the sports each coach is qualified to teach (table `coach_sports`). - Primary Key: user_id, sport - Foreign Key: user_id → users
13	CodeCounter	Store the last number used for each kind of code and year, so codes have no gaps (table `code_counters`). - Primary Key: kind, yyyy - Foreign Key: None
14	ContactLog	Store follow-up contacts made with members at risk of leaving (table `contact_log`). - Primary Key: id - Foreign Key: user_id → users, created_by → users
15	CourtBooking	Store a court-time owned by a member or a walk-in, from hold to completed (table `court_bookings`). - Primary Key: id - Foreign Key: court_id → courts, user_id → users, occupancy_id → occupancies, promo_id → promotions, series_id → booking_series
16	Court	Store a playable court for one sport, with a status of ready, maintenance or closed (table `courts`). - Primary Key: id - Foreign Key: pair_court_id → courts
17	DayPass	Store day passes sold for entry on one day, with their payment and use (table `day_passes`). - Primary Key: id - Foreign Key: user_id → users, payment_id → payments, issued_by → users
18	Enrollment	Store each member's place in a class, confirmed or on the waitlist (table `enrollments`). - Primary Key: id - Foreign Key: class_id → classes, user_id → users
19	EquipmentItem	Store rentable gear with stock on hand (table `equipment_items`). - Primary Key: id - Foreign Key: None
20	EquipmentLoan	Store gear lent out, with its quantity, due time and return (table `equipment_loans`). - Primary Key: id - Foreign Key: item_id → equipment_items, booking_id → court_bookings
21	FeatureFlag	Store the on or off switch of each optional module (table `feature_flags`). - Primary Key: key - Foreign Key: None
22	Homework	Store homework set by a coach for a class or one student (table `homework`). - Primary Key: id - Foreign Key: coach_id → users, class_id → classes, user_id → users

#	Entity	Description
23	HomeworkRecipient	Store each student's progress on a homework checklist (table `homework_recipients`). - Primary Key: homework_id, user_id - Foreign Key: homework_id → homework, user_id → users
24	IdempotencyKey	Store the first response to each request sent with an Idempotency-Key, so a replay returns it (table `idempotency_keys`). - Primary Key: key - Foreign Key: None
25	InvoiceLine	Store the lines of each e-invoice (table `invoice_lines`). - Primary Key: id - Foreign Key: invoice_id → invoices
26	Invoice	Store the receipt issued for a payment, with a gapless code (table `invoices`). - Primary Key: id - Foreign Key: payment_id → payments
27	LoyaltyAdjustment	Store changes to a member's loyalty points, such as points redeemed for a code (table `loyalty_adjustments`). - Primary Key: id - Foreign Key: user_id → users, promo_id → promotions, created_by → users
28	MemberLevel	Store a member's skill level in each sport, set by a coach (table `member_levels`). - Primary Key: user_id, sport - Foreign Key: user_id → users, assessed_by → users
29	MembershipPlan	Store a plan on sale: price, duration, court hours or sessions, sport scope (table `membership_plans`). - Primary Key: id - Foreign Key: None
30	Occupancy	Store each court time and its single owner: a booking, a class session or a maintenance block (table `occupancies`). - Primary Key: id - Foreign Key: court_id → courts
31	OtpChallenge	Store hashed one-time codes for sign-up and password reset, with their expiry and attempts (table `otp_challenges`). - Primary Key: id - Foreign Key: None
32	Notification	Store queued and in-app notifications with their delivery attempts and read time (table `outbox`). - Primary Key: id - Foreign Key: user_id → users
33	Payment	Store money in or out, with method, amount, VAT, shift and capture mode (table `payments`). - Primary Key: id - Foreign Key: user_id → users, shift_id → cashier_shifts, created_by → users, promo_id → promotions
34	PriceRule	Store price per hour by sport, court class, day type and hour band (table `price_rules`). - Primary Key: id - Foreign Key: court_id → courts

#	Entity	Description
35	ProgressReview	Store an append-only review scoring technique, fitness and attitude (table `progress_reviews`). - Primary Key: id - Foreign Key: user_id → users, coach_id → users
36	PromotionRedemption	Store each use of a promo code against a payment, and its return after a full refund (table `promotion_redemptions`). - Primary Key: id - Foreign Key: promo_id → promotions, user_id → users, payment_id → payments
37	Promotion	Store a promo code with its discount, validity and usage limits (table `promotions`). - Primary Key: id - Foreign Key: plan_id → membership_plans, created_by → users
38	ProviderSequence	Store the last order code sent to each payment provider (table `provider_sequences`). - Primary Key: provider - Foreign Key: None
39	SessionResult	Store each student's result for a session (table `session_results`). - Primary Key: id - Foreign Key: session_id → sessions, user_id → users, recorded_by → users
40	ClassSession	Store one dated meeting of a class (table `sessions`). - Primary Key: id - Foreign Key: class_id → classes, court_id → courts, occupancy_id → occupancies
41	AuthSession	Store hashed session tokens of signed-in users and their expiry (table `sessions_auth`). - Primary Key: id - Foreign Key: user_id → users
42	Subscription	Store a member's copy of a plan, with start and end dates and balances (table `subscriptions`). - Primary Key: id - Foreign Key: user_id → users, plan_id → membership_plans, promo_id → promotions
43	SupportTicket	Store members' support requests and the desk's replies (table `tickets`). - Primary Key: id - Foreign Key: user_id → users, replied_by → users
44	TrainingPlanVersion	Store each saved version of a training plan (table `training_plan_versions`). - Primary Key: id - Foreign Key: plan_id → training_plans, edited_by → users
45	TrainingPlan	Store up to 12 timed blocks for a class, a session or one student (table `training_plans`). - Primary Key: id - Foreign Key: class_id → classes, user_id → users, session_id → sessions, created_by → users, parent_id → training_plans

#	Entity	Description
46	TrainingProfile	Store each member's personal training goal (table `training_profiles`). - Primary Key: user_id - Foreign Key: user_id → users
47	User	Store every person with an account, with one role: member, receptionist, coach or manager (table `users`). - Primary Key: id - Foreign Key: None
48	WaitlistOffer	Store seat offers made to waitlisted members and their expiry (table `waitlist_offers`). - Primary Key: id - Foreign Key: enrollment_id → enrollments
49	WorkOrder	Store a maintenance job that can block a court while it is open (table `work_orders`). - Primary Key: id - Foreign Key: court_id → courts, reported_by → users

3. Detailed Design

3.1 Booking and Payment

3.1.1 Class Diagram

Figure 28 and Figure 29 refine the ERD for this subsystem, with attributes, operations, enumerations and multiplicities: Figure 28 draws courts, price rules, occupancies, bookings, series and day passes, and Figure 29 draws payments, shifts, promotions, redemptions and invoices. Together they cover Court, PriceRule, Occupancy, CourtBooking, BookingSeries, DayPass, Payment, CashierShift, Promotion, PromotionRedemption, Invoice and InvoiceLine. CourtBooking, DayPass and Subscription realise the Payable interface.

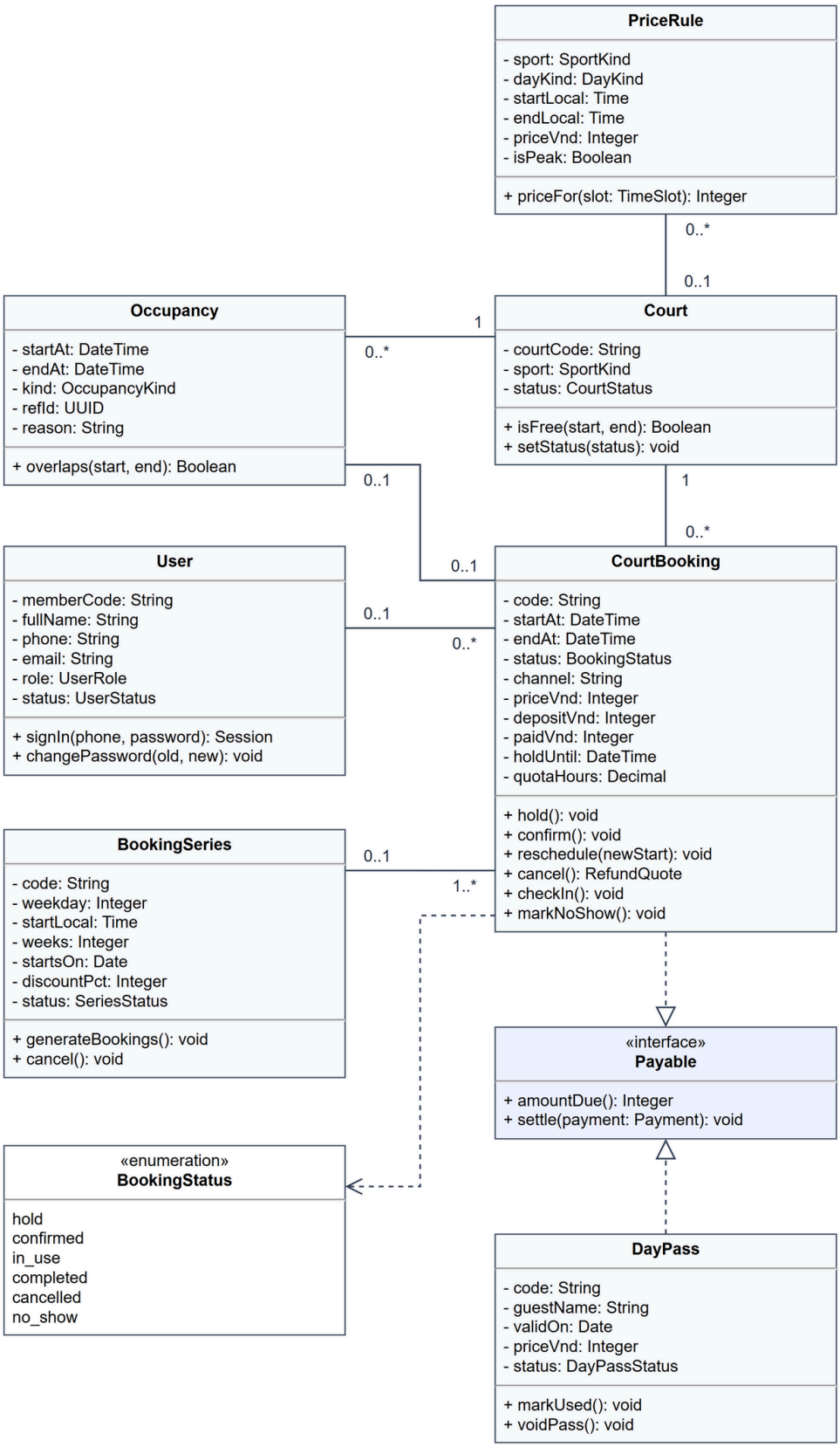


Figure 28. *Class diagram: courts and bookings*

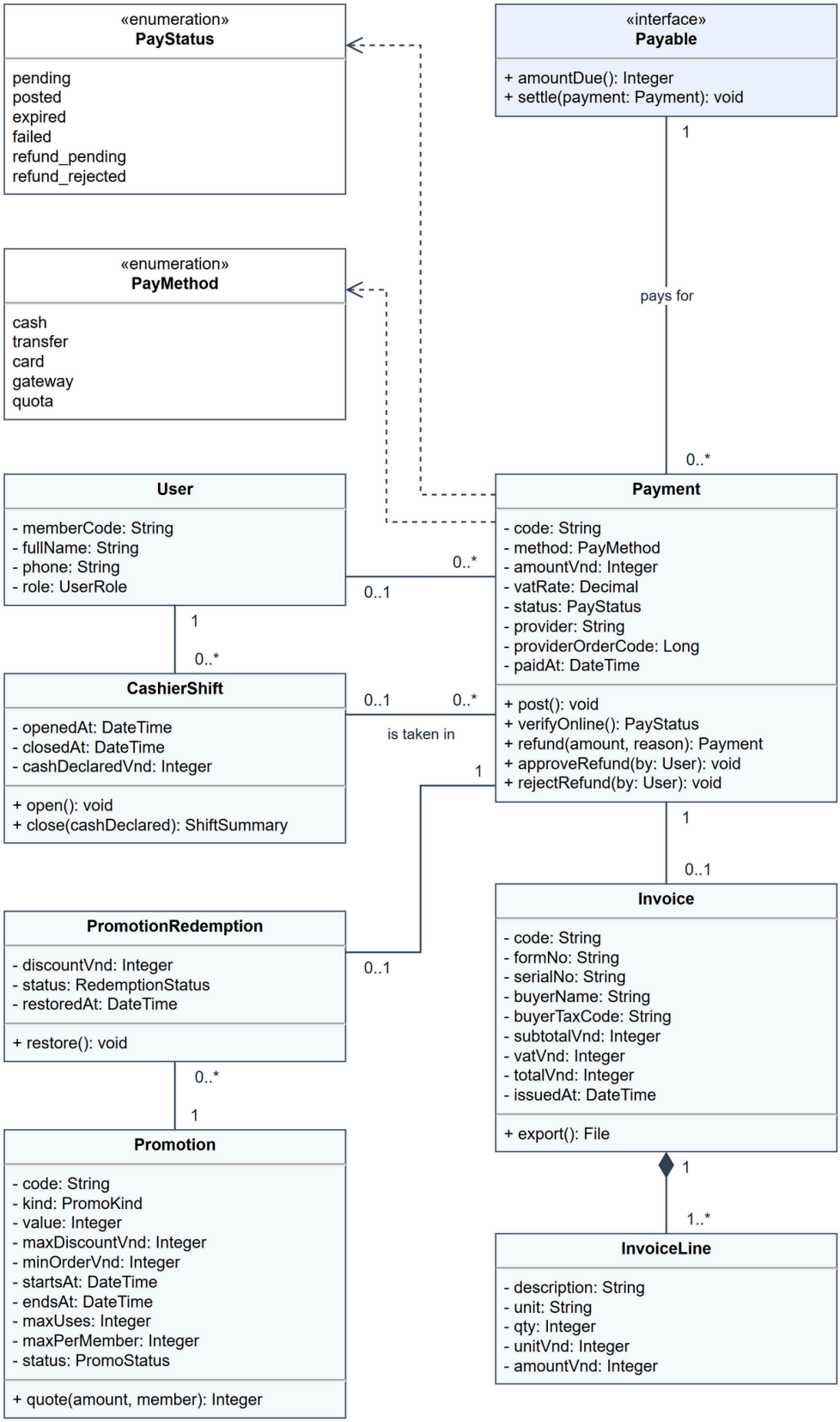


Figure 29. *Class diagram: payments*

3.1.2 Sequence Diagram

Figure 30 shows how a member books a court and pays by payOS QR, with the webhook and the polling check running side by side.

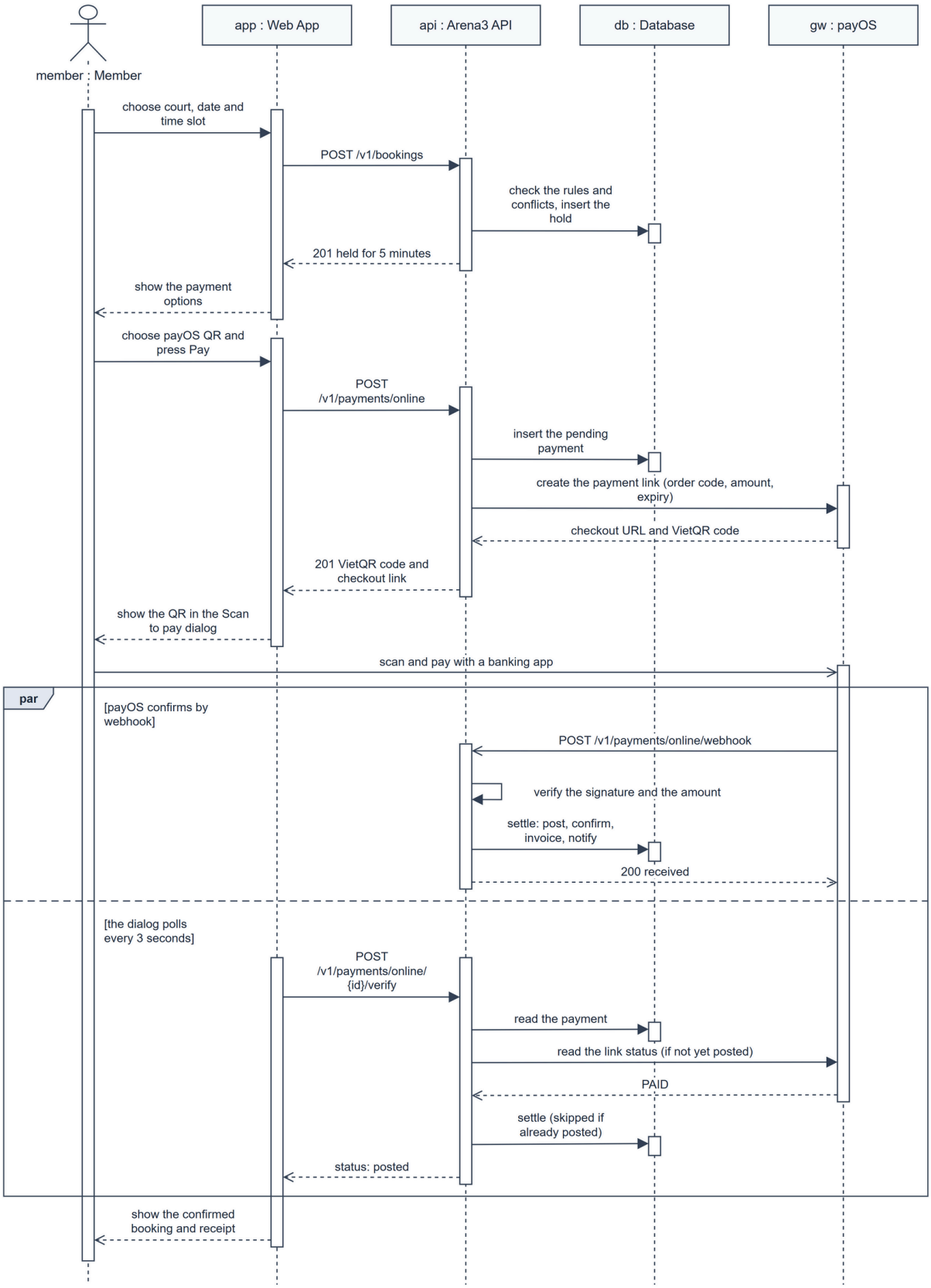


Figure 30. *Sequence diagram: book a court and pay by payOS QR*

Figure 31 shows how a refund is raised at the desk and, above the limit, approved by another manager.

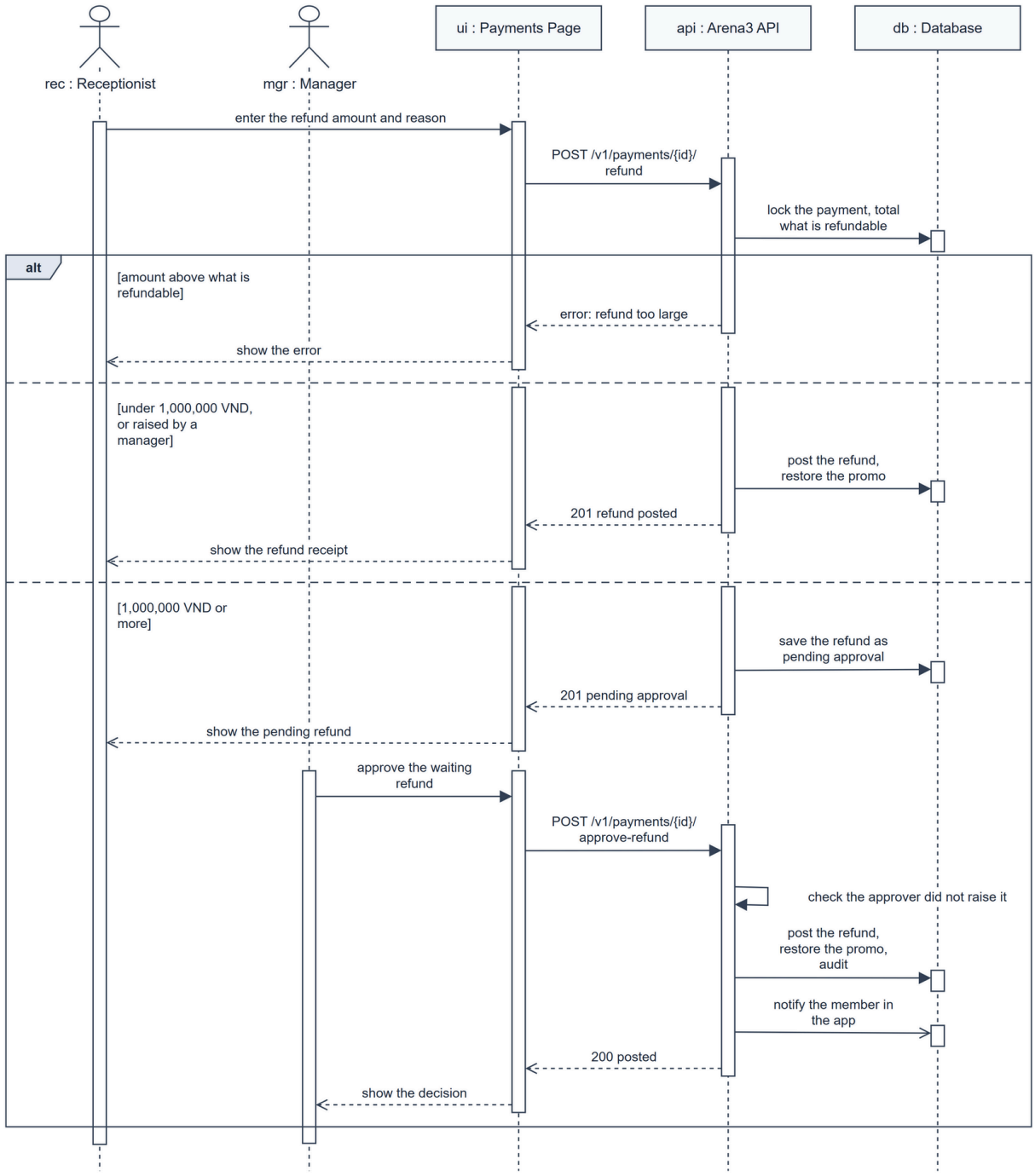


Figure 31. *Sequence diagram: raise and approve a refund*

3.2 Membership and Classes

3.2.1 Class Diagram

Figure 32 and Figure 33 refine the ERD for this subsystem, with attributes, operations, enumerations and multiplicities: Figure 32 draws membership plans and subscriptions, and Figure 33 draws classes, sessions, enrolments, waitlist offers and attendance. Together they cover MembershipPlan, Subscription, TrainingClass, ClassSession, Enrollment, WaitlistOffer and Attendance, with their status enumerations.

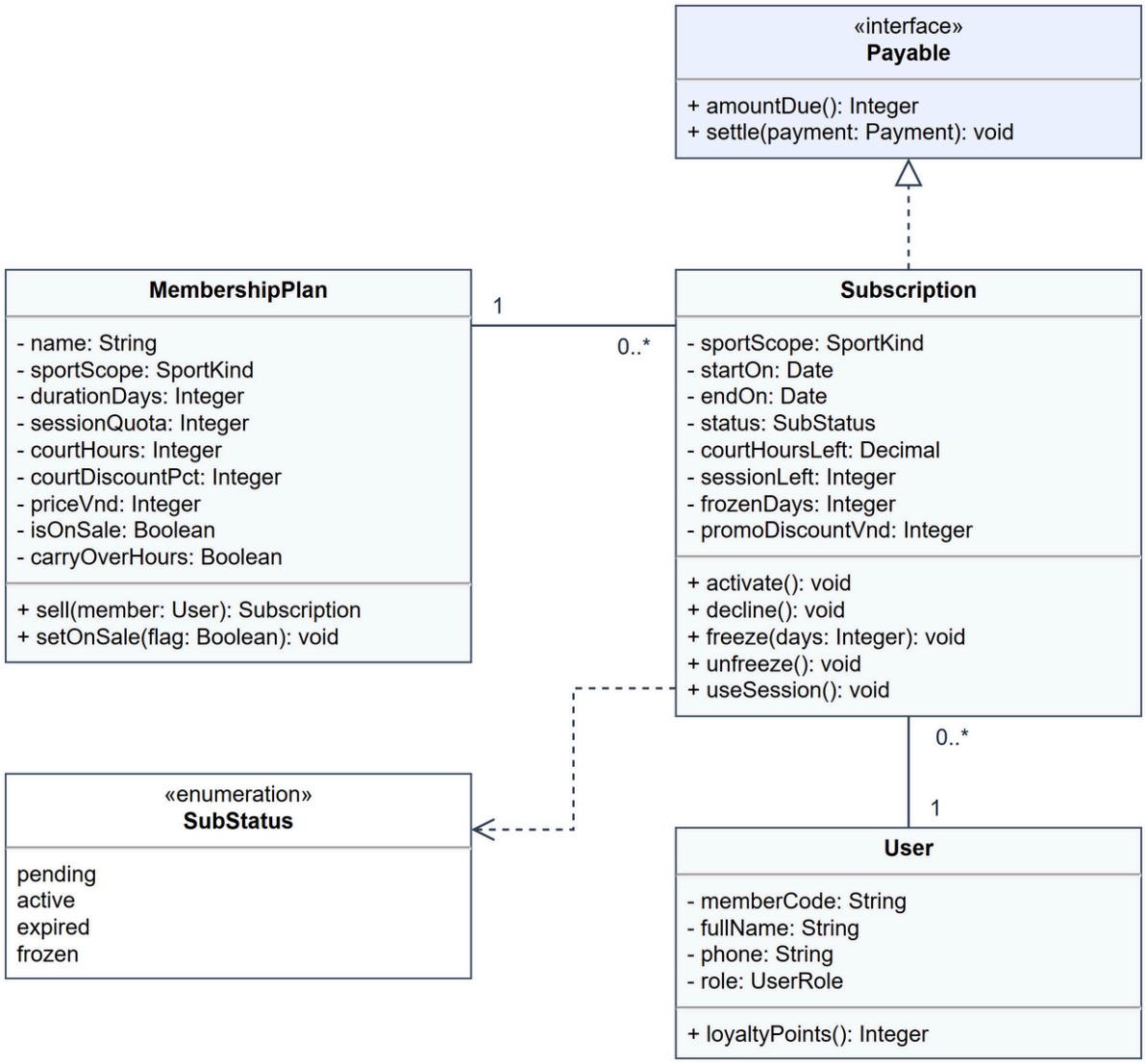


Figure 32. *Class diagram: membership plans*

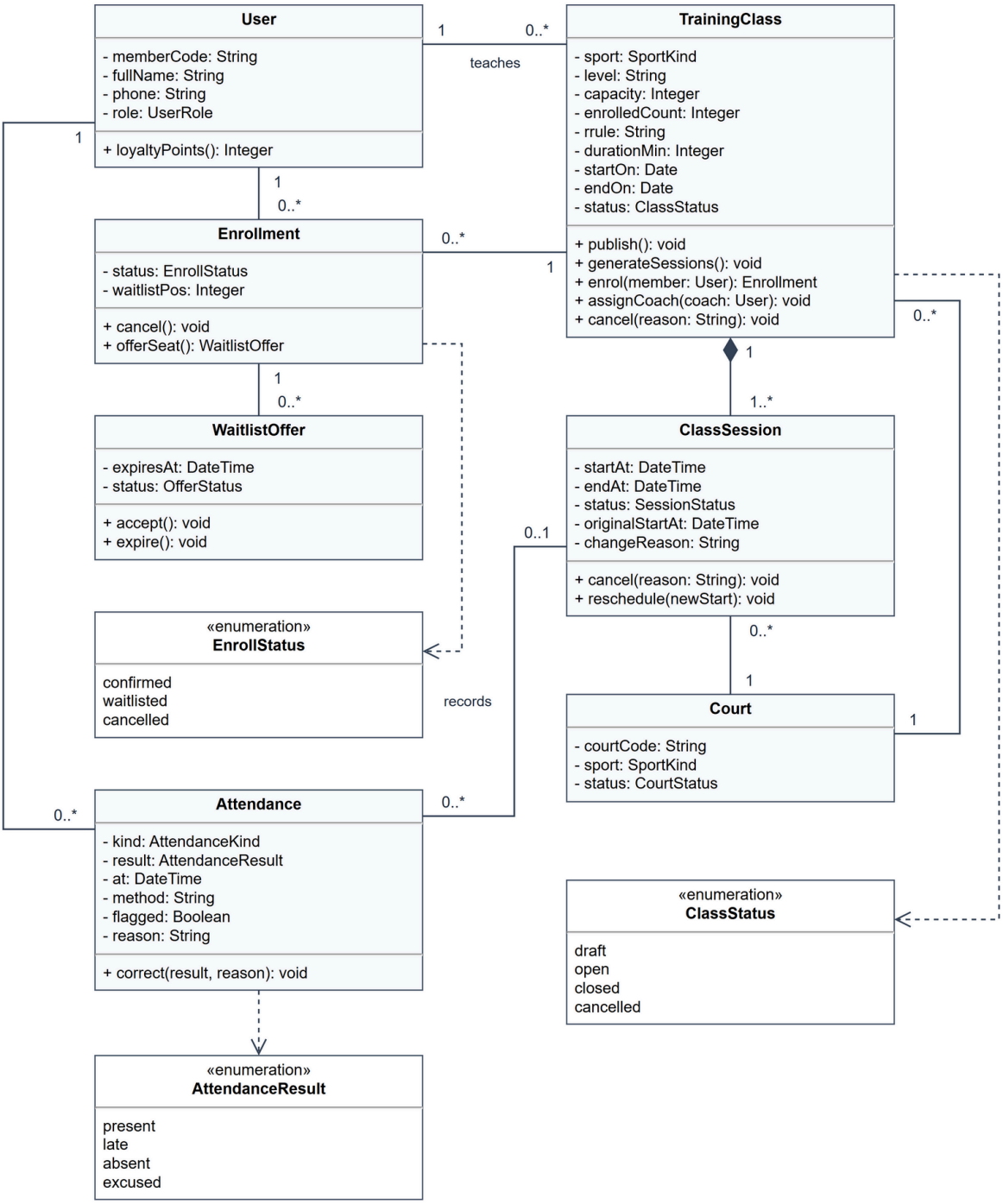


Figure 33. *Class diagram: classes and attendance*

3.2.2 Sequence Diagram

Figure 34 shows how a member enrols in a class, including the waitlist branch when the class is full.

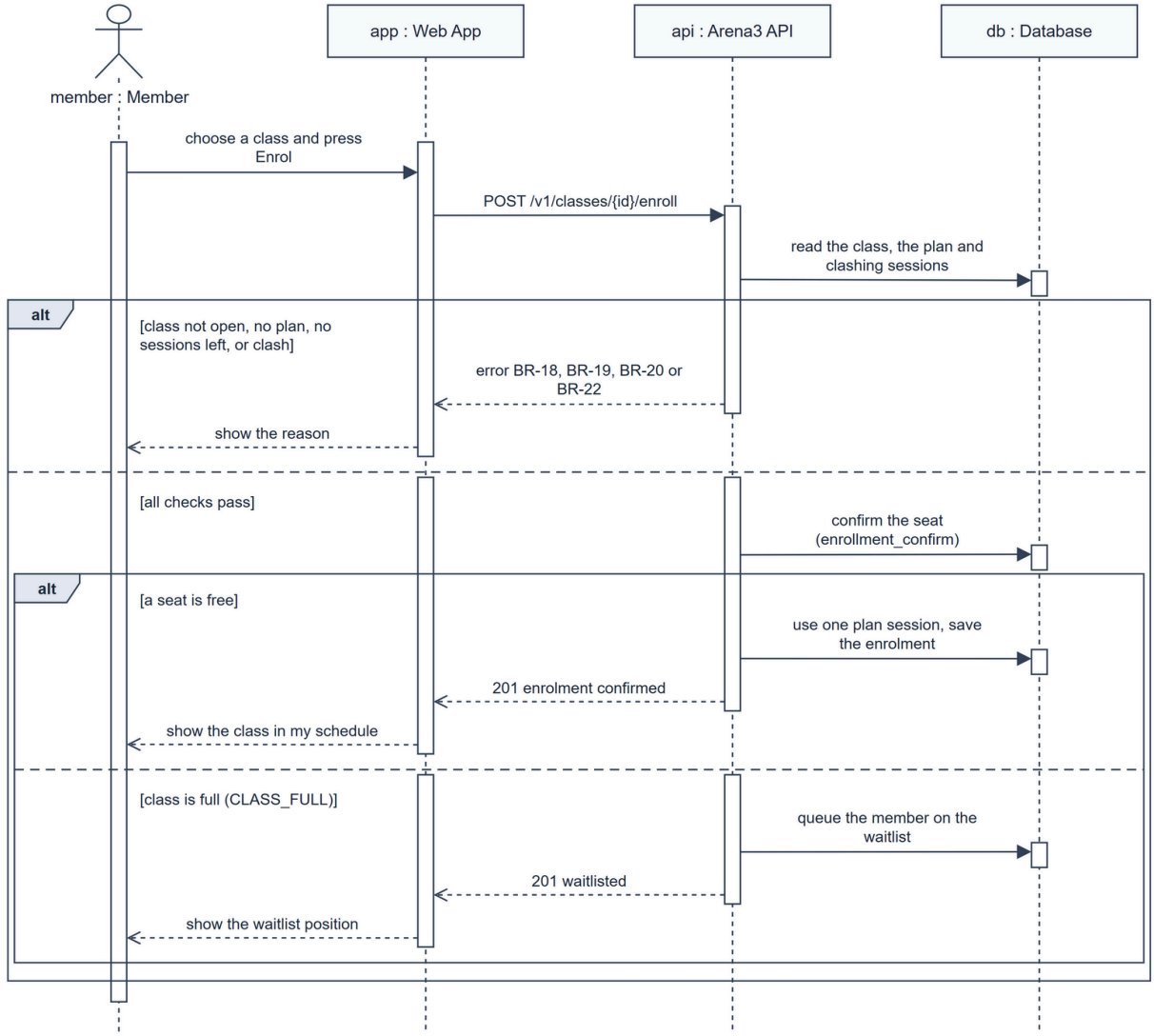


Figure 34. *Sequence diagram: enrol in a class*

3.3 Coaching and Training

3.3.1 Class Diagram

Figure 35 and Figure 36 refine the ERD for this subsystem, with attributes, operations, enumerations and multiplicities: Figure 35 draws the coach and member roles with notes, reviews, levels and results, and Figure 36 draws coach pay, training plans and homework. Together they specialise User into Coach and Member. They cover CoachRate, CoachPayout, MemberLevel, SessionResult, TrainingPlan, TrainingPlanVersion, Homework, HomeworkRecipient, ProgressReview and CoachNote.

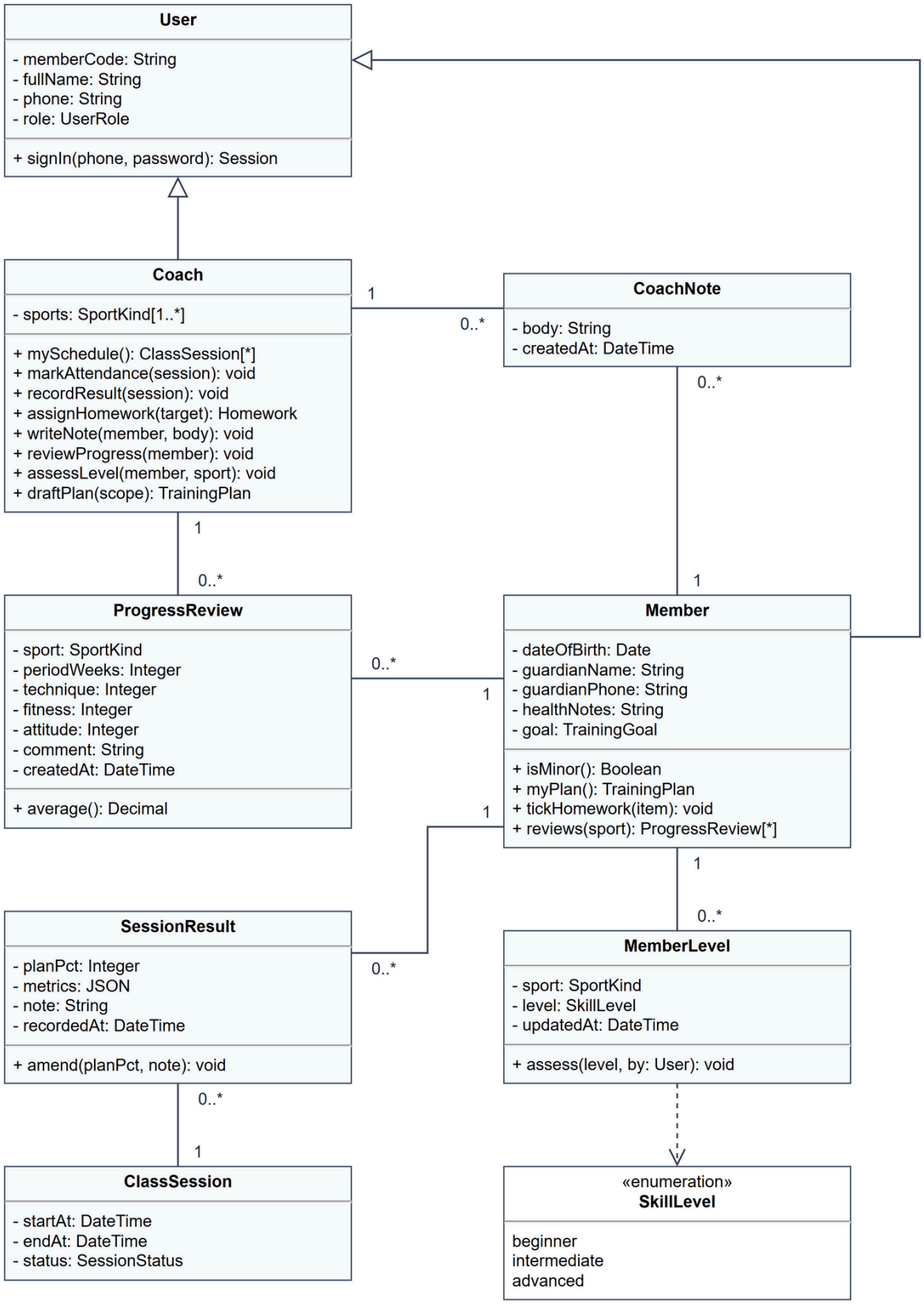


Figure 35. *Class diagram: coach and member roles*

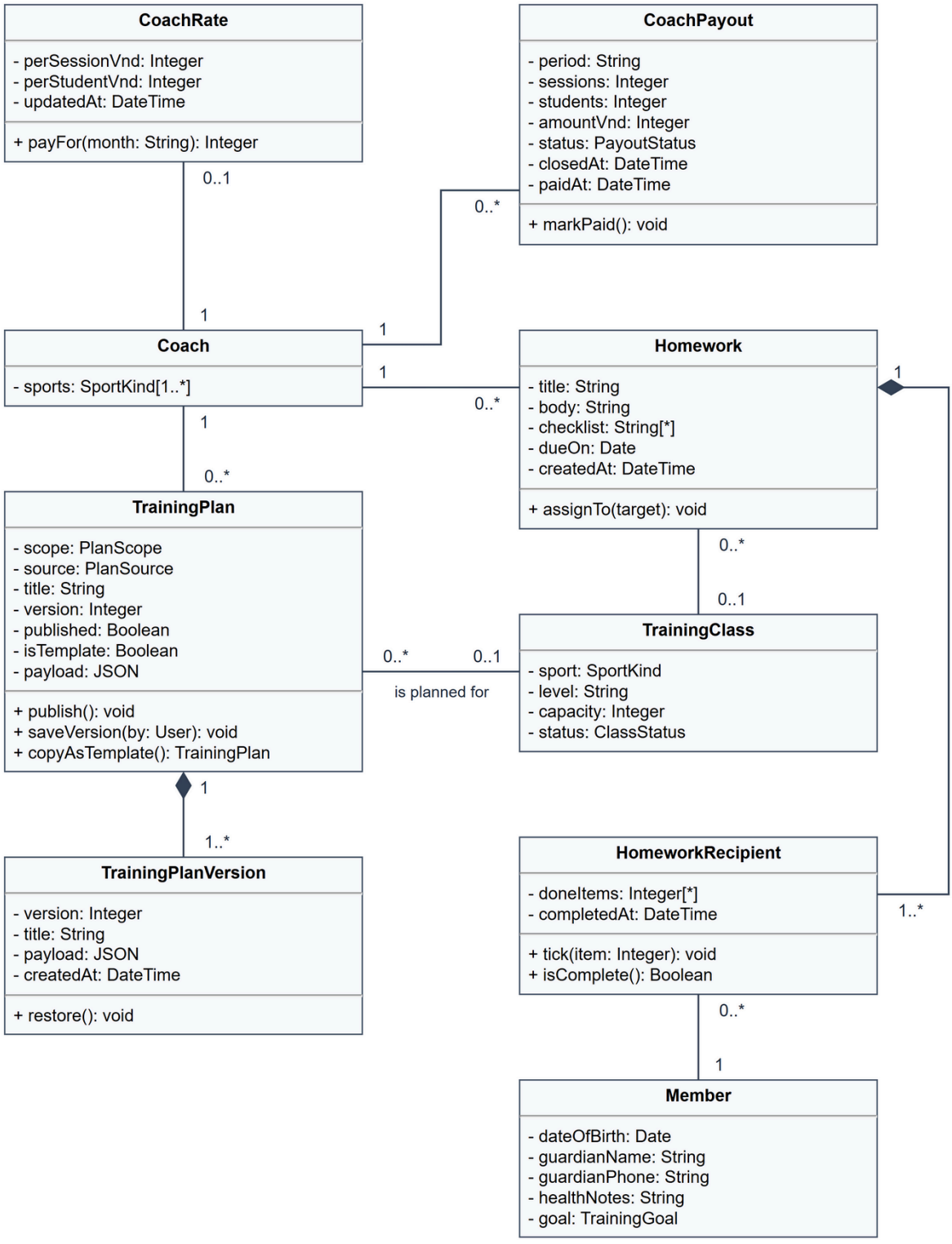


Figure 36. *Class diagram: coach pay, plans and homework*

3.3.2 Sequence Diagram

Figure 37 shows how a coach takes a session register and what happens once it is locked.

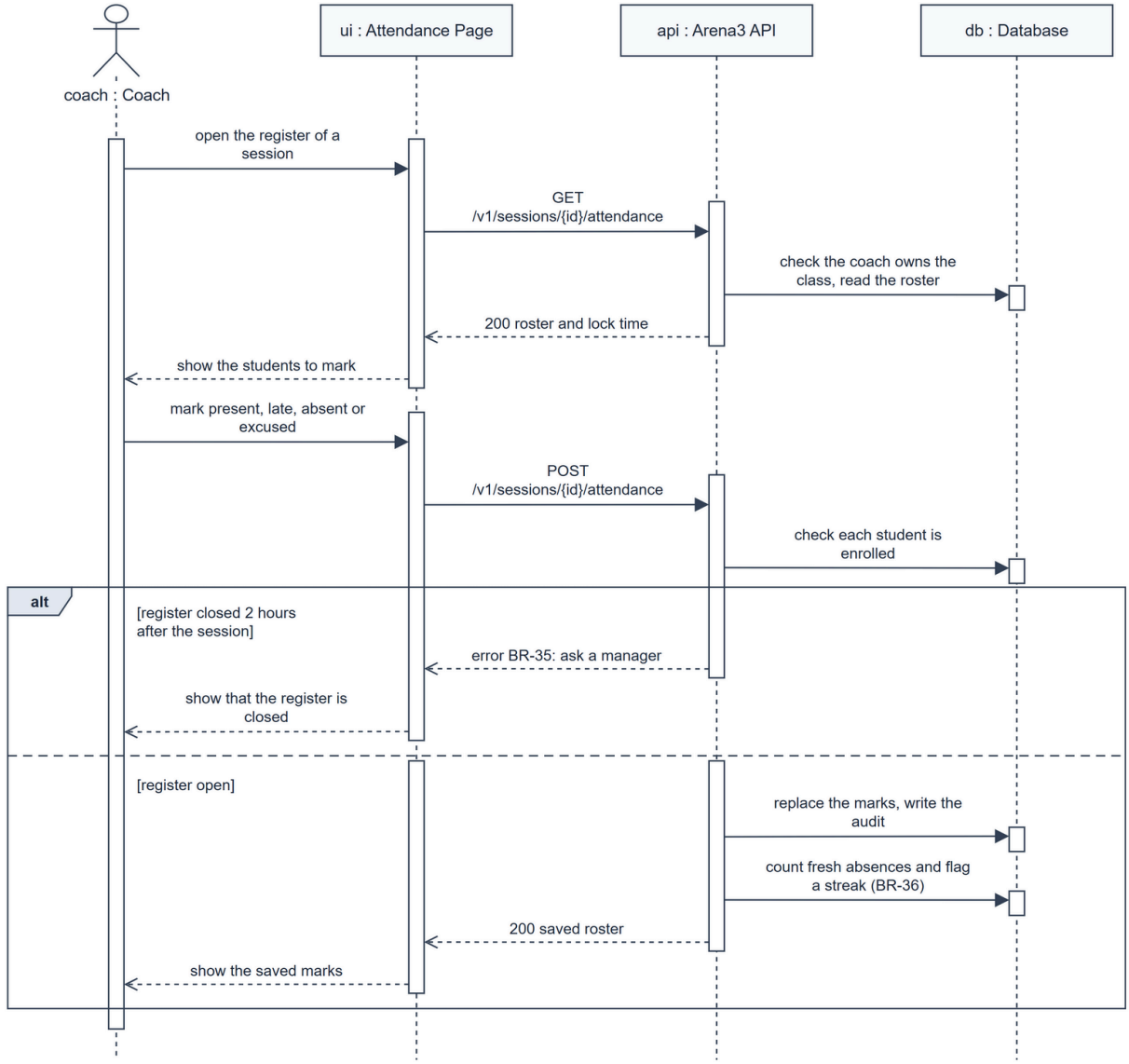


Figure 37. *Sequence diagram: take session attendance*

3.4 Centre Operations

3.4.1 Class Diagram

Figure 38 and Figure 39 refine the ERD for this subsystem, with attributes, operations, enumerations and multiplicities: Figure 38 draws accounts, gate tokens, notifications, support, loyalty and the audit log, and Figure 39 draws settings, feature flags, courts, work orders and gear. Together they cover CenterSettings, FeatureFlag, OtpChallenge, CheckinToken, Notification, SupportTicket, ContactLog, LoyaltyAdjustment, AuditLog, WorkOrder, EquipmentItem and EquipmentLoan.

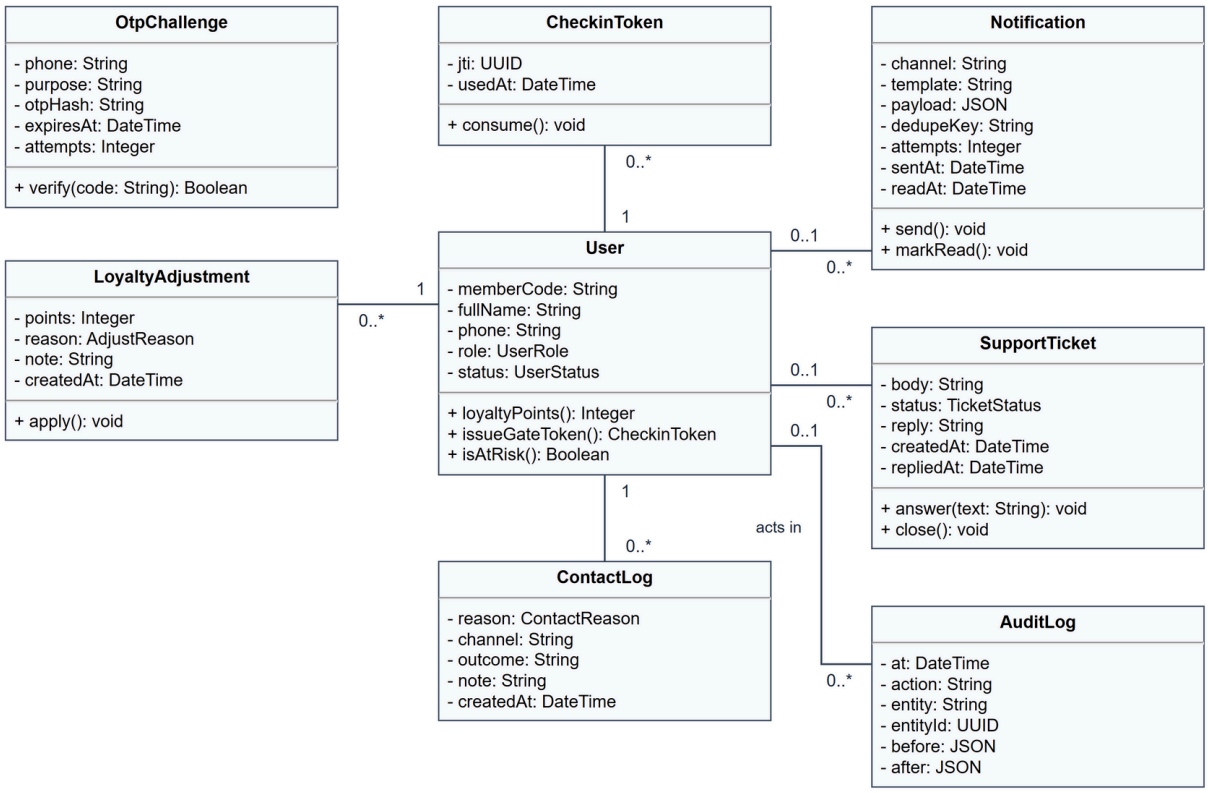


Figure 38. *Class diagram: accounts and member care*

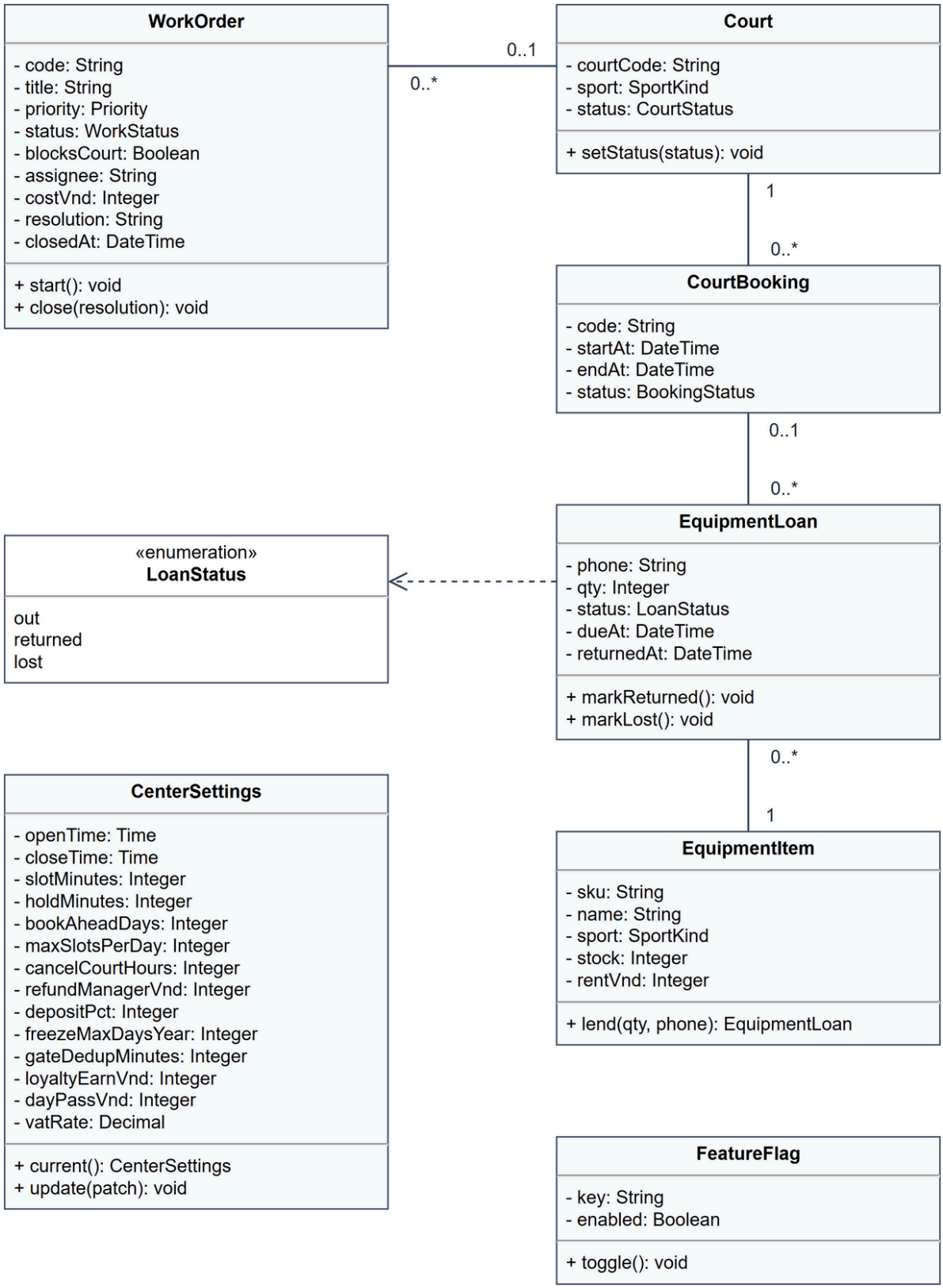


Figure 39. *Class diagram: centre and equipment*

3.4.2 Sequence Diagram

Figure 40 shows how a guest registers and proves the email address with a one-time code.

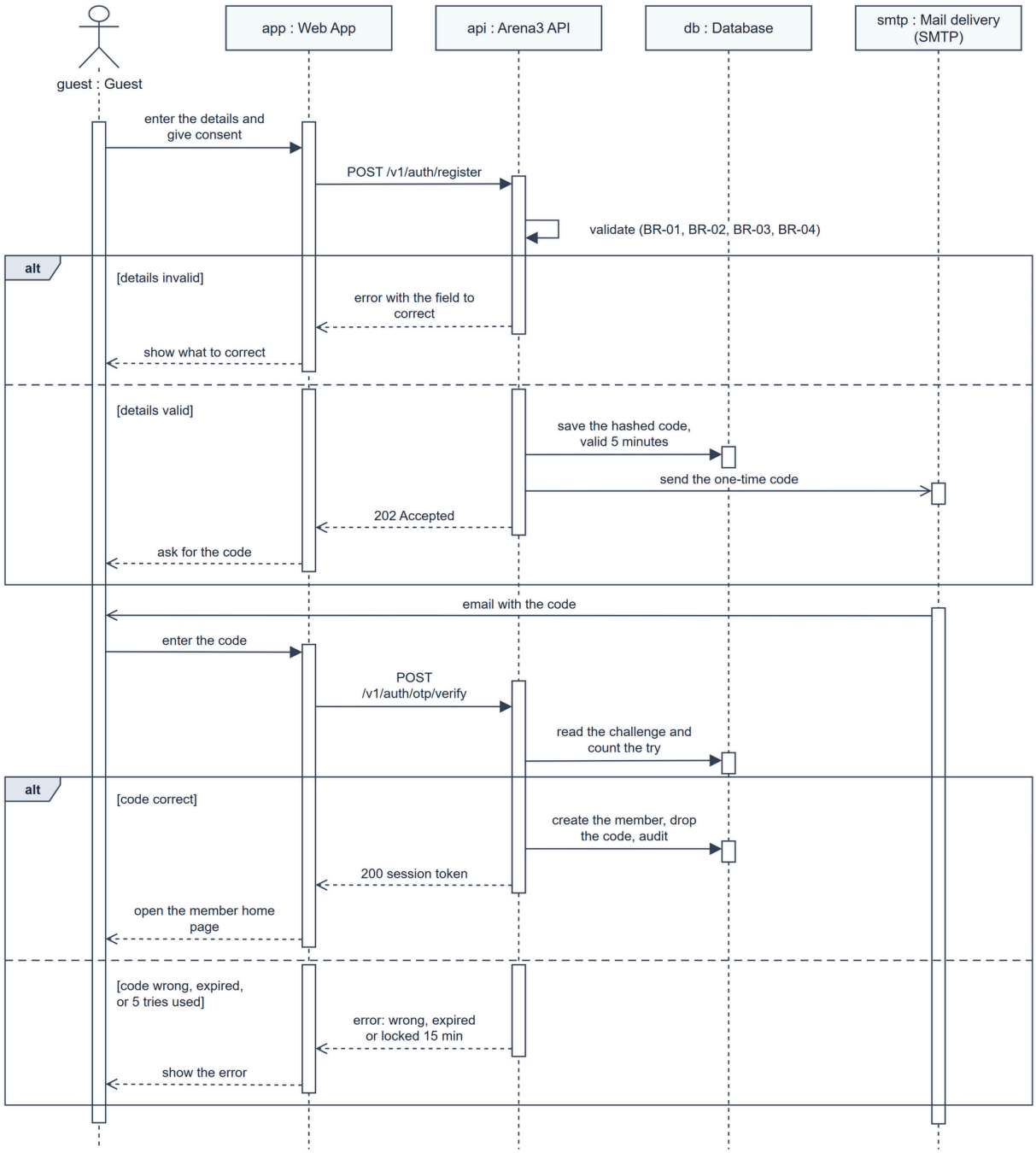


Figure 40. *Sequence diagram: register a member account*

Figure 41 shows how a member is checked in at the gate, with the override path.

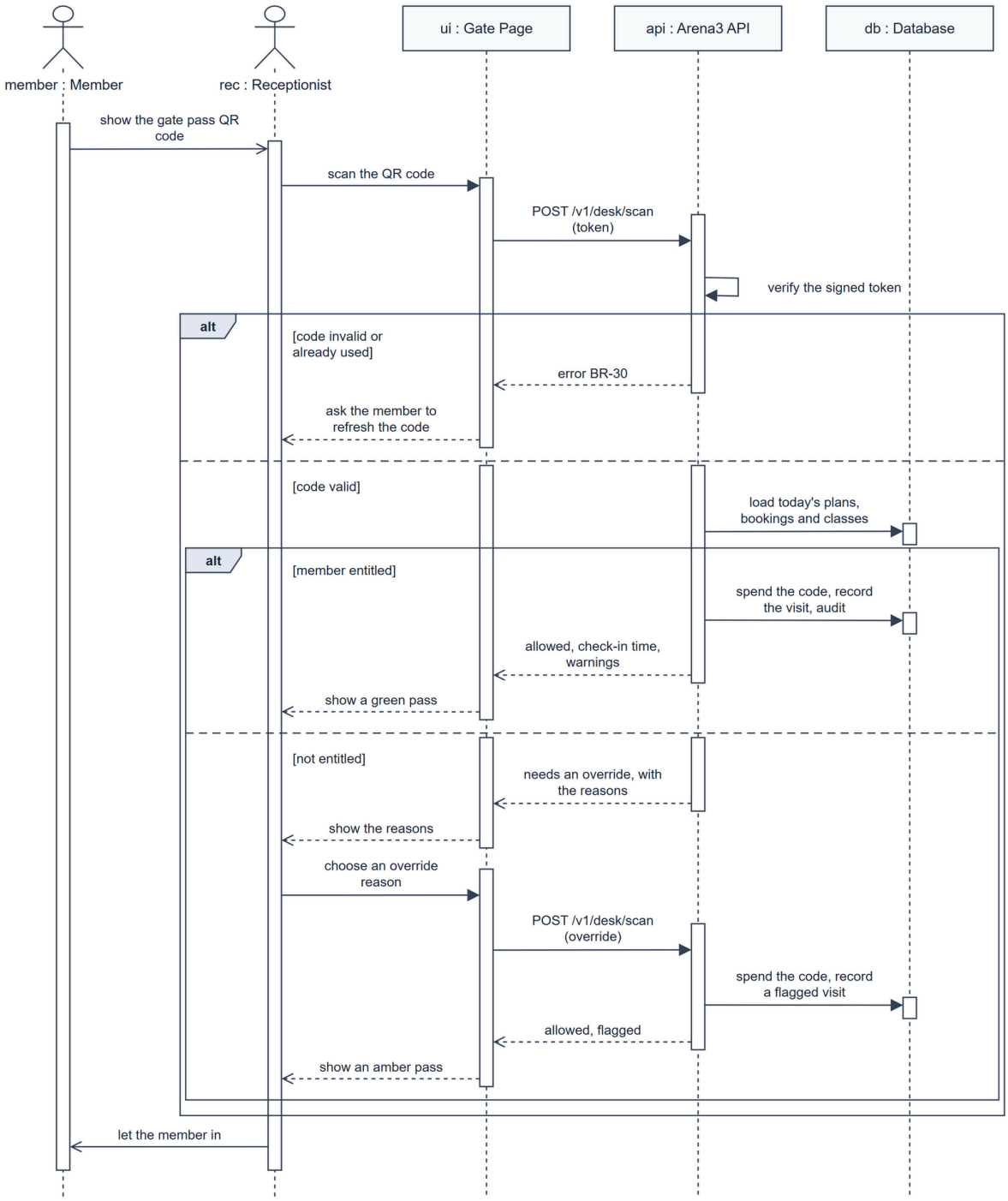


Figure 41. *Sequence diagram: check a member in at the gate*